

数字系统设计

Digital System Design

-----触发器与存储电路

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任课教师

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Office Hours: 玉泉信电楼 308室 (可以钉钉、微信或邮件联系) .

Prerequisites预修课程

- 电子电路基础
- 电子线路
- C语言
- How to learn this Course?
 - Not only listening, thinking and waiting
 - But Exercise, Simulation, Practice!

课程简介

- 课程代码: **671C0050**
- 参考书
 - 阎石,《数字电子技术基础》,第6版,高等教育出版社,2016.
 - 王金明著,《数字系统设计与Verilog HDL》,电子工业出版社,第8版
 - 《数字系统设计-补充讲义》,浙江大学自编讲义
 - Stanford 大学 108A课程notes.
 - R.H.Katz, G.Borriello, *Contemporary Logic Design*, second edition,电子工业出版社,2005.
 - M.M.Mano, 数字设计(第四版),电子工业出版社,2010.
 - » <http://www.prenhall.com/mano>

Other Course Info

- **Website:**

- <http://mypage.zju.edu.cn/wdwd/教学工作/>
- **学在浙大: 数字系统设计**
- Check frequently

- **答疑**

- 玉泉信电楼308室/周一周四下午2:30-5:30
- 上课课间、课后均可
- Email, 钉钉/微信群/数字系统设计, 短信均可

Grading (考核)

Final grades will be computed approximately as follows:

- 平时（含课程作业、期中考试、小测验、Project、出勤等） **40%**

- 期中考试 **10%**
- Homework Sets **15% (12~20)%**
- Project **5% (5~7)%**
- Class Room Check+小测验等 **10% (7~10)%**
- 作业每周一上交, **截止期为课后一周内**有效。

- **Project**

- Project-1必做 (1 members)
- Project-2可选(成绩加分1%~5%, 但不超过平时成绩范围)

- Finial Exam期末**闭卷**考试 - **60%**

授课时间和地点:

- 2025年春夏学期,
 - 周一上午, 第1~2节(8:00-9:35)
 - 星期四上午, 第6~7节(13:25-13:00)
- 地点: 紫金港西2-415(多)
- <http://mypage.zju.edu.cn/wdwd/教学工作/>
- 学在浙大/数字系统设计
- 钉钉/微信/数字系统设计-答疑

课程结构

- 数字理论知识(必备)
 - 数字系统和编码、逻辑代数、门电路
- 数字电路分析与设计
 - 组合逻辑电路
 - 触发器、半导体存贮器、可编程器件
 - 时序逻辑电路
- 脉冲电路与接口
- 控制器与数字系统
 - 状态机
 - 控制器
 - 微码控制器
 - 测试和验证
- 微处理器简介
 - 指令集
 - **4位CPU**介绍

第五章 时序电路基础之 —— 触发器

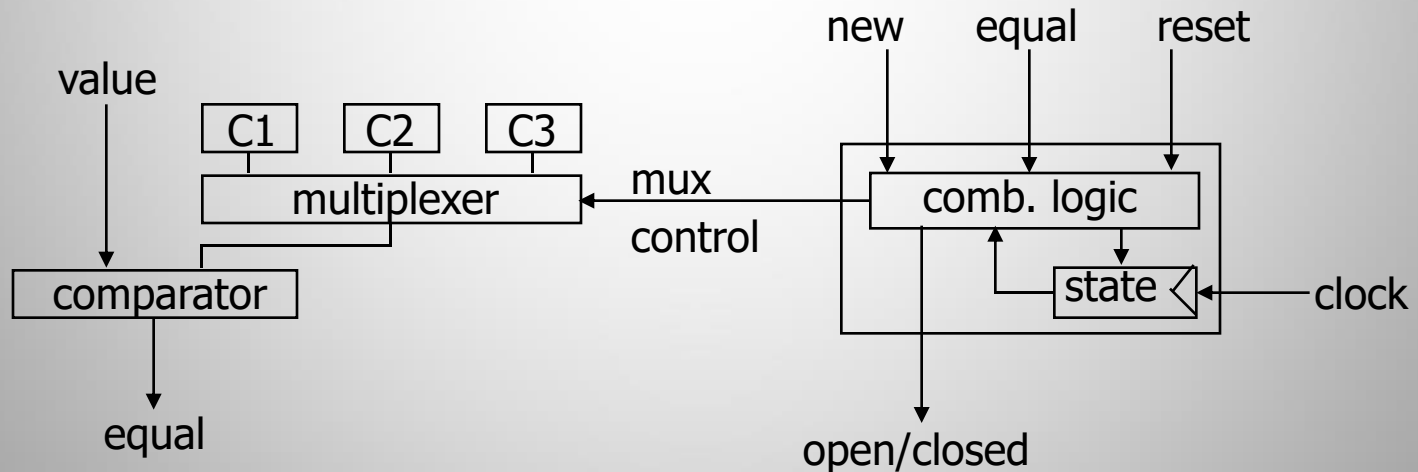
Sequential Logic时序逻辑

- **Sequential Circuits时序**
 - Simple circuits with feedback
 - Latches记忆元件
 - Edge-triggered flip-flops啪嗒作响
- **Timing Methodologies定时**
 - Cascading级联 flip-flops for proper operation
 - Clock skew时钟偏移
- Any prior input-level conditions have no effect on the present outputs because combinational logic circuits have no memory.
- Most digital systems consist of both combinational circuits and memory elements.

Sequential Circuits

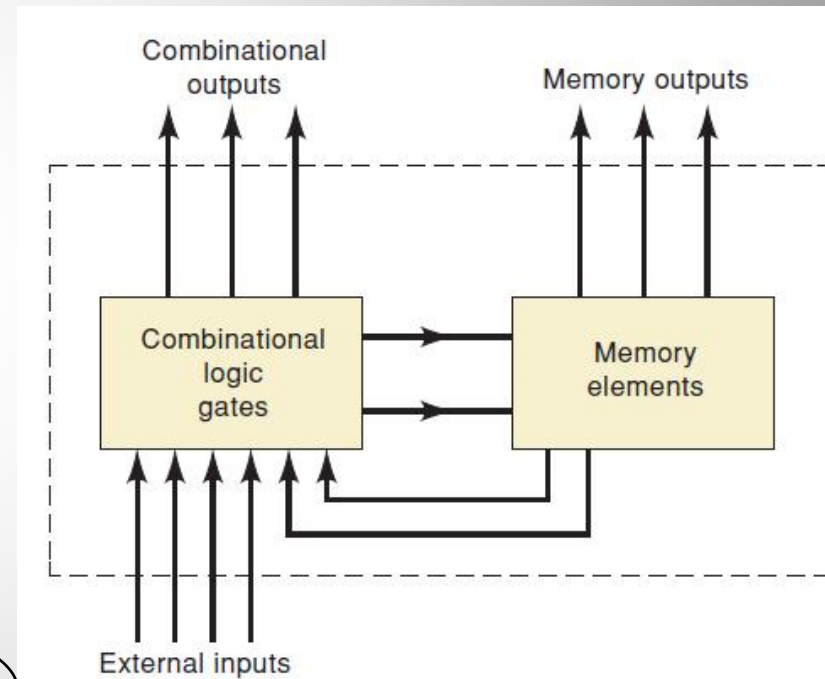
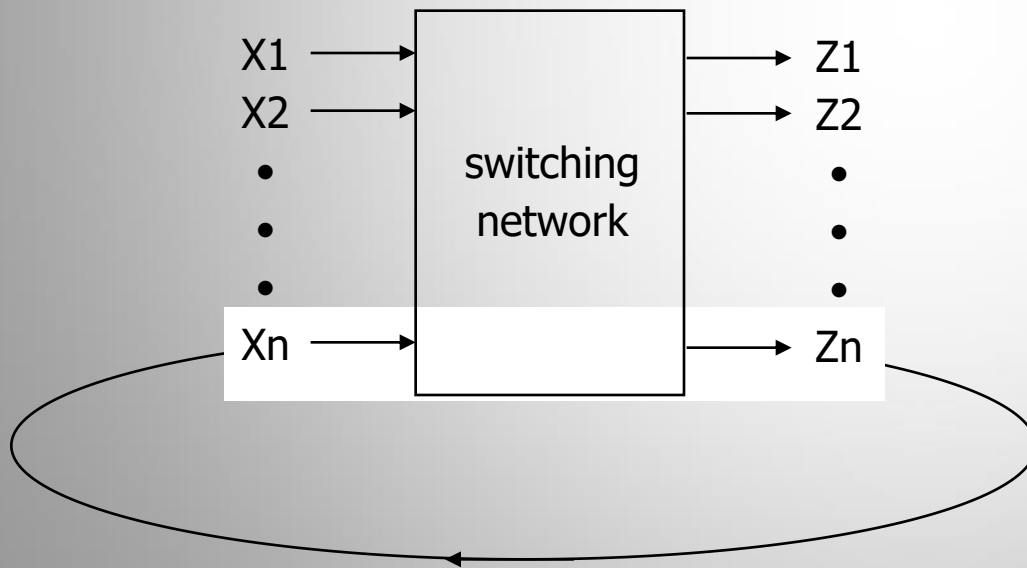
- **Circuits with Feedback**

- Outputs = $f(\text{inputs, past inputs, past outputs})$
- Basis for building "**memory**" into logic circuits
- Door combination lock is an example of a sequential circuit
 - » State is memory
 - » State is an "output" and an "input" to combinational logic
 - » Combination storage elements are also memory



Circuits with Feedback

- **How to control feedback?**
 - What stops values from cycling around endlessly



触发器 flip-flop(F-F)

5.1 概述

一、用于记忆1位二进制信号

1. 有两个能自行保持的状态
2. 根据输入信号可以置成0或1

二、分类

1. 按触发方式(电平, 脉冲, 边沿)
2. 按逻辑功能(RS, JK, D, T)

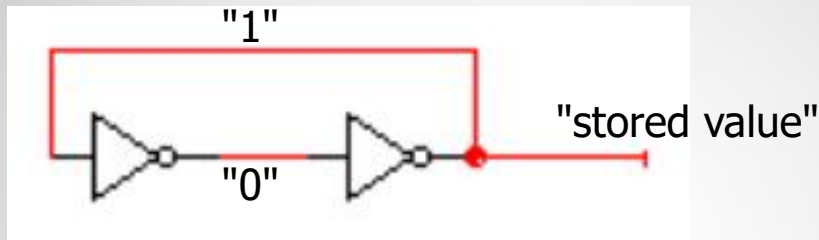
三、存储器件

ROM、SRAM、DRAM、FLASHROM

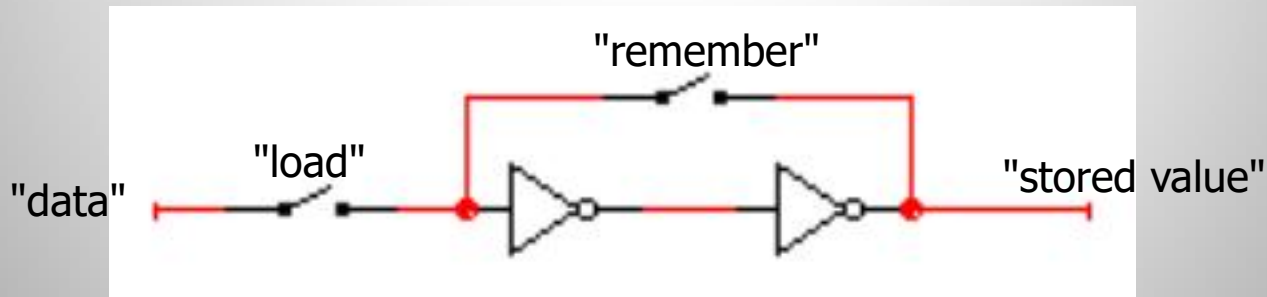
The most important memory element is the **flip-flop**, which is made up of an assembly of logic gates.

Simplest Circuits with Feedback

- **Two inverters form a static memory cell**
 - Will hold value as long as it has power applied



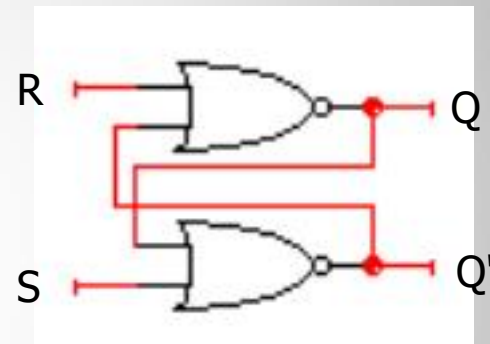
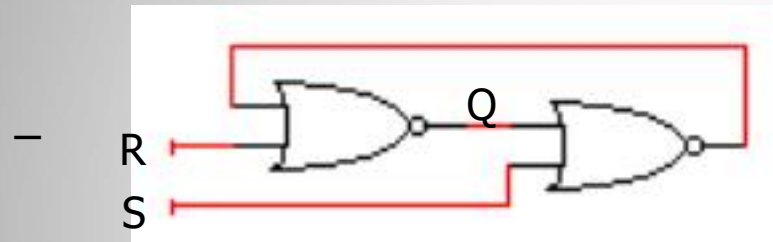
- **How to get a new value into the memory cell?**
 - Selectively break feedback path
 - Load new value into cell



Memory with Cross-coupled Gates

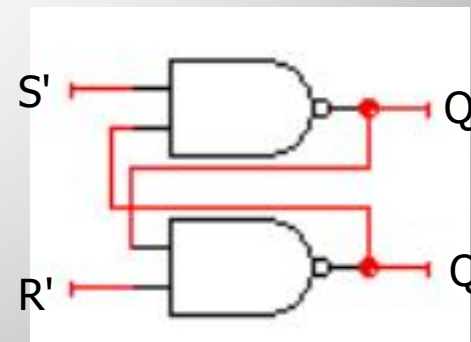
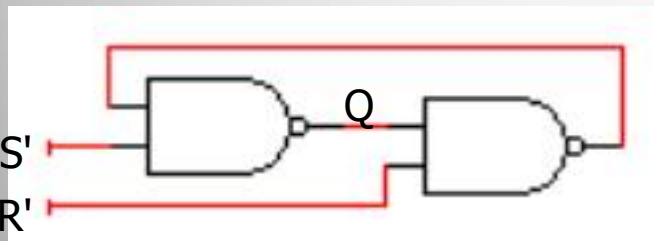
- **Cross-coupled NOR gates**

- Similar to inverter pair, with capability to force output to 0 (reset=1) or 1 (set=1)



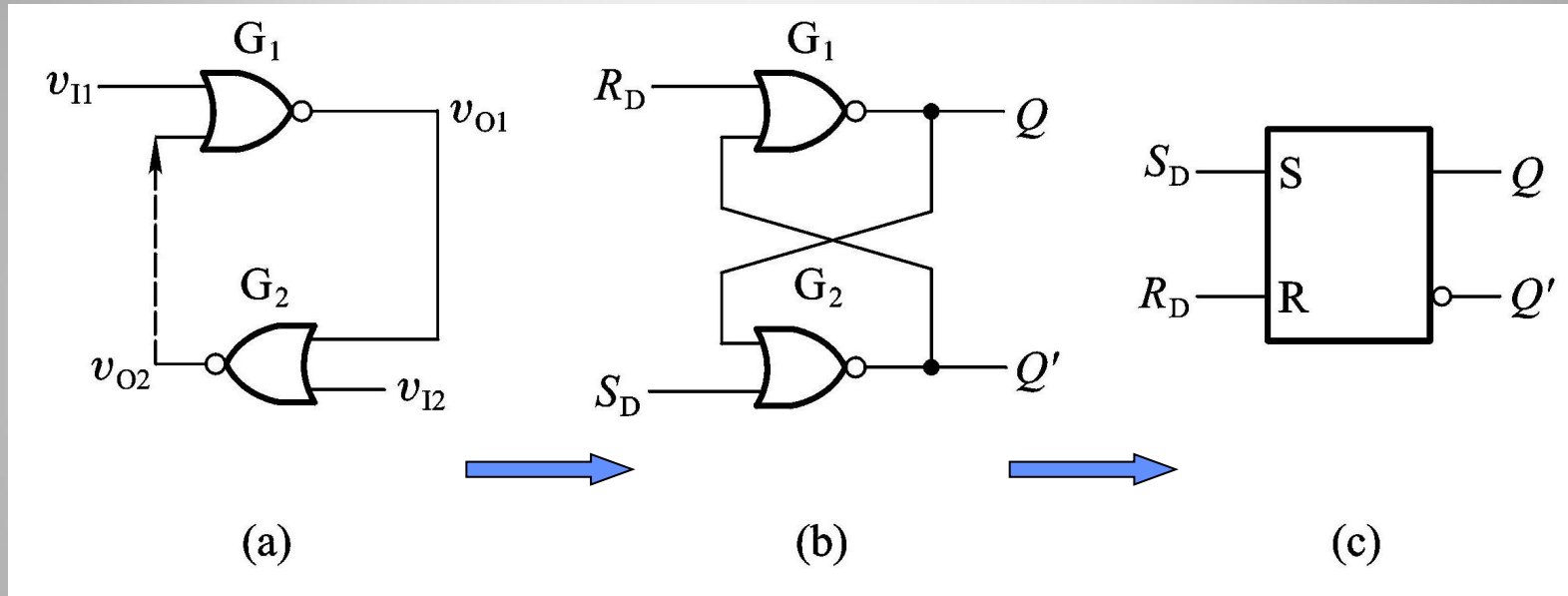
- **Cross-coupled NAND gates**

- Similar to inverter pair, with capability to force output to 0 (reset=0) or 1 (set=0)



5.2 SR锁存器Latch

一、电路结构与工作原理



1.工作原理

两个或非门接成反馈，引出输入端用来置0,1

定义： $Q = 1, Q' = 0$ 为“1”状态

$Q = 0, Q' = 1$ 为“0”状态

R_D 为置0输入端， S_D 为置1输入端

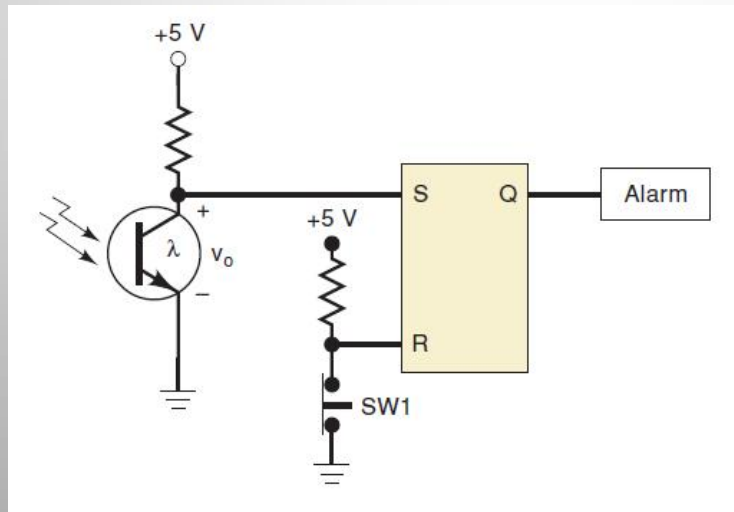
2.根据工作原理得到真值表

S_D 和 R_D 的“1”信号同时消失后， Q^* 不定 ①

所以正常工作下，应遵循 $S_D R_D = 0$ 的约束条件。

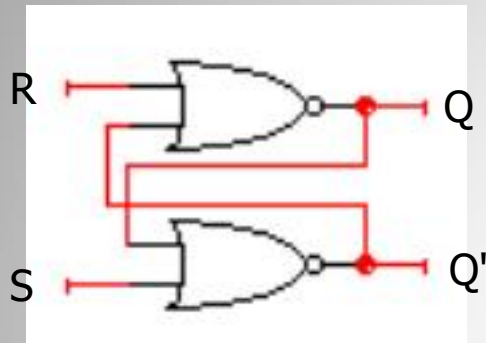
S_D	R_D	Q	Q^*
0	0	0	0
0	0	1	1
1	0	0	1
1	0	1	1
0	1	0	0
0	1	1	0
1	1	0	0 ^①
1	1	1	0 ^①

应用：



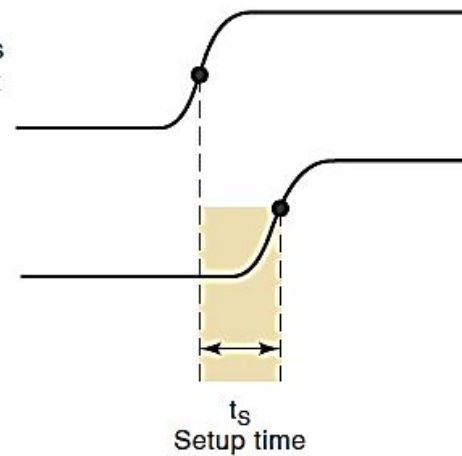
Timing Behavior

Setup and Hold Times



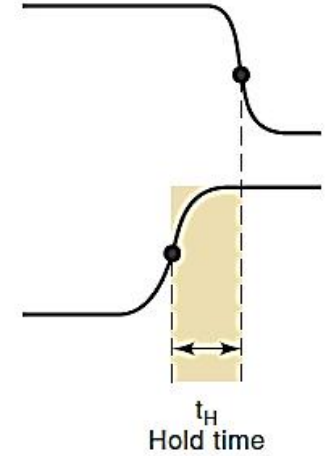
Synchronous control input

Clock input



t_s
Setup time

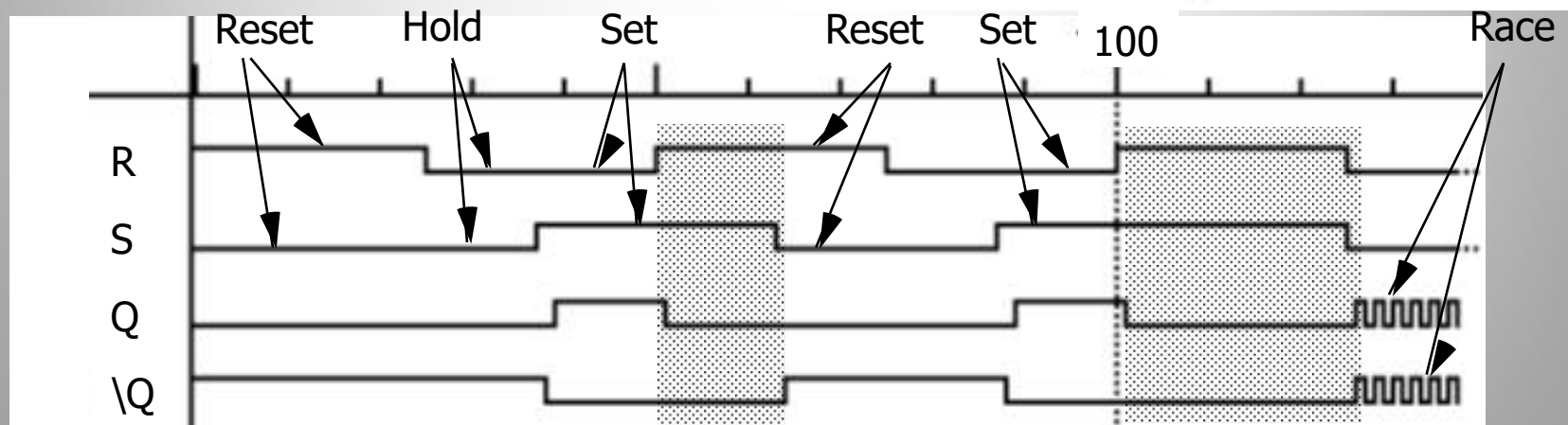
(a)



t_H
Hold time

(b)

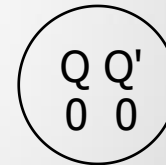
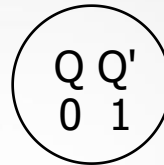
(b)



State Behavior of R-S latch

- Truth table of R-S latch behavior

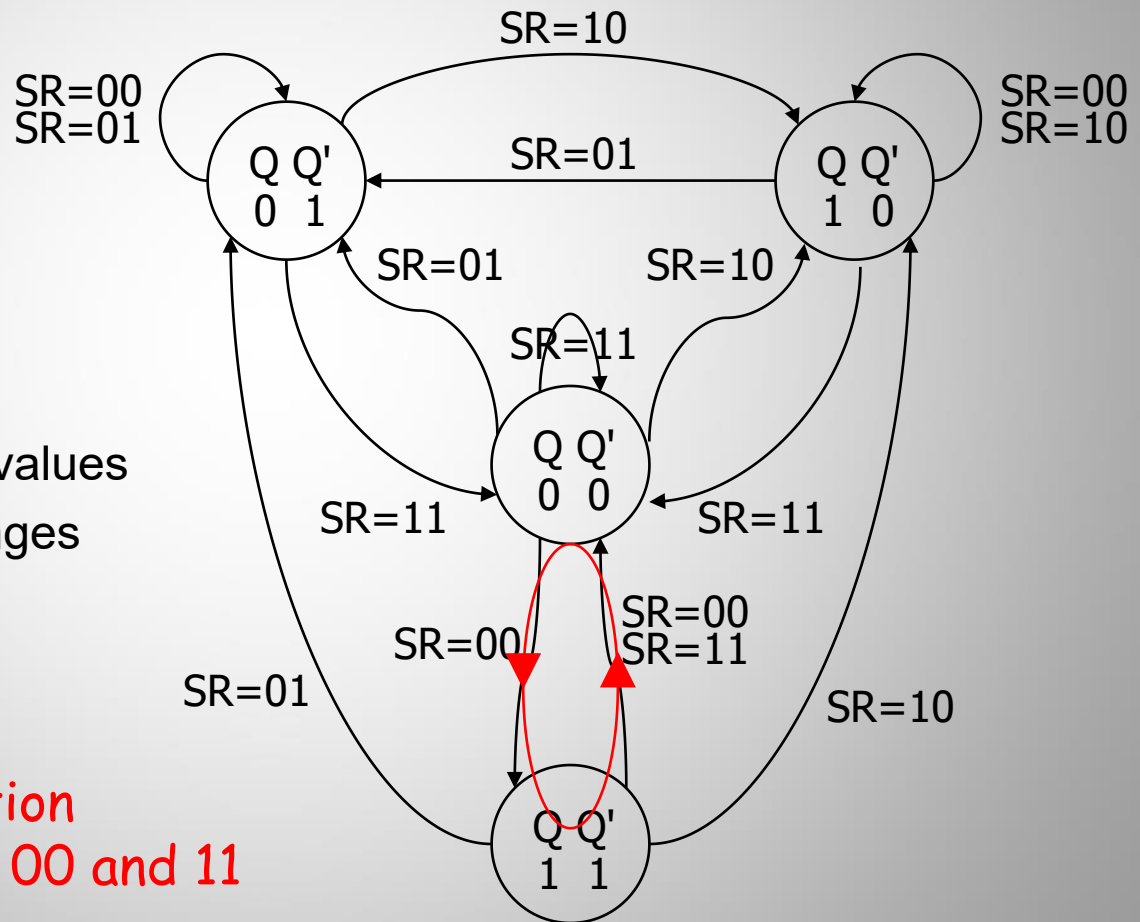
S	R	Q
0	0	hold
0	1	0
1	0	1
1	1	unstable



Theoretical R-S Latch Behavior

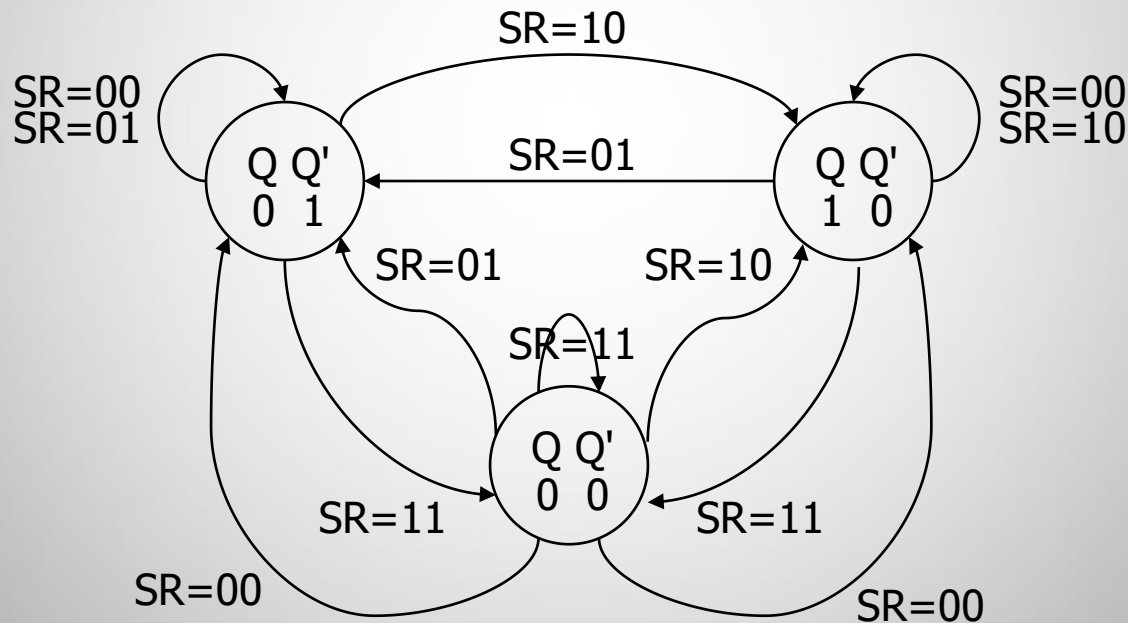
- State Diagram
 - States: possible values
 - Transitions: changes based on inputs

possible oscillation
between states 00 and 11



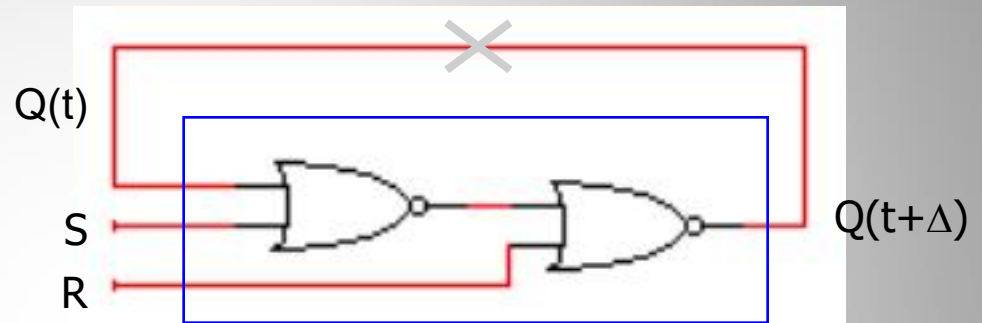
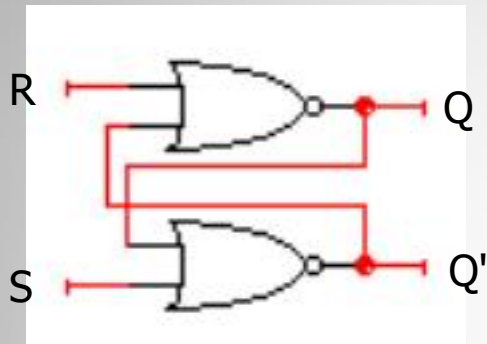
Observed R-S Latch Behavior

- Very difficult to observe R-S latch in the 1-1 state
 - One of R or S usually changes first
- **Ambiguously returns to state 0-1 or 1-0**
 - A so-called "race condition"
 - Or non-deterministic transition



R-S Latch Analysis

- Break feedback path



S	R	Q(t)	Q(t+Δ)	
0	0	0	0	hold
0	0	1	1	
0	1	0	0	reset
0	1	1	0	
1	0	0	1	set
1	0	1	1	
1	1	0	X	not allowed
1	1	1	X	

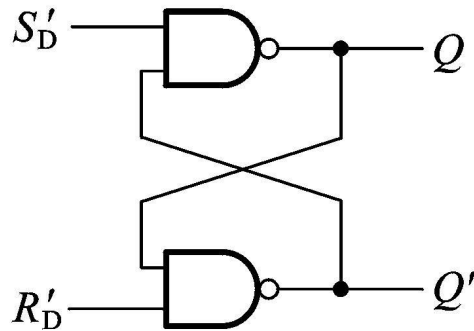
		S	
Q(t)	0	0	X
	1	0	X
		R	
		0	1
		1	1

characteristic equation
 $Q(t+\Delta) = S + R' Q(t)$

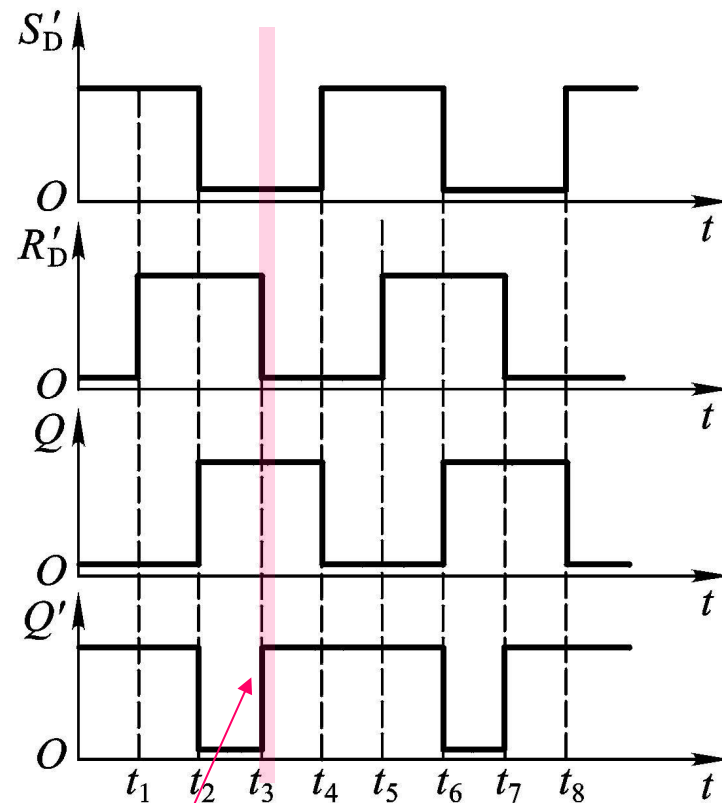
二、动作特点

在任何时刻，输入都能直接改变输出的状态。

例：



(a)



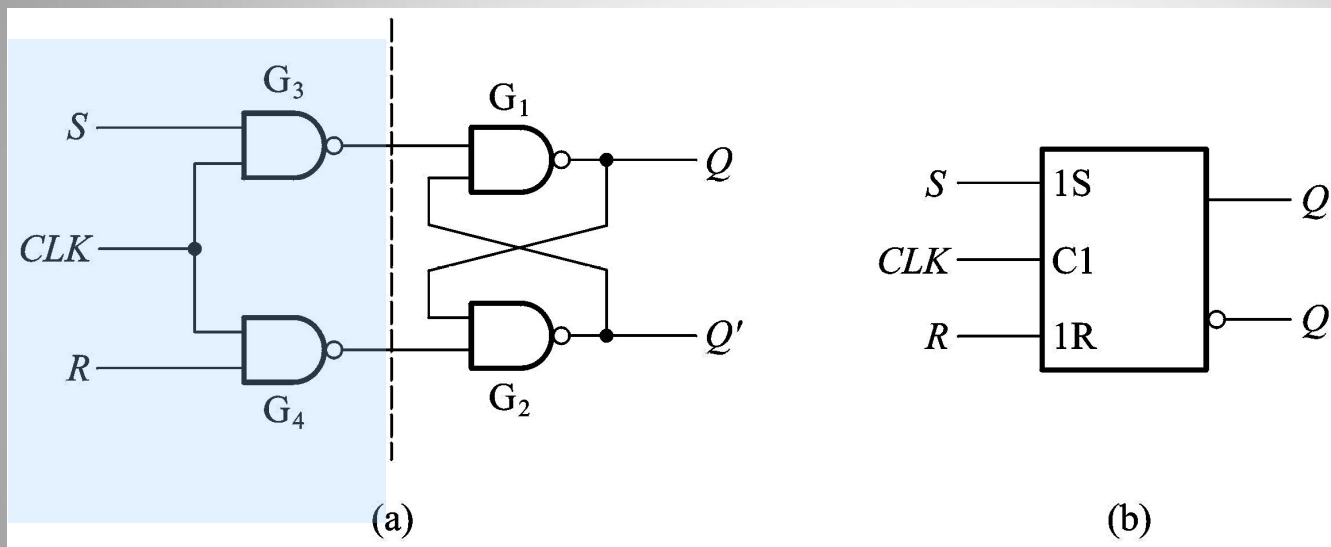
(b)

S'_D 和 R'_D 同时为 0 $\Rightarrow Q, Q'$ 同为 1

5.3 触发器

5.3.1 电平触发的触发器

一、电路结构与工作原理



CLK	S	R	Q	Q^*
0	X	X	0	0
0	X	X	1	1
1	0	0	0	0
1	0	0	1	1
1	1	0	0	1
1	1	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	1	0	1*
1	1	1	1	1*

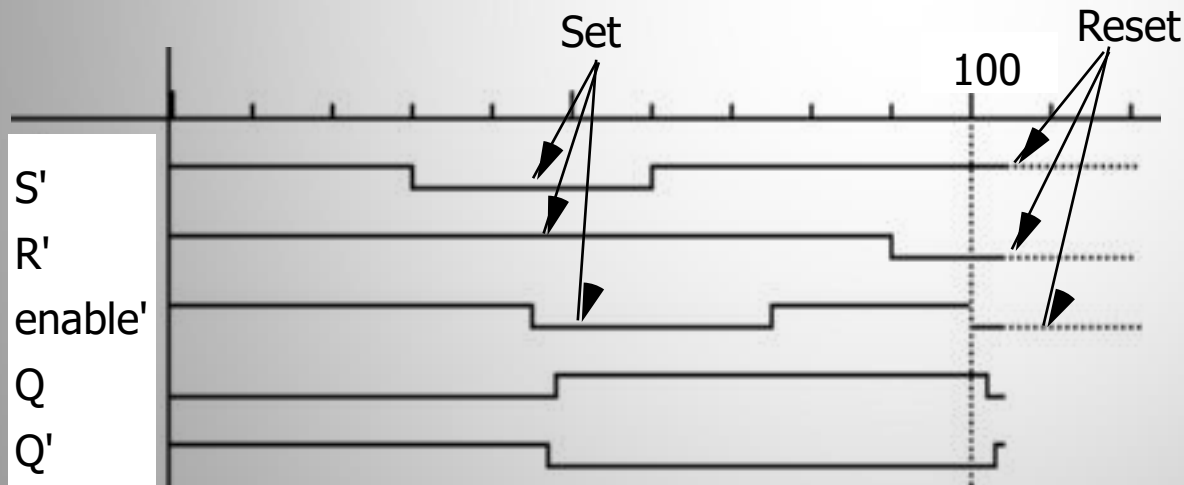
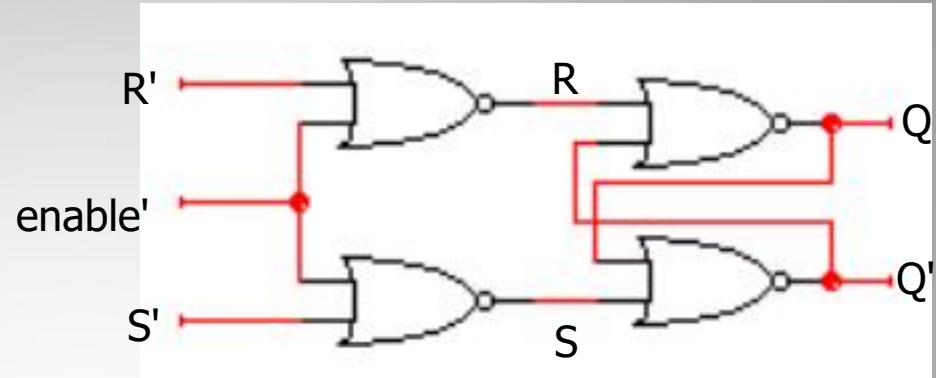
输入控制门 + 基本 **RS** 触发器

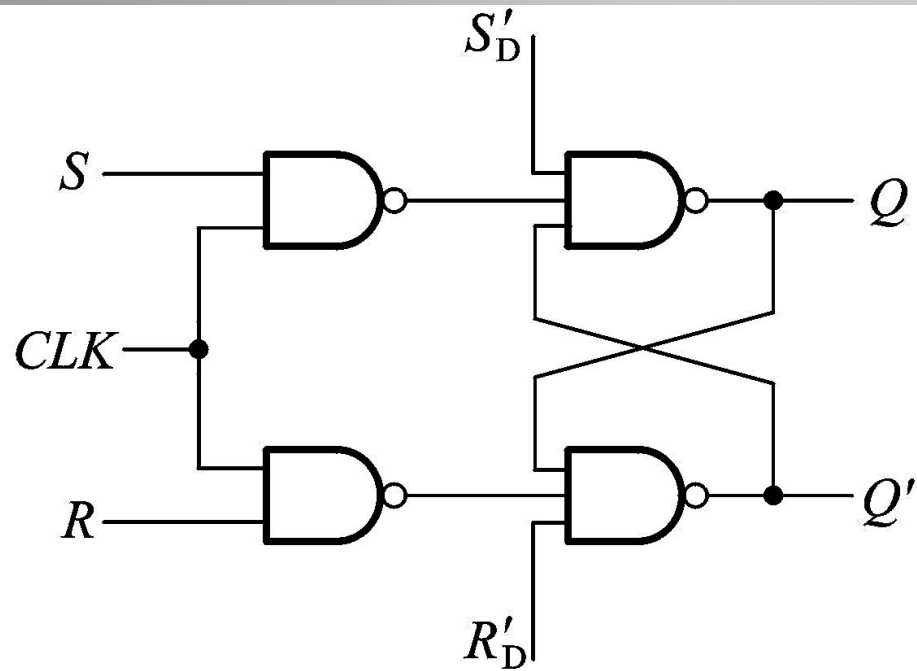
只有触发信号 **CLK** 到达，**S** 和 **R** 才起作用。

Gated R-S Latch

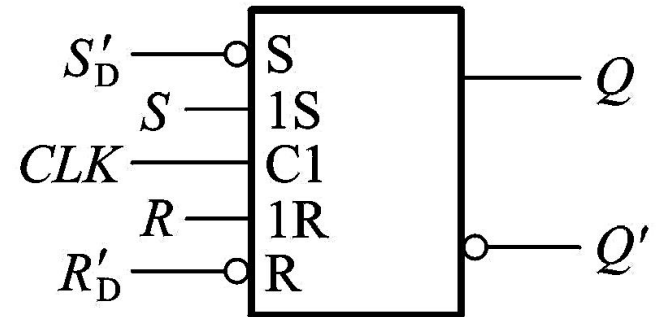
- **Control when R and S inputs matter**

- Otherwise, the slightest glitch on R or S while enable is low could cause change in value stored





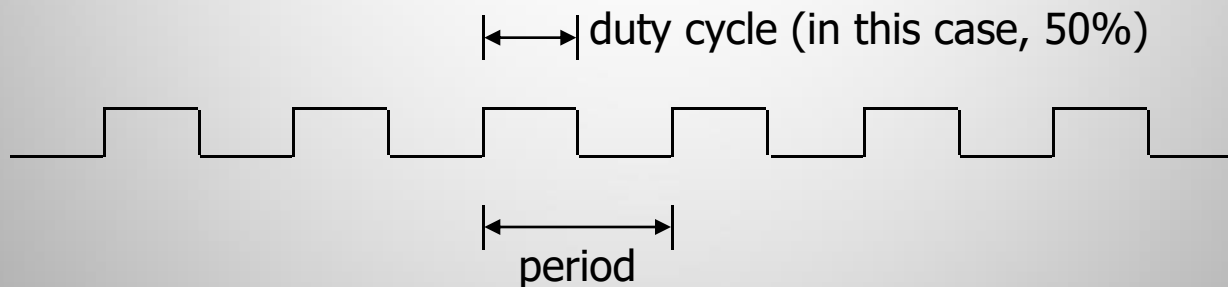
(a)



(b)

Clocks

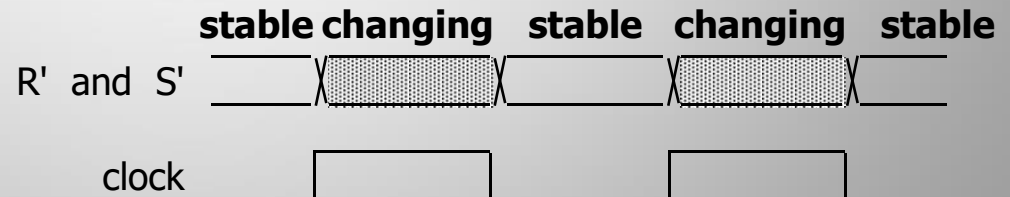
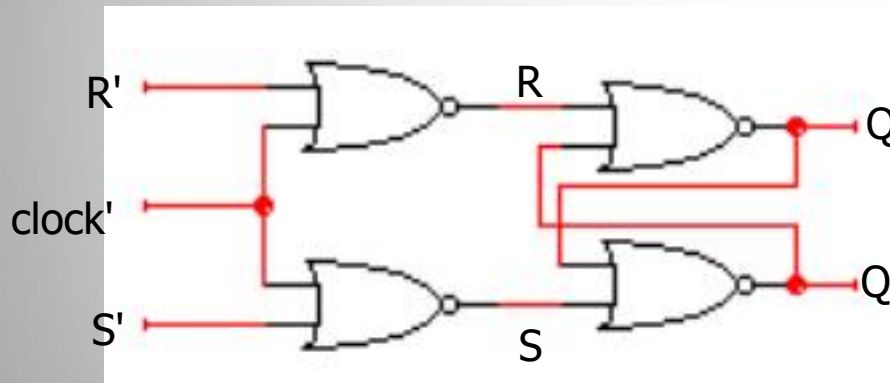
- **Used to keep time**
 - Wait long enough for inputs (R' and S') to settle
 - Then allow to have effect on value stored
- **Clocks are regular periodic signals**
 - Period (time between ticks)
 - Duty-cycle (time clock is high between ticks - expressed as % of period)



Clocks (cont'd)

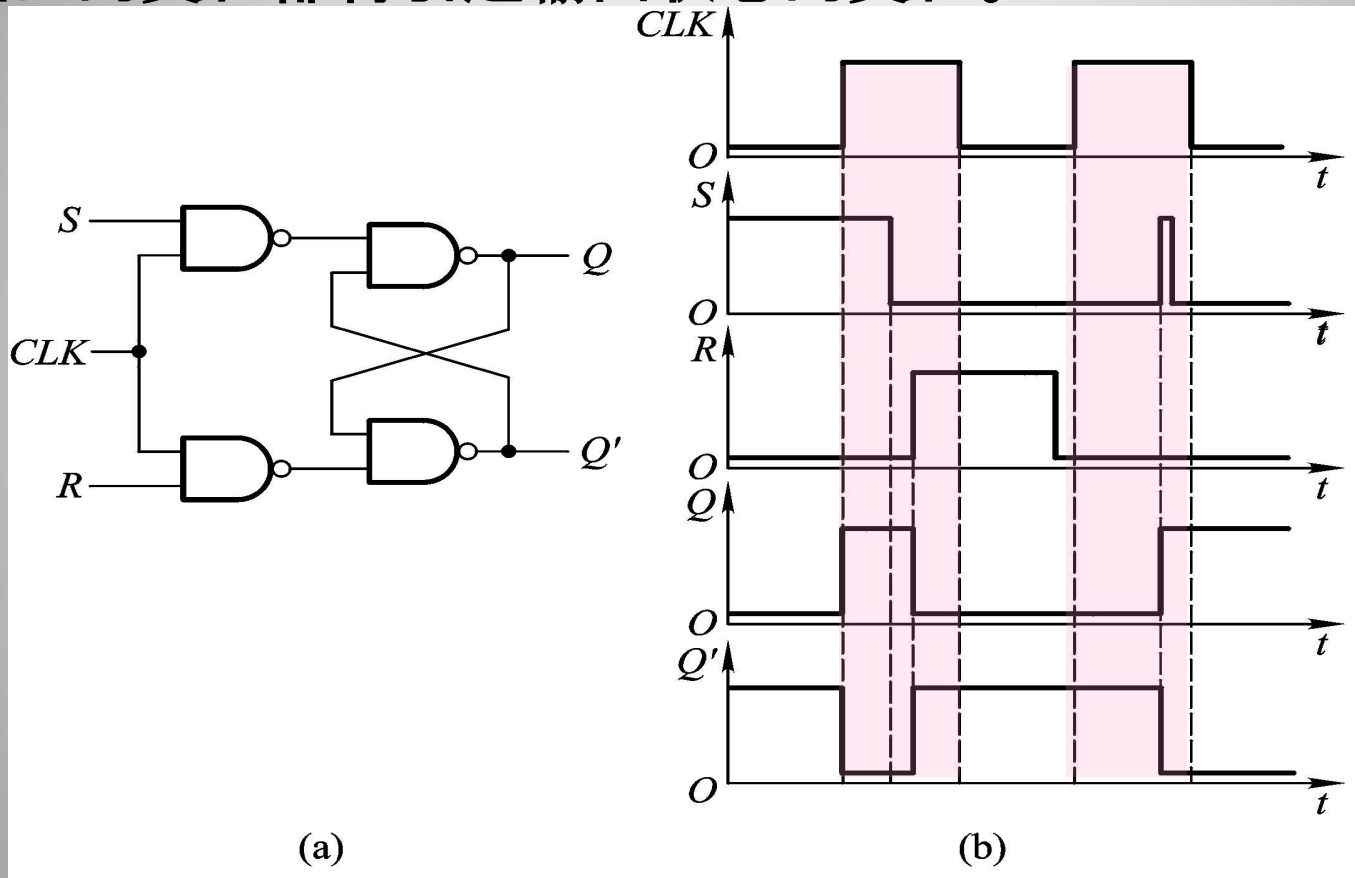
- **Controlling an R-S latch with a clock**

- Can't let R and S change while clock is active (allowing R and S to pass)
- Only have half of clock period for signal changes to propagate
- Signals must be stable for the other half of clock period



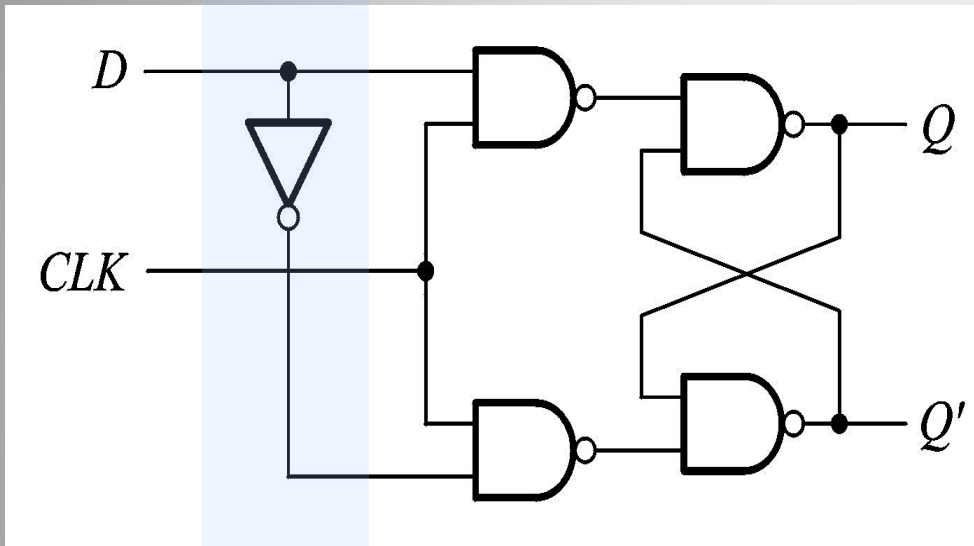
二、动作特点

在 $CLK=1$ 的全部时间里，
 S 和 R 的变化都将引起输出状态的变化。



在 $CLK = 1$ 期间， Q 和 Q' 可能随 S 、 R 变化多次翻转

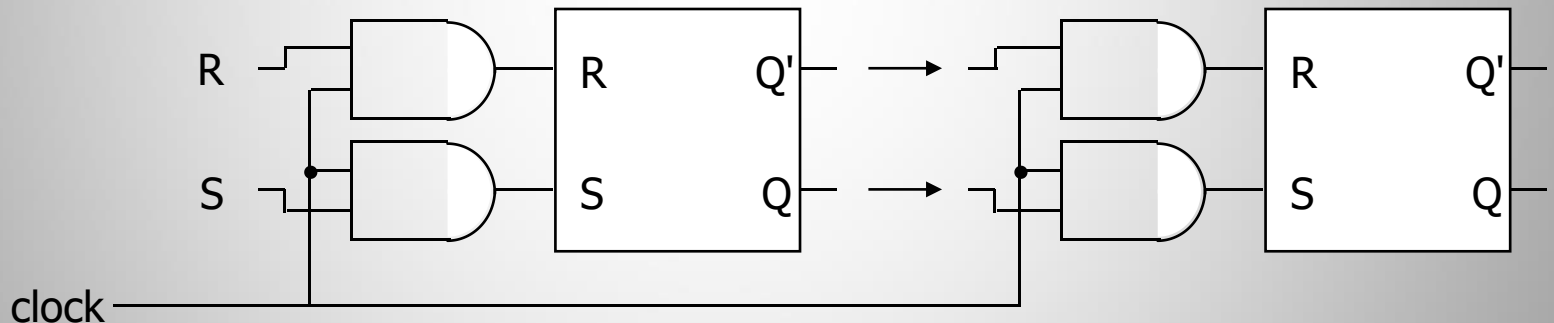
D触发器



CLK	S	R	Q	Q^*
0	X	X	0	0
0	X	X	1	1
1	0	0	0	0
1	0	0	1	1
1	1	0	0	1
1	1	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	1	0	1*
1	1	1	1	1*

Cascading Latches

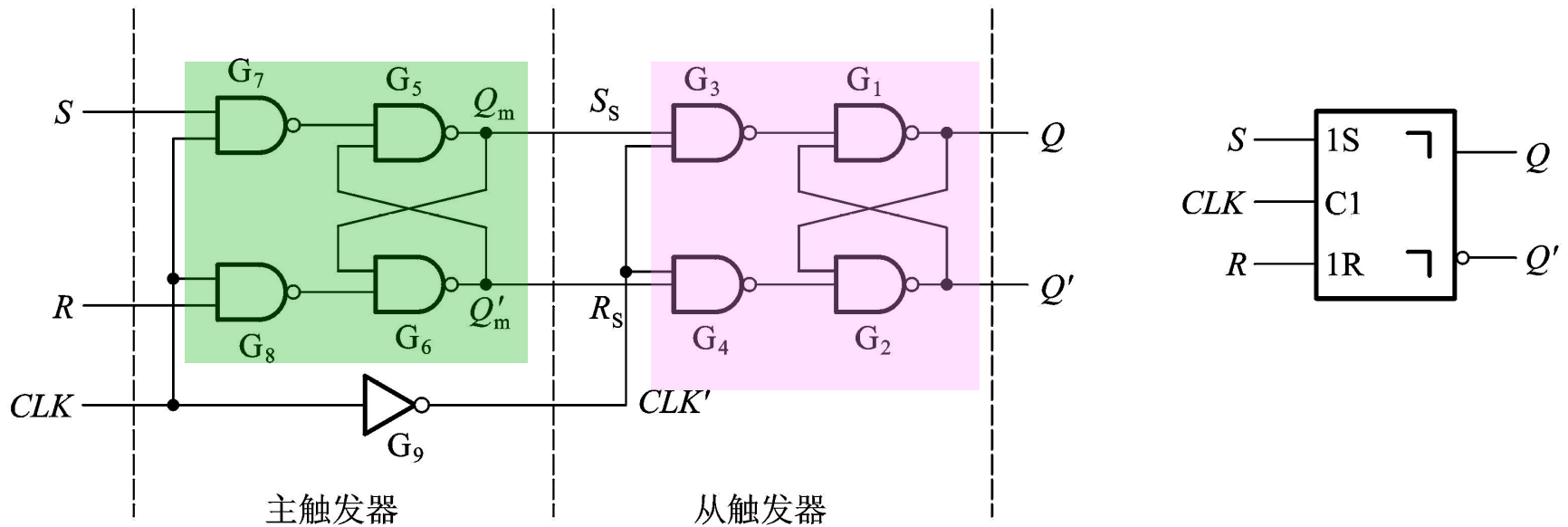
- **Connect output of one latch to input of another**
- **How to stop changes from racing through chain?**
 - Need to control flow of data from one latch to the next
 - Advance from one latch per clock period
 - Worry about logic between latches (arrows) that is too fast



5.3.3 脉冲触发的触发器

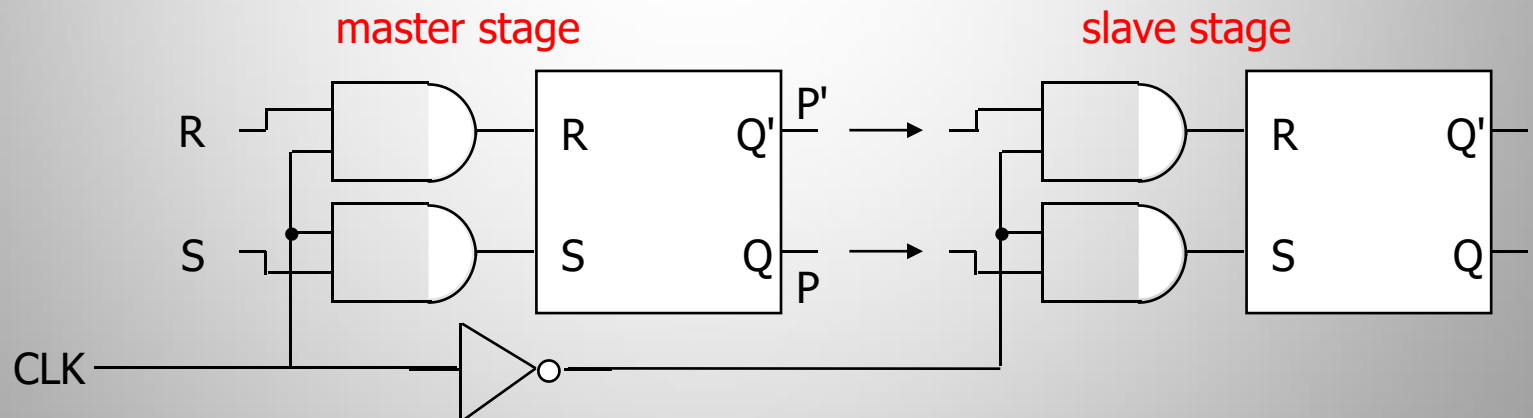
提高可靠性，要求每个**CLK**周期
输出状态只能改变1次

一、电路结构与工作原理



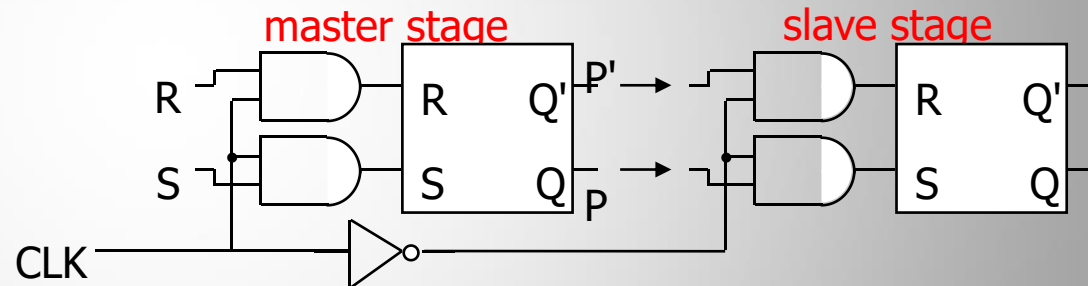
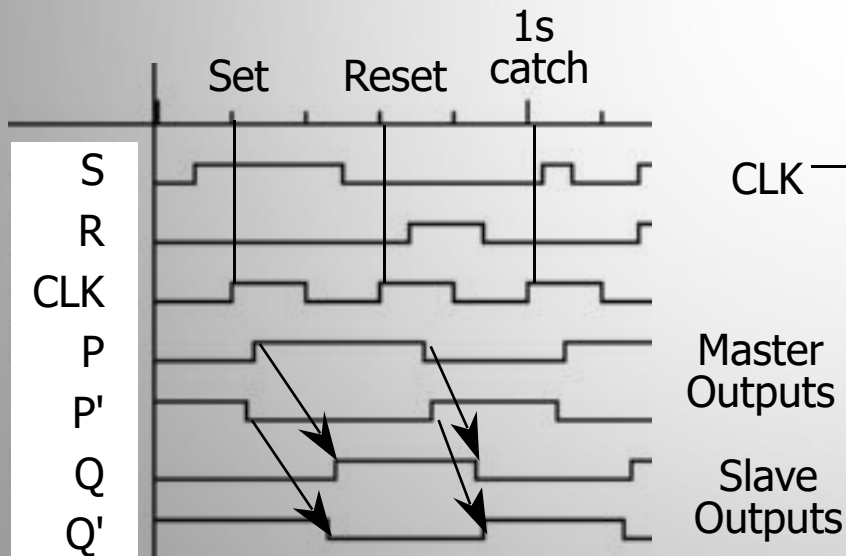
Master-Slave Structure

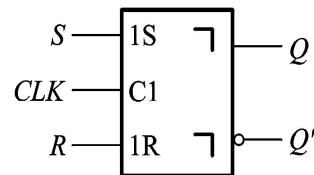
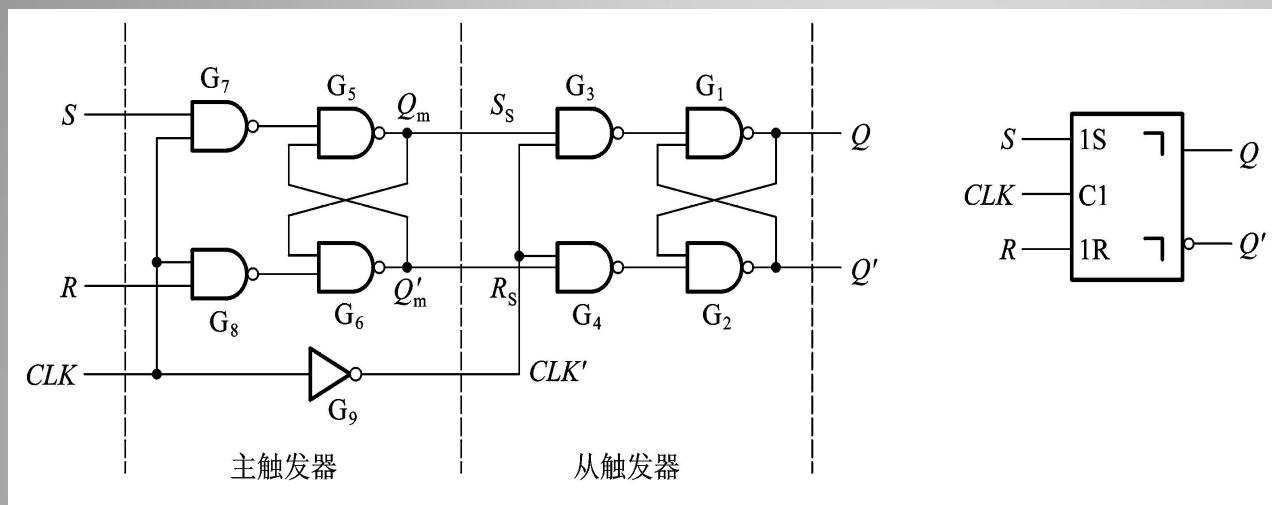
- **Break flow by alternating clocks (like an air-lock)**
 - Use positive clock to latch inputs into one R-S latch
 - Use negative clock to change outputs with another R-S latch
- **View pair as one basic unit**
 - master-slave flip-flop
 - twice as much logic
 - output changes a few gate delays after the falling edge of clock but does not affect any cascaded flip-flops



The 1s Catching Problem

- In first R-S stage of master-slave FF
 - 0-1-0 glitch on R or S while clock is high “caught” by master stage 捕捉
 - Leads to constraints on logic to be hazard-free





CLK	S	R	Q	Q^*
X	X	X	X	Q^n
	0	0	0	0
	0	0	1	1
	1	0	0	1
	1	0	1	1
	0	1	0	0
	0	1	1	0
	1	1	0	1*
	1	1	1	1*

1. 主从 SR 触发器

(1) $clk = 1$ 时，“主”按 S, R 翻转，“从”保持

(2) clk 下降沿到达时，“主”保持，

“从”根据“主”的状态翻转

所以每个 clk 周期，输出状态只可能改变一次

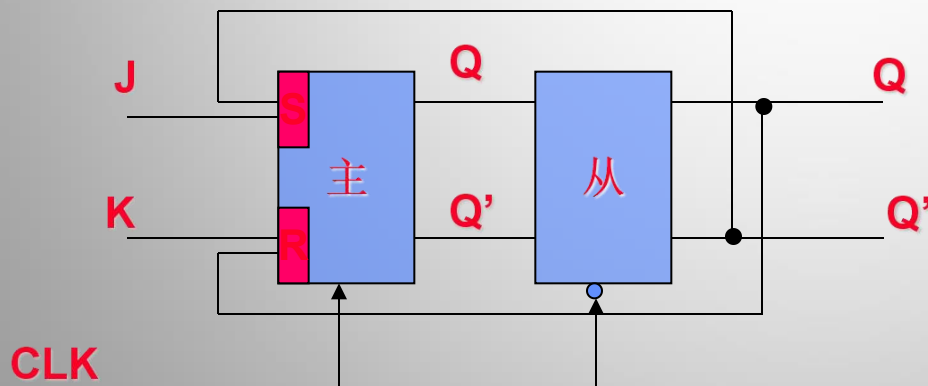
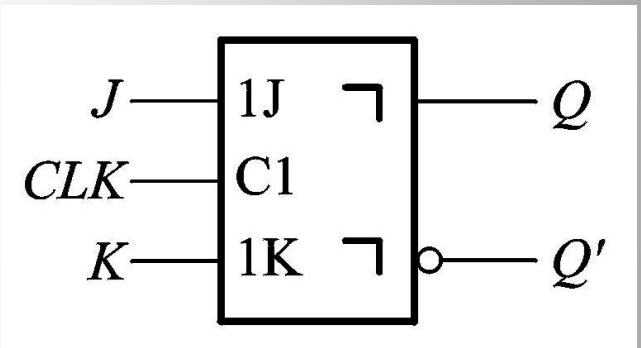
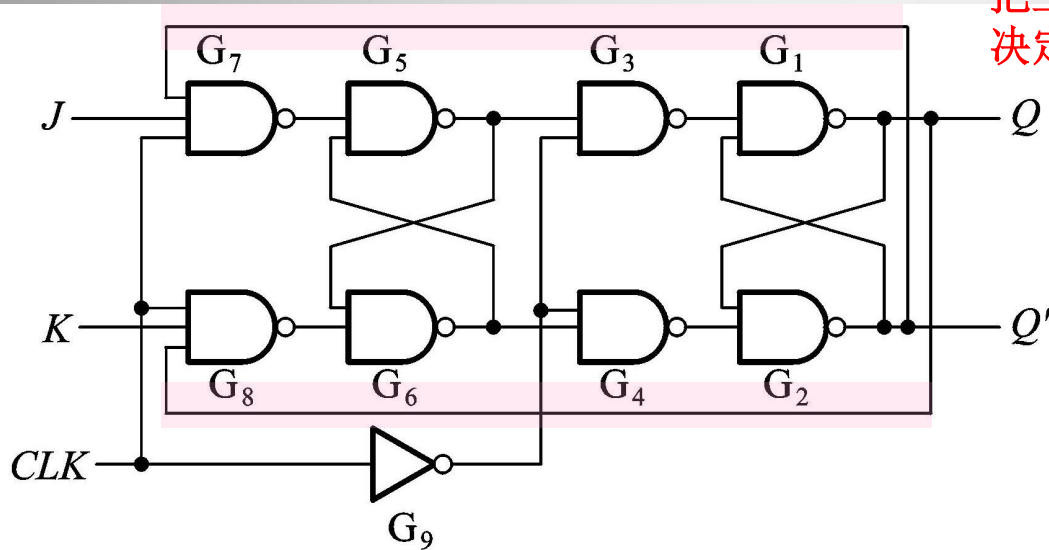
问：在 $CLK=1$ 时，如果 $S=R=1$ ，主输出“11”，在 CLK 下降沿到来后，主输出会停在“??”

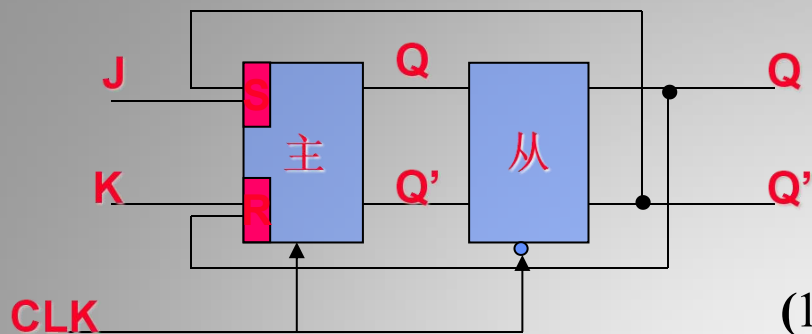
2. 主从JK触发器

为解除约束

即使出现 $S = R = 1$ 的情况下， Q^* 也是确定的

把上次的输出加入反馈，
决定输入





(1) 若 $J = 1, K = 0$ 则 $clk = 1$ 时,

$$\begin{cases} Q^* = 1, \text{“主” 保持} \\ Q^* = 0, \text{“主”} = 1 \end{cases} \Rightarrow clk \downarrow \text{后, “从”} = 1$$

(2) 若 $J = 0, K = 1$ 则 $clk = 1$ 时,

$$\begin{cases} Q^* = 1, \text{“主”} = 0 \\ Q^* = 0, \text{“主” 保持} \end{cases} \Rightarrow clk \downarrow \text{后, “从”} = 0$$

(3) 若 $J = K = 0$ 则 $clk = 1$ 时,

$$\begin{cases} Q^* = 1 \\ Q^* = 0 \end{cases} \text{“主” 保持} \Rightarrow clk \downarrow \text{后, “从” 保持}$$

(4) 若 $J = K = 1$ 则 $clk = 1$ 时,

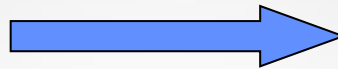
$$\begin{cases} \text{若 } Q^* = 1, \text{则 “主” 置} 0 \\ \text{若 } Q^* = 0, \text{则 “主” 置} 1 \end{cases}$$

加入反馈后的效果

$$clk \downarrow \text{后, “从”} = (Q^*)'$$

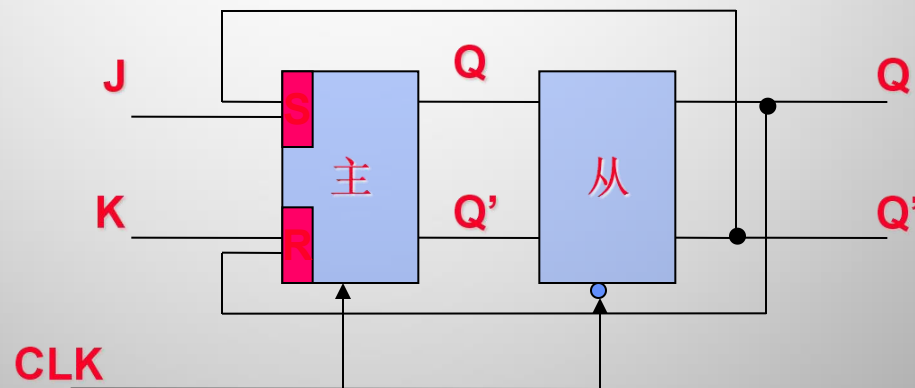
(5) 列出真值表

CLK	S	R	Q	Q^*
X	X	X	X	Q^*
	0	0	0	0
	0	0	1	1
	1	0	0	1
	1	0	1	1
	0	1	0	0
	0	1	1	0
	1	1	0	1*
	1	1	1	1*



CLK	J	K	Q	Q^*
X	X	X	X	Q^*
	0	0	0	0
	0	0	1	1
	1	0	0	1
	1	0	1	1
	0	1	0	0
	0	1	1	0
	1	1	0	1
	1	1	1	0

CLK=1时的脉冲期间，
输入的中间变化会影响输出；
要分析整个CLK=1的输入，
才能知道输出会是什么。



二、脉冲触发方式的动作特点

1. 分两步动作：

第一步 $clk = 1$ 时，“主”接收信号，“从”保持

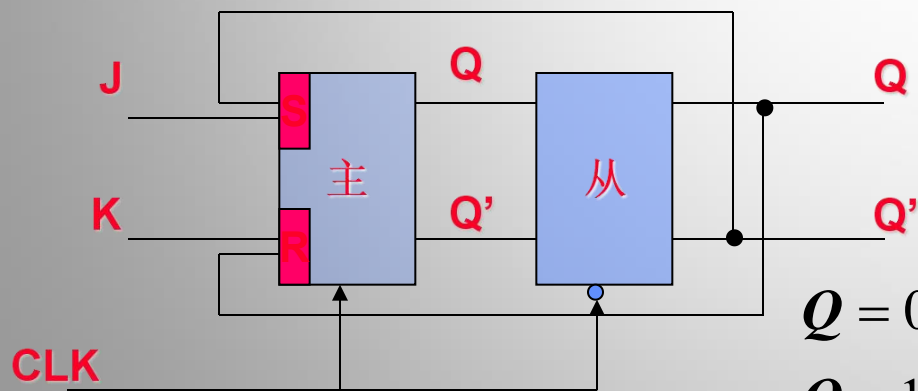
第二步 $clk \downarrow$ 到达后，“从”按“主”状态翻转

$\therefore$ 输出状态只能改变一次

2. 主从 SR ，“主”为同步 SR ， $clk = 1$ 的全部时间里输入信号对“主”都起控制作用

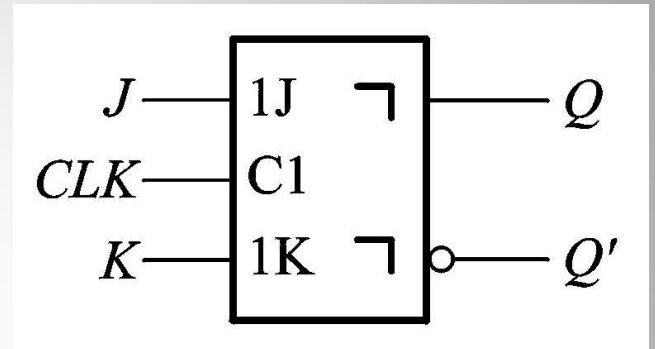
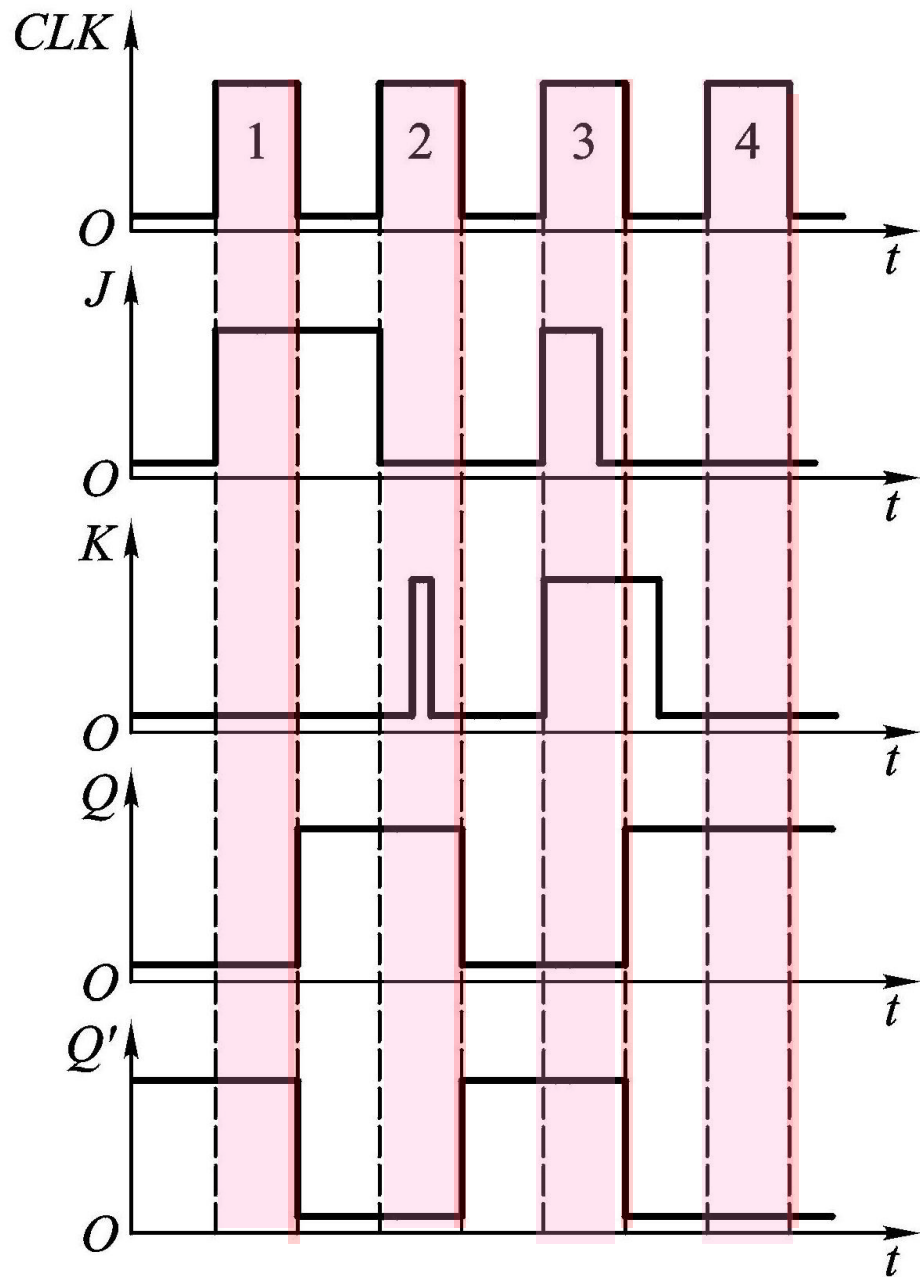
* 但主从 JK 在 clk 高电平期间，“主”只可能翻转一次

$\therefore$ 在 $clk = 1$ 期间里输入发生变化时，要找出 $clk \downarrow$ 前 Q' 最后的状态，决定 Q^* 。



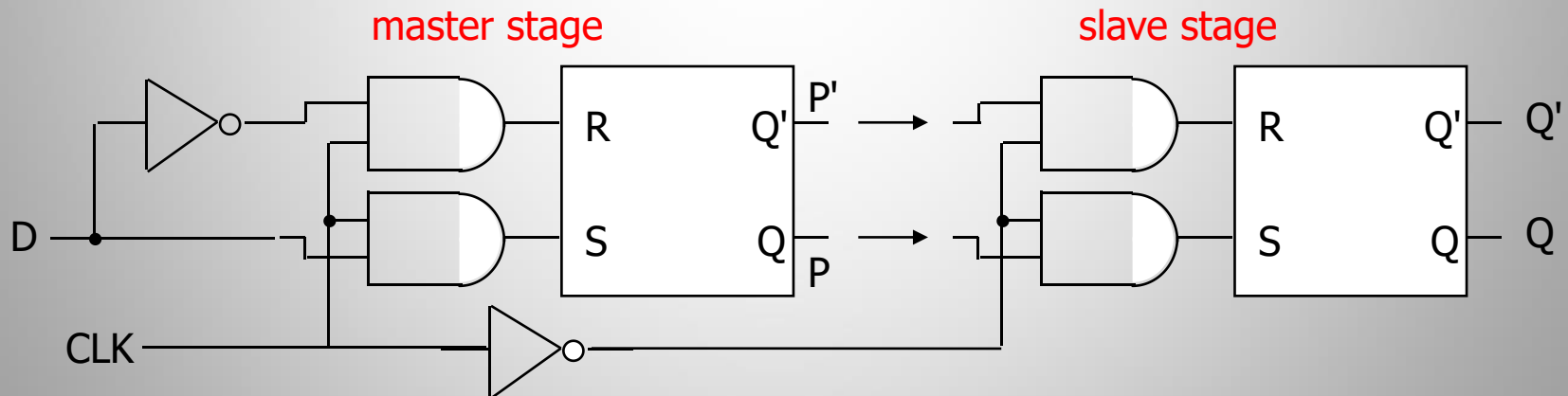
$Q = 0$ 时，只允许 $J = 1$ 的信号进入主触发器

$Q = 1$ 时，只允许 $K = 1$ 的信号进入主触发器



D Flip-Flop

- Make S and R complements of each other
 - Eliminates 1s catching problem
 - Can't just hold previous value (must have new value ready every clock period)
 - Value of D just before clock goes low is what is stored in flip-flop
 - Can make R-S flip-flop by adding logic to make $D = S + R' Q$



10 gates

5.3.2边沿触发的触发器

为了提高可靠性, 增强抗干扰能力,
希望触发器的次态仅取决于CLK的下降沿(或上升沿)到来
时的输入信号状态, 与在此前、后输入的状态没有关系。

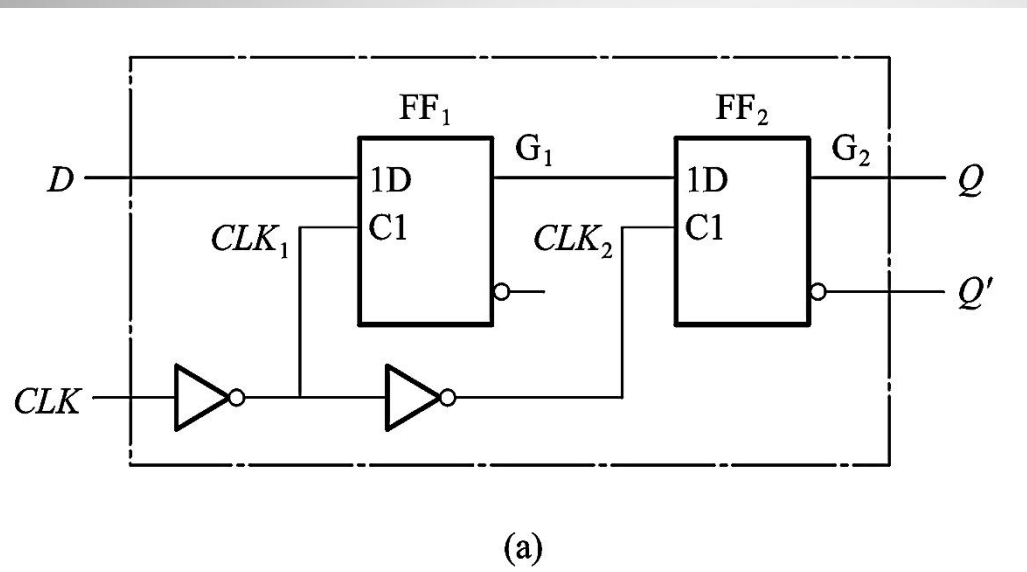
这样我们不需要分析整个**CLK=1**期间的输入变化,
只需要分析**CLK边沿几个时刻**的输入情况。

实现方式分类:

- 1)用CMOS传输门的边沿触发器
- 2)维持阻塞触发器
- 3)用门电路tpd的边沿触发器
- ...

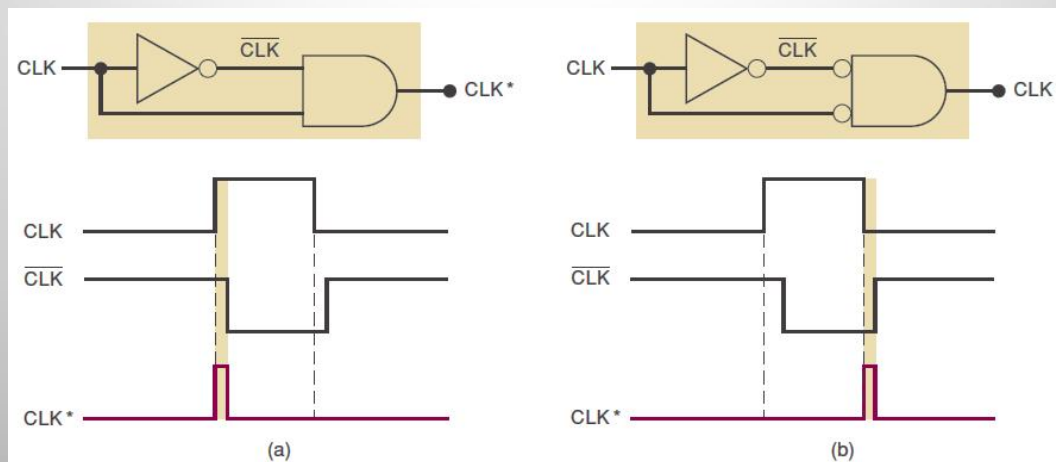
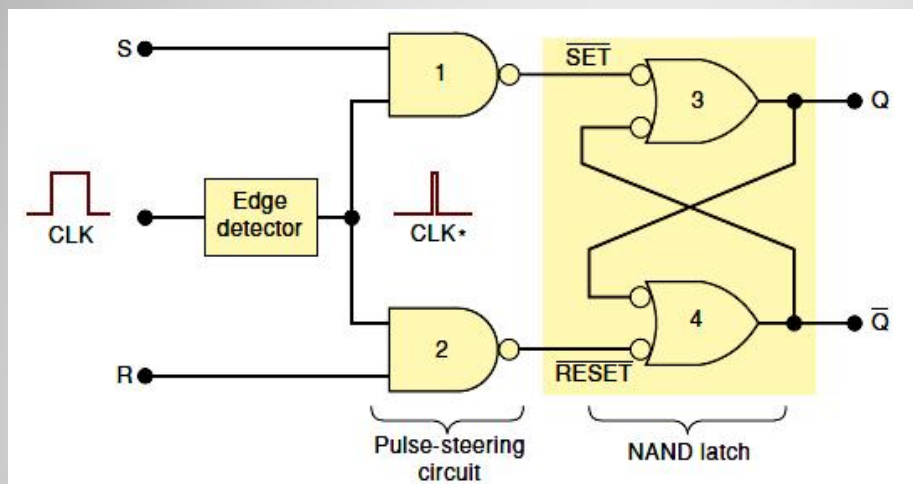
一、电路结构和工作原理

1、用两个电平触发D触发器组成的边沿触发器

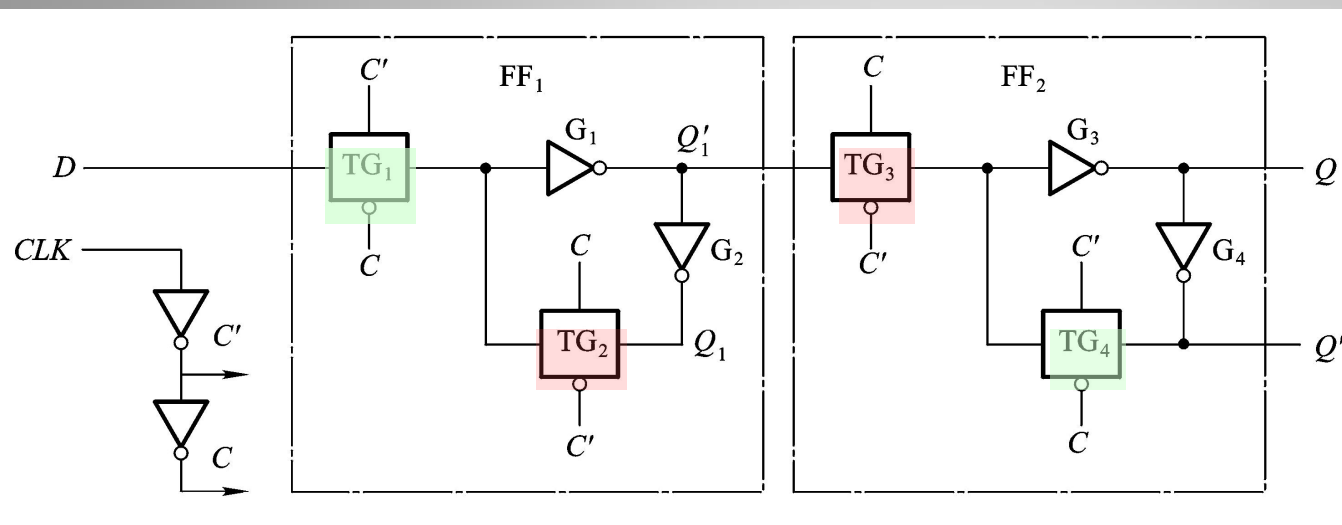


边沿触发器

2、用门电路tpd的边沿触发器



3.利用CMOS传输门的边沿触发器



(4)列出真值表

CLK	D	Q	Q^*
X	X	X	Q
$\uparrow$	0	X	0
$\downarrow$	1	X	1

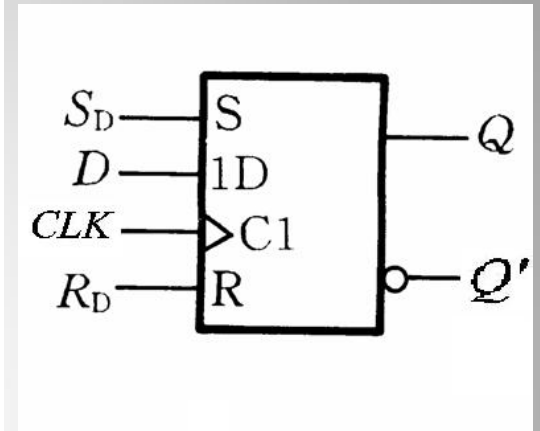
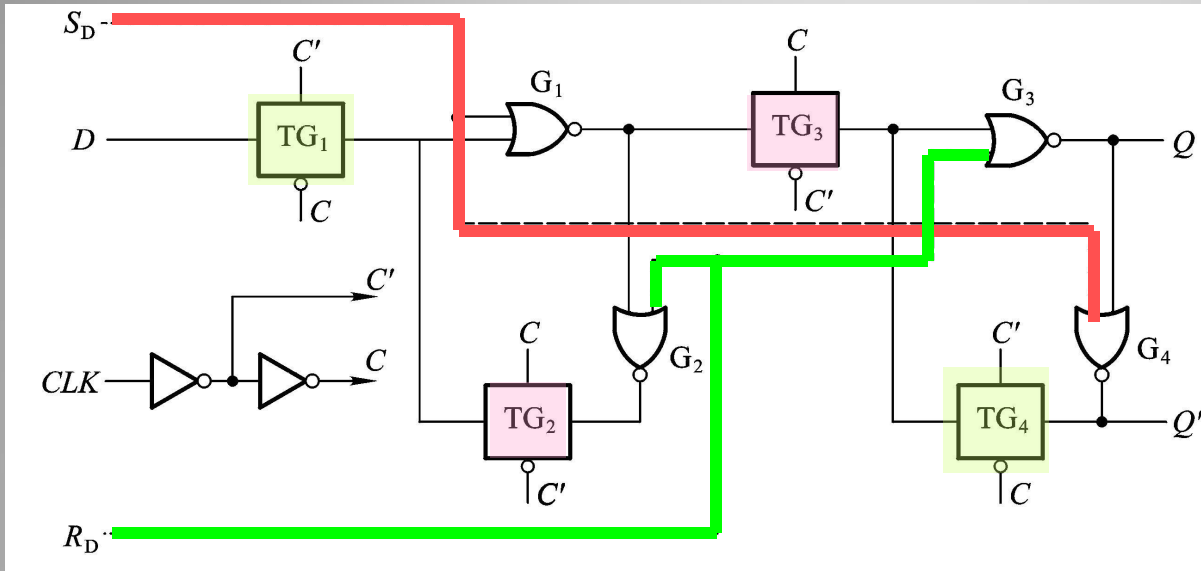
(1) $clk = 0$ 时, $\begin{cases} TG_1 \text{通}, TG_2 \text{断} \rightarrow Q' = D, Q' \text{随着} D \text{而变化} \\ TG_3 \text{断}, TG_4 \text{通} \rightarrow Q \text{保持, 反馈通路接通, 自锁} \end{cases}$

(2) $clk \uparrow$ 后, $\begin{cases} TG_1 \text{断}, TG_2 \text{通} \rightarrow \text{“主” 保持此前的状态} D \\ TG_3 \text{通}, TG_4 \text{断} \rightarrow Q' = Q, \text{反馈不通} \end{cases}$

(3) $clk \downarrow$ $\begin{cases} TG_1 \text{通}, TG_2 \text{断} \rightarrow Q' = D, \text{接收新的输入} \\ TG_3 \text{断}, TG_4 \text{通} \rightarrow Q \text{保持, 反馈通路接通} \end{cases}$

直到下个 $clk \uparrow$ 后, 输出才能变化。

(5)有异步置 1, 置 0端



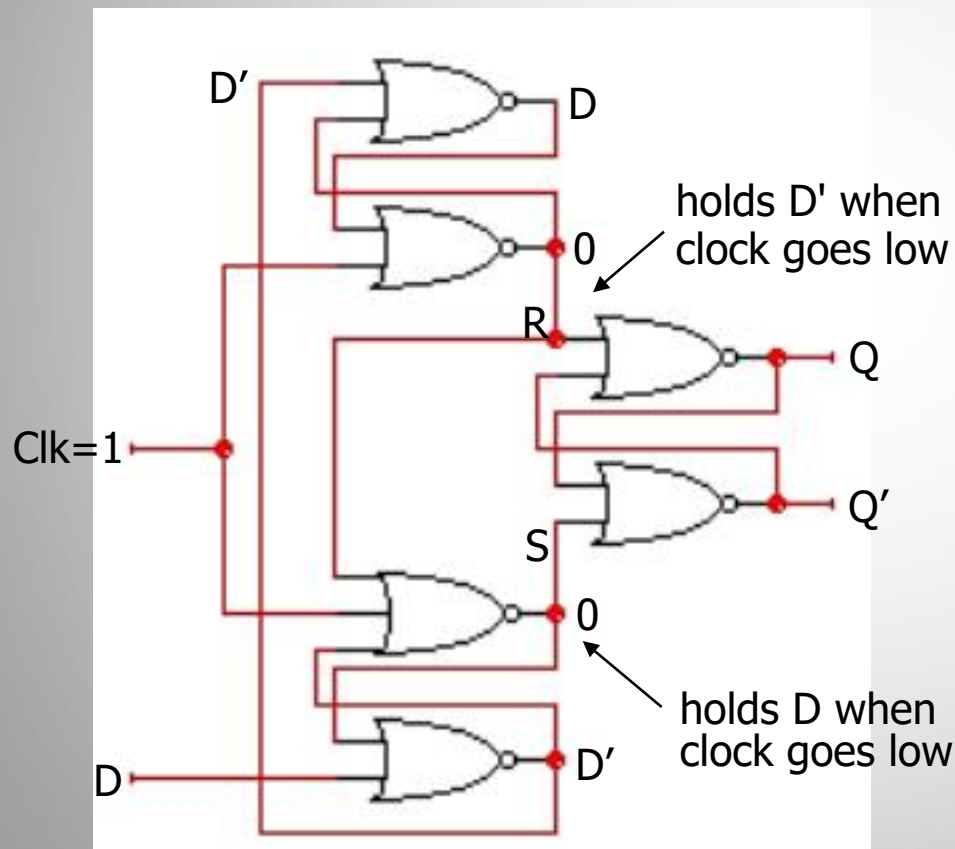
二、动作特点

Q^* 变化发生在 clk 的上升沿（或下降沿），

Q^* 仅取决于上升沿到达时输入的状态，而与此前、后的状态无关

4. Edge-Triggered Flip-Flops 维持阻塞型

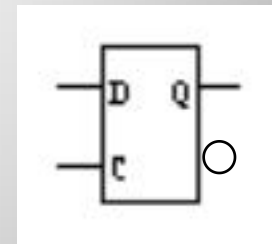
- More efficient solution: only 6 gates
 - sensitive to inputs only near edge of clock signal (not while high)



negative edge-triggered D
flip-flop (D-FF)

4-5 gate delays

**must respect setup and hold time
constraints to successfully
capture input**

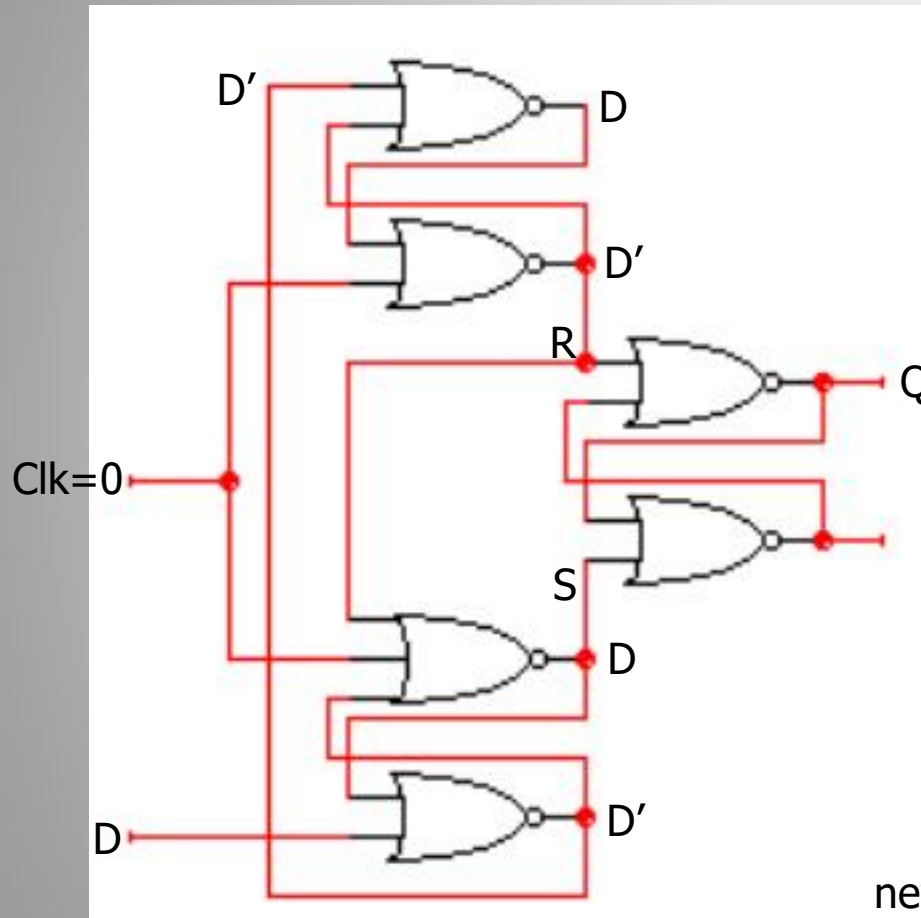


characteristic equation

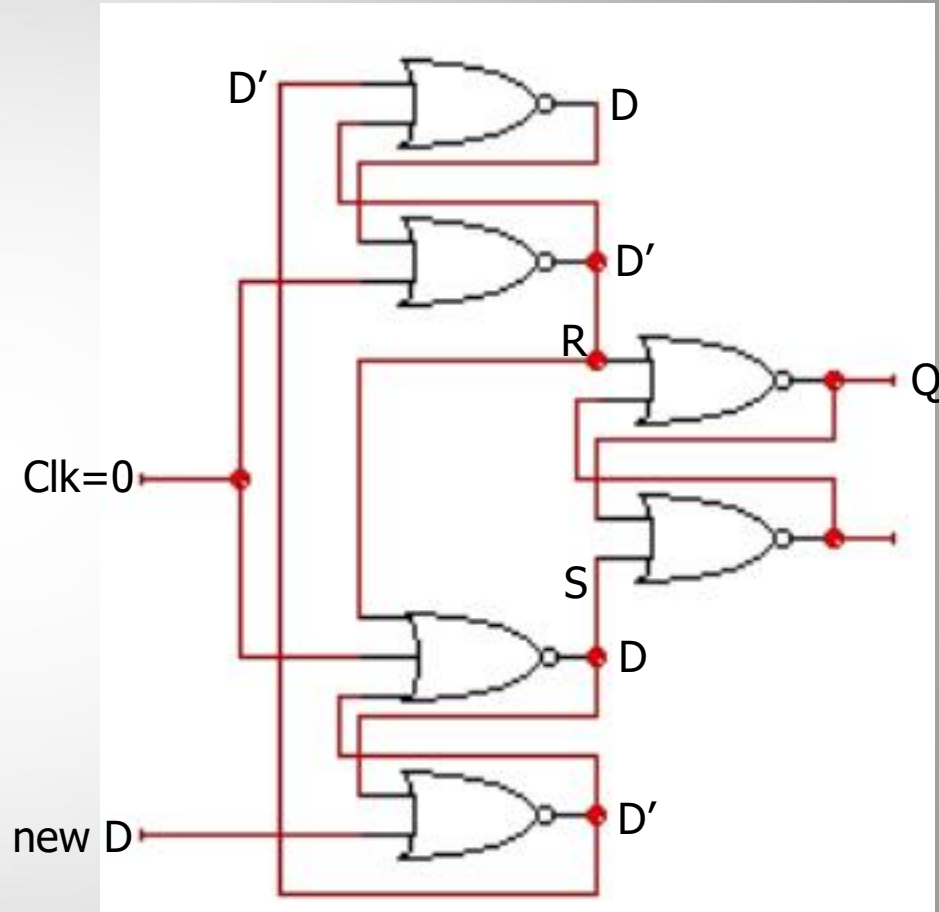
$$Q(t+1) = D$$

Edge-Triggered Flip-Flops (cont'd)

- Step-by-step analysis



when clock goes high-to-low
data is latched

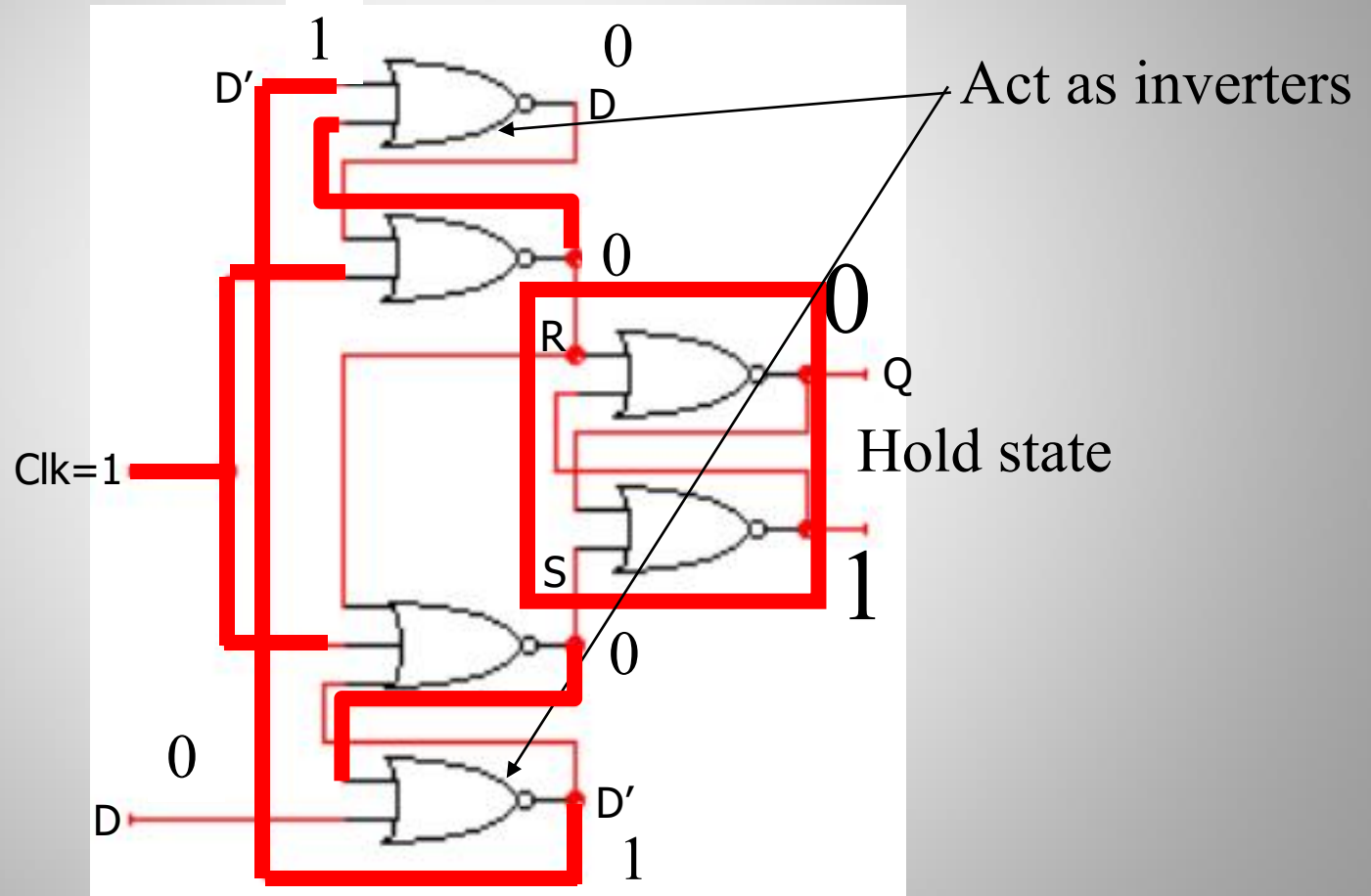


new D $\neq$ old D

when clock is low
data is held

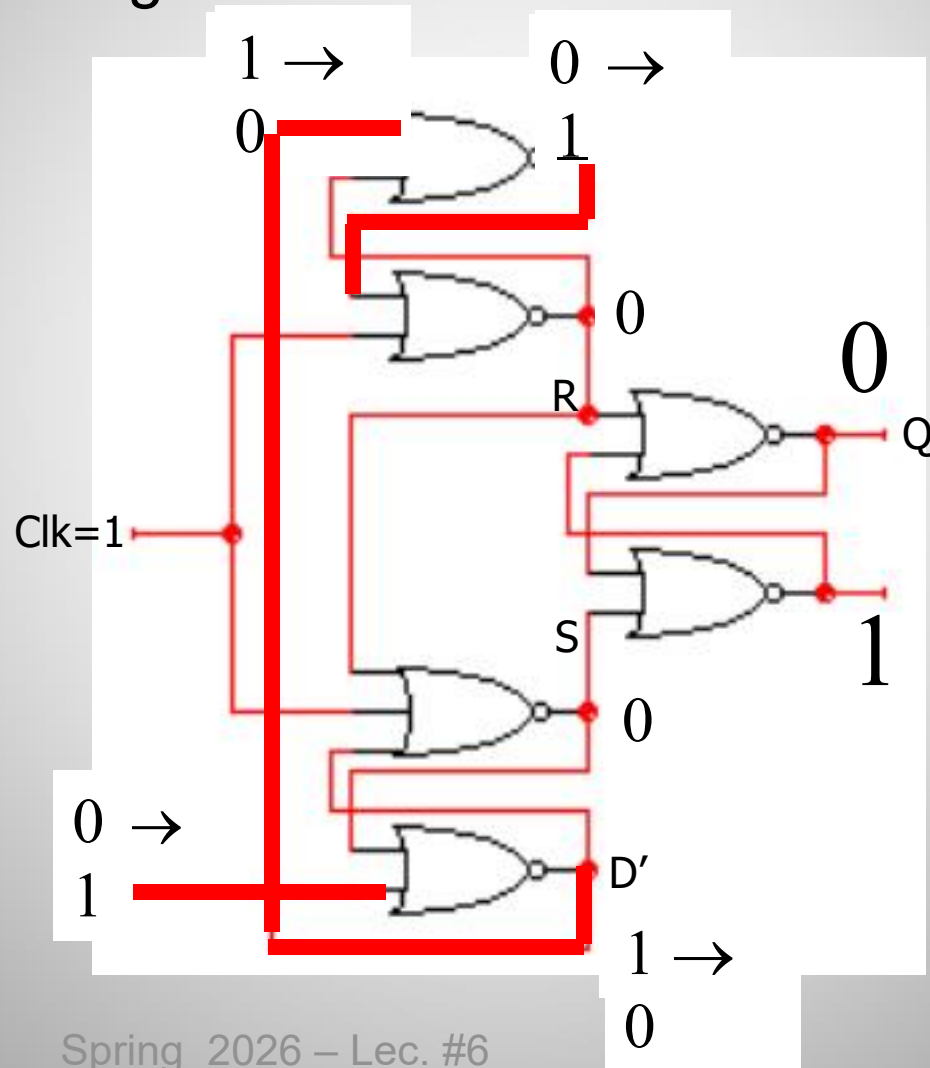
Edge-Triggered Flip-Flops (cont'd)

- $D = 0$, Clk High



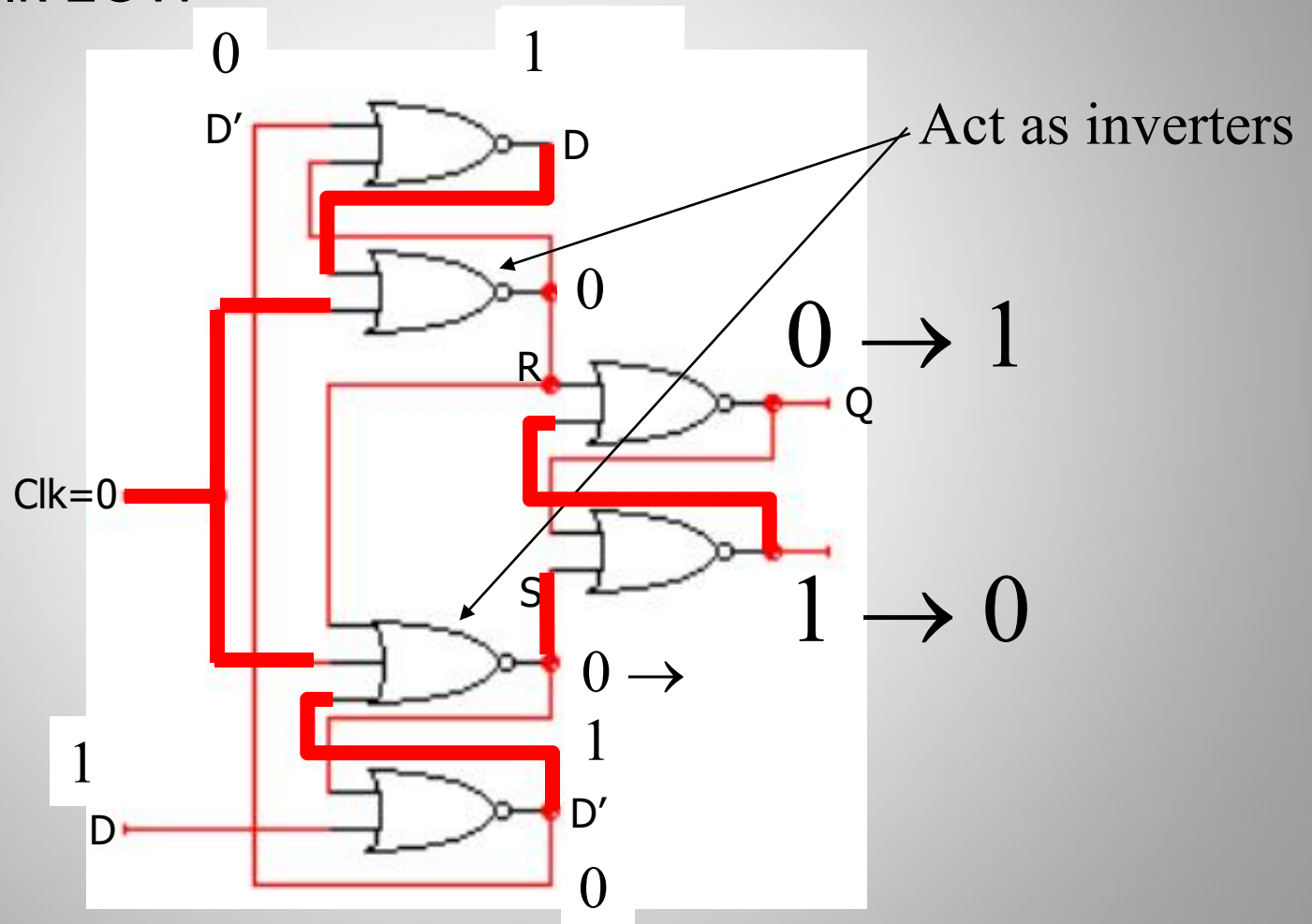
Edge-Triggered Flip-Flops (cont'd)

- $D = 1$, Clk High



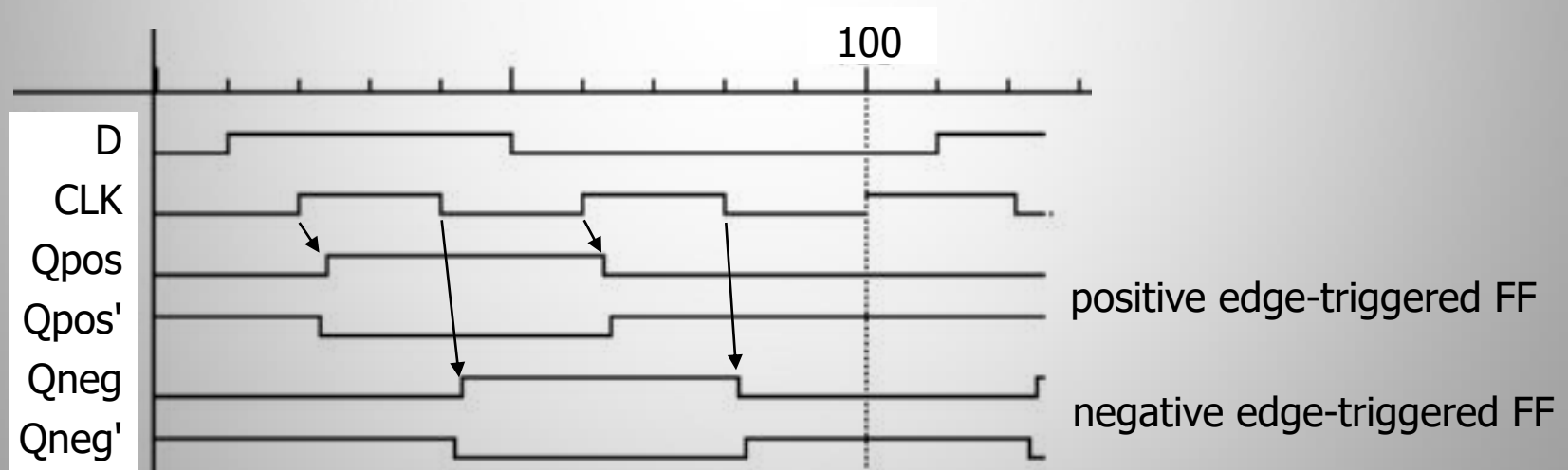
Edge-Triggered Flip-Flops (cont'd)

- $D = 1$, Clk LOW



Edge-Triggered Flip-Flops (cont'd)

- **Positive edge-triggered**
 - Inputs sampled on rising edge; outputs change after rising edge
- **Negative edge-triggered flip-flops**
 - Inputs sampled on falling edge; outputs change after falling edge



5. Negative Edge Trigger FF in Verilog

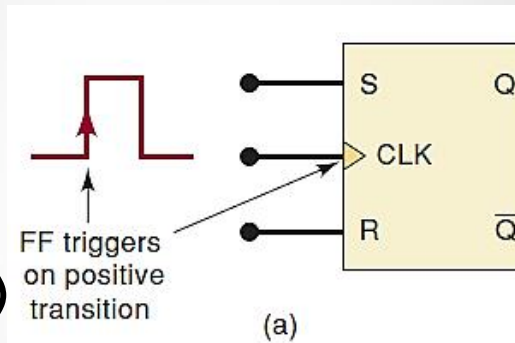
```

module d_ff (q, q_bar, data, clk);
    input  data, clk;
    output q, q_bar;
    reg    q;

    assign q_bar = ~q;

    always @(negedge clk)
        begin
            q <= data;
        end
endmodule

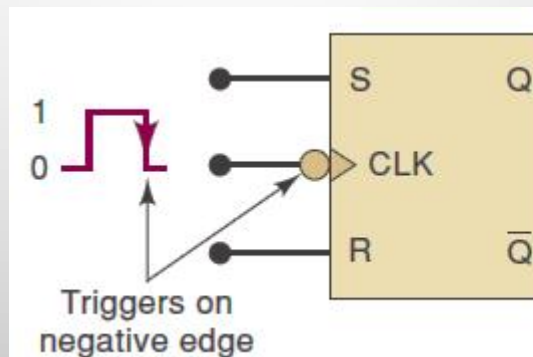
```



Inputs			Output
S	R	CLK	Q
0	0	↑	Q_0 (no change)
1	0	↑	1
0	1	↑	0
1	1	↑	Ambiguous

Q_0 is output level prior to ↑ of CLK.
↓ of CLK produces no change in Q.

(b)



Inputs			Output
S	R	CLK	Q
0	0	↓	Q_0 (no change)
1	0	↓	1
0	1	↓	0
1	1	↓	Ambiguous

5.3.4 触发器的逻辑功能分类

5.6.1 触发器按逻辑功能的分类

时钟控制的触发器中

由于输入方式不同(单端, 双端输入)、次态()随输入变化的规则不同

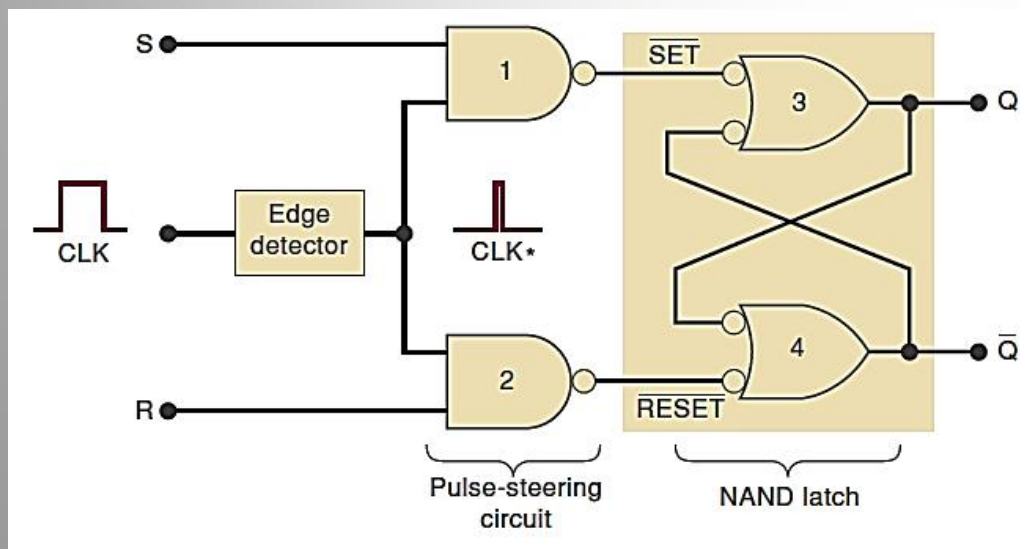
注：如果没有特别指明，以后说的有时钟的触发器默认都是指边沿触发的触发器

一、SR触发器

1. 定义, 凡在时钟信号作用下, 具有如下功能的触发器称为SR触发器

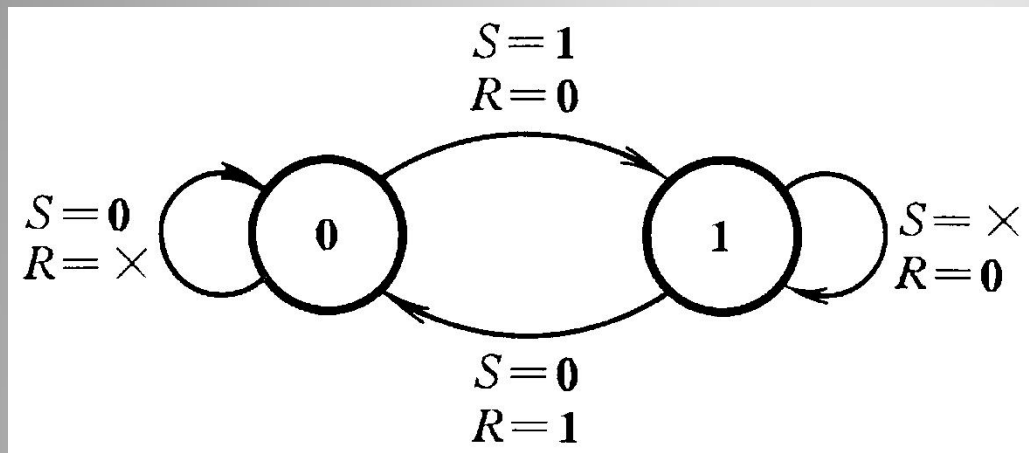
2. 特性方程

$$\begin{cases} Q^* = S'R'Q + SR'Q' + SR'Q = S'R'Q + SR' = S + R'Q \\ SR = 0 \quad \text{约束条件} \end{cases}$$

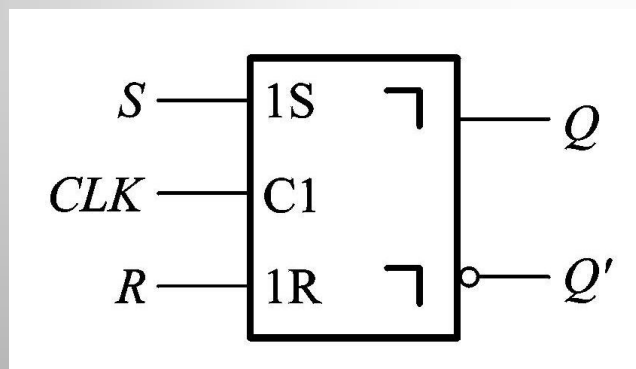


S	R	Q	Q^*
0	0	0	0
0	0	1	1
1	0	0	1
1	0	1	1
0	1	0	0
0	1	1	0
1	1	0	1*
1	1	1	1*

3. 状态转换图



4. 符号



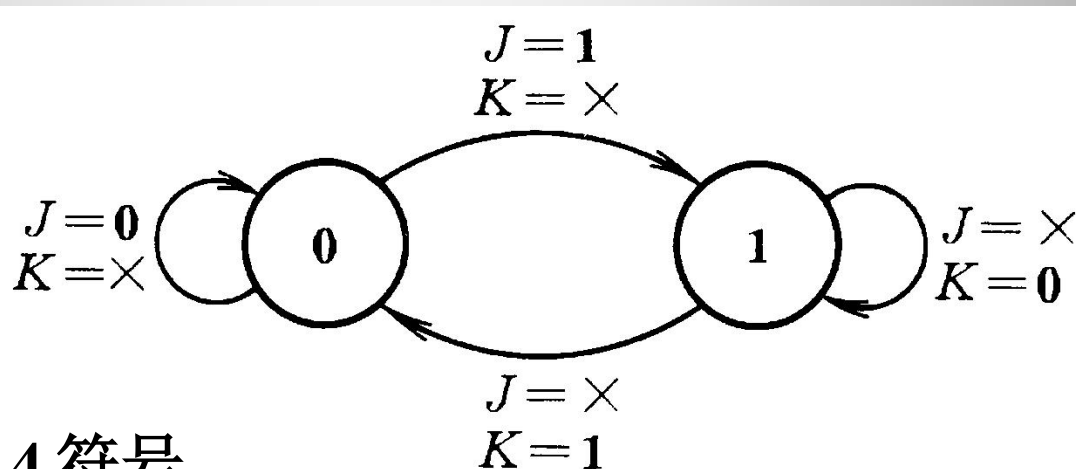
二、JK触发器

1.定义

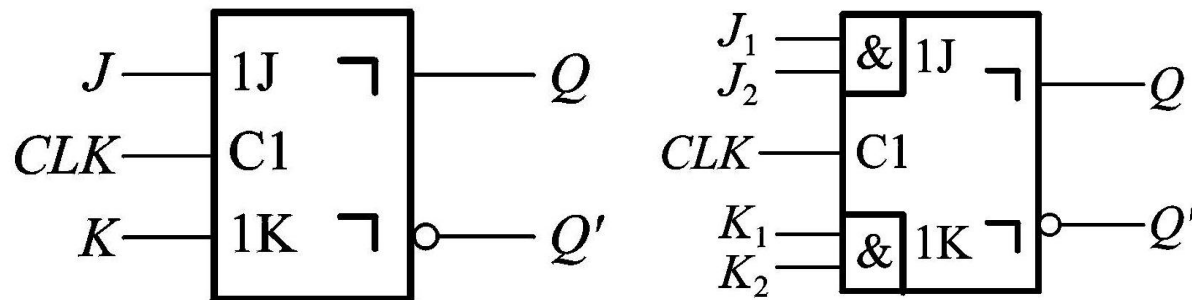
J	K	Q	Q^*
0	0	0	0
0	0	1	1
1	0	0	1
1	0	1	1
0	1	0	0
0	1	1	0
1	1	0	1
1	1	1	0

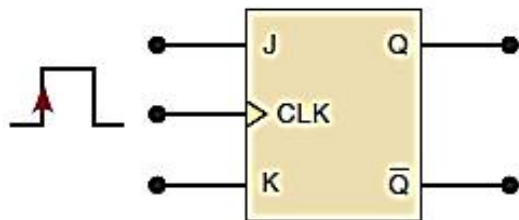
2.特性方程： $Q^* = JQ' + K'Q$

3.状态转换图



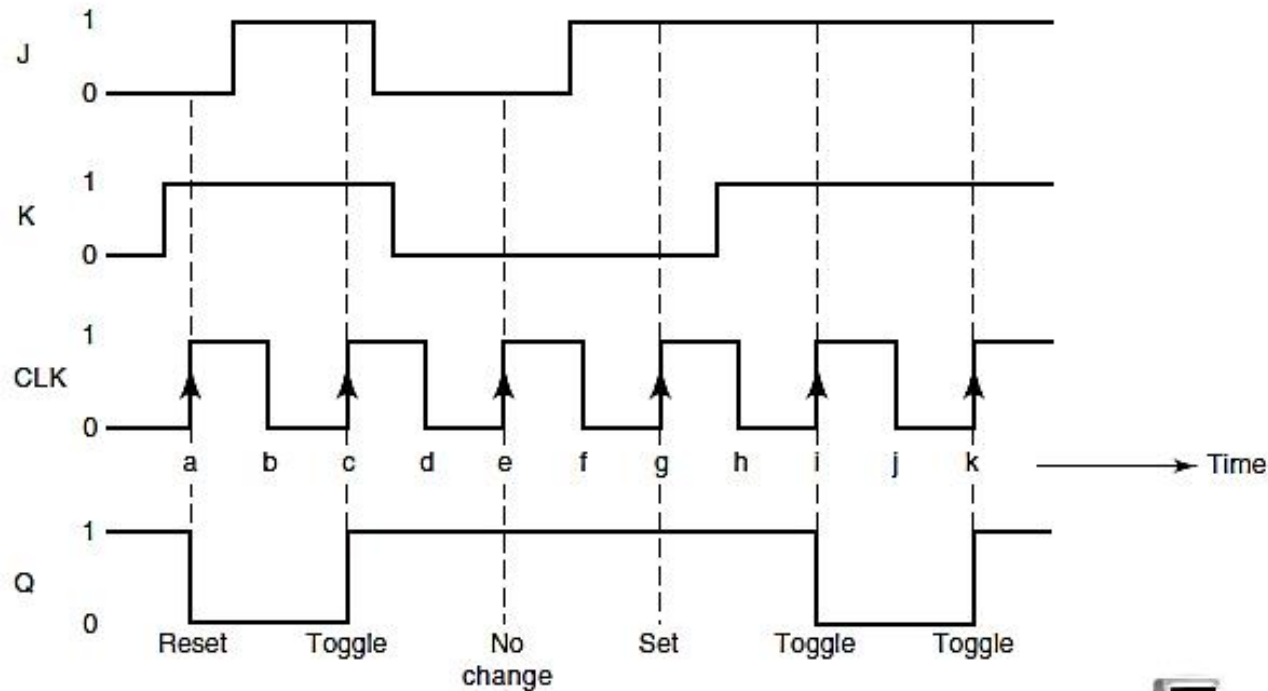
4.符号





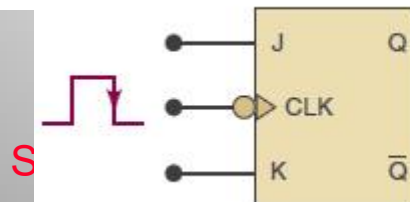
J	K	CLK	Q
0	0	↑	Q_0 (no change)
1	0	↑	1
0	1	↑	0
1	1	↑	$\overline{Q_0}$ (toggles)

(a)



(b)

FIGURE 5-23 (a) Clocked J-K flip-flop that responds only to the positive edge of the clock; (b) waveforms.



J	K	CLK	Q
0	0	↓	Q_0 (no change)
1	0	↓	1
0	1	↓	0
1	1	↓	$\overline{Q_0}$ (toggles)

三、T触发器

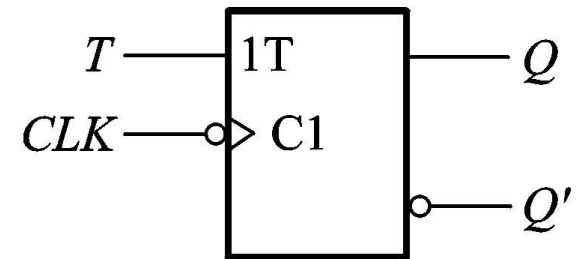
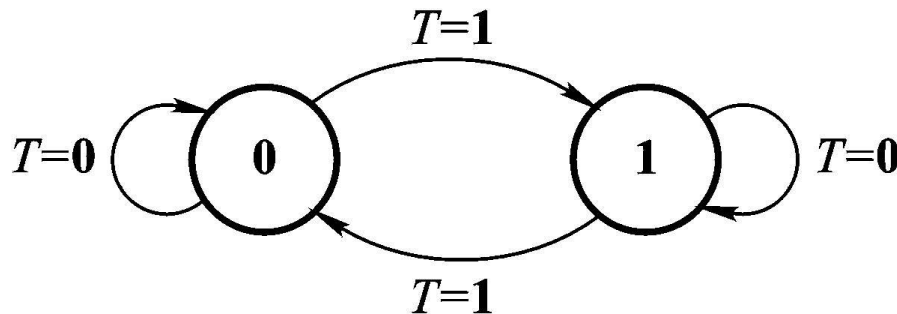
1. 定义：凡在时钟信号作用下，具有如下功能的触发器

T	Q	Q^*
0	0	0
0	1	0
1	0	1
1	1	0

2. 特性方程： $Q^* = TQ' + T'Q$

3. 状态转换图

4. 符号



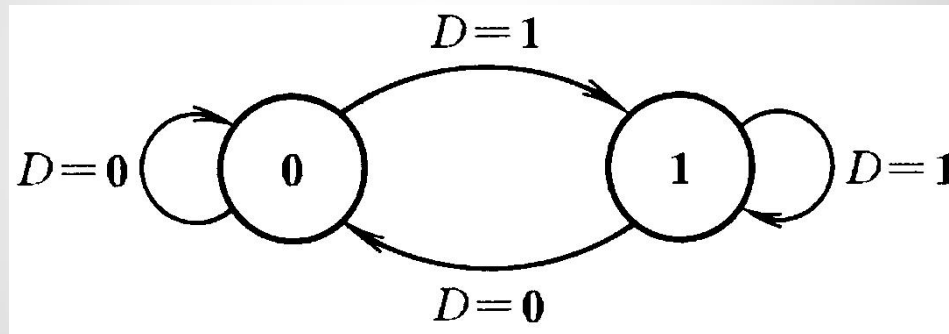
四、D触发器

1. 定义：凡在时钟信号作用下，具有如下功能的触发器

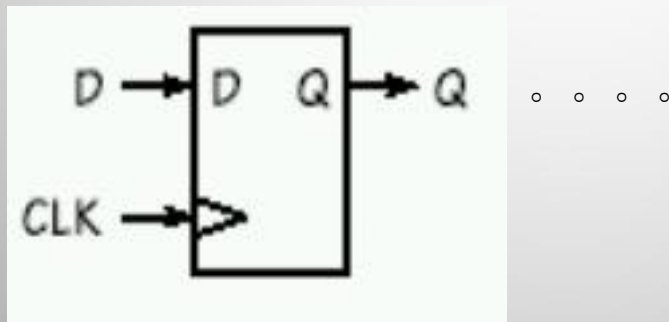
D	Q	Q^*
0	0	0
0	1	0
1	0	1
1	1	1

2. 特性方程： $Q^* = D$

3. 状态转换图

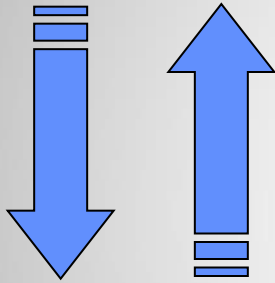


4. 符号



逻辑功能:

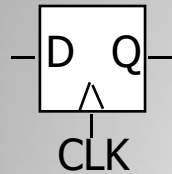
是 Q^* 与输入及 Q 在CLK作用后稳态之间的关系
(RS, JK, D, T)



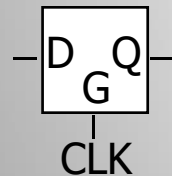
电路结构形式:

具有不同的动作特点(转换状态的动态过程)
(同步, 主从, 边沿)

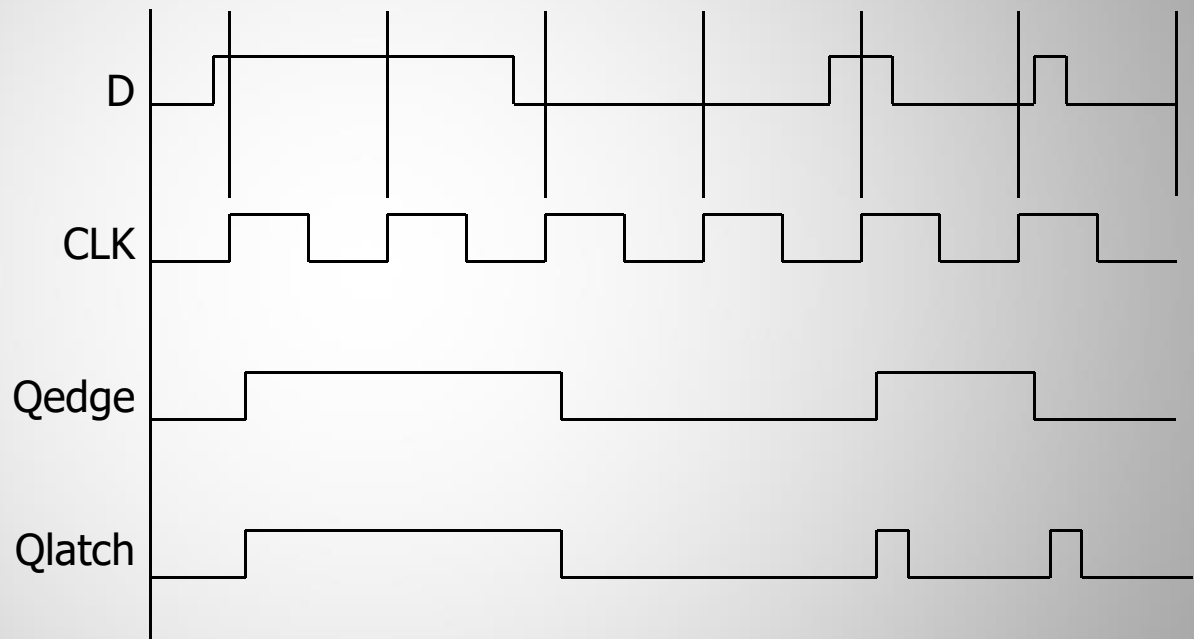
Comparison of Latches and Flip-Flops



positive
edge-triggered
flip-flop



transparent
(**level-sensitive**)
latch



behavior is the same unless input changes
while the clock is high

Comparison of Latches and Flip-Flops (cont'd)

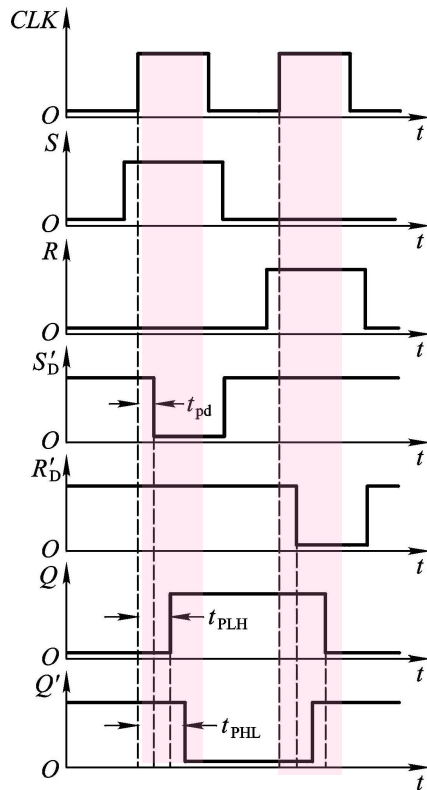
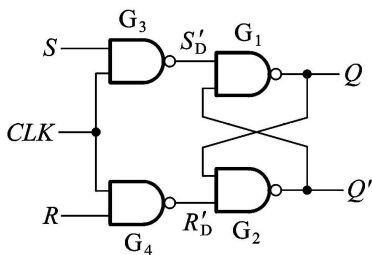
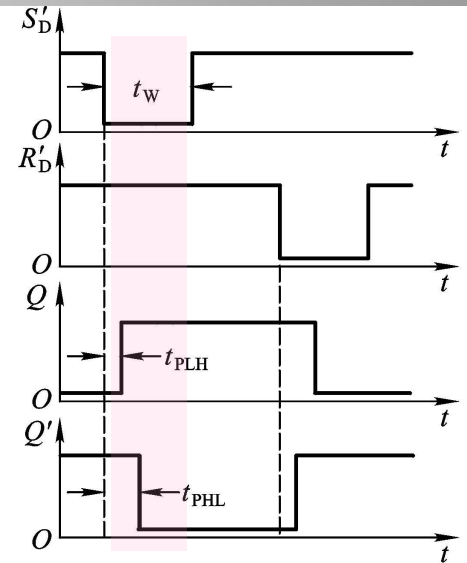
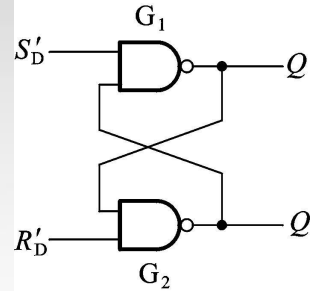
<u>Type</u>	<u>When inputs are sampled</u>	<u>When output is valid</u>
unclocked latch	always	propagation delay from input change
level-sensitive latch	clock high (T_{su}/T_h around falling edge of clock)	propagation delay from input change or clock edge (whichever is later)
master-slave flip-flop	clock high (T_{su}/T_h around falling edge of clock)	propagation delay from falling edge of clock
negative edge-triggered flip-flop	clock hi-to-lo transition (T_{su}/T_h around falling edge of clock)	propagation delay from falling edge of clock

5.3.5 触发器的动态特性

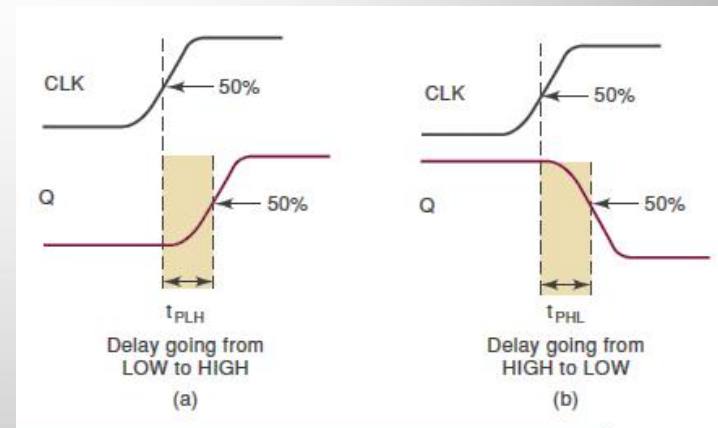
一、输入信号宽度

二、传输延迟时间

t_{PLH}, t_{PHL}



假设门传输延时时间为 t_{pd}



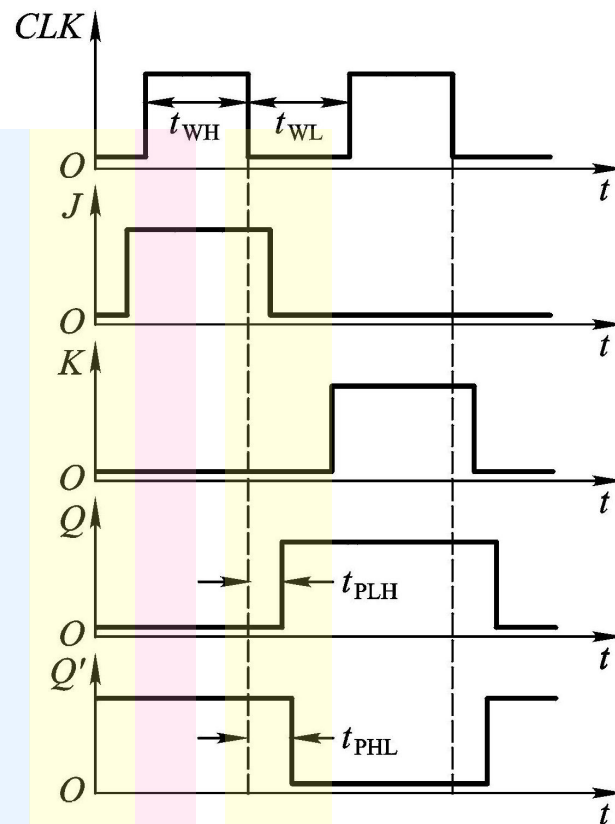
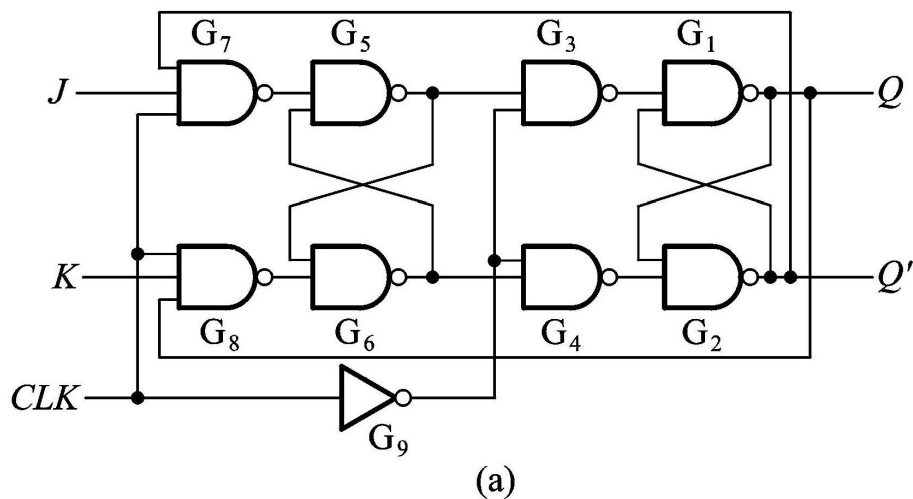
假设门传输延时时间为 t_{pd}



一、建立时间 t_{SETUP}

二、保持时间 t_{HOLD}

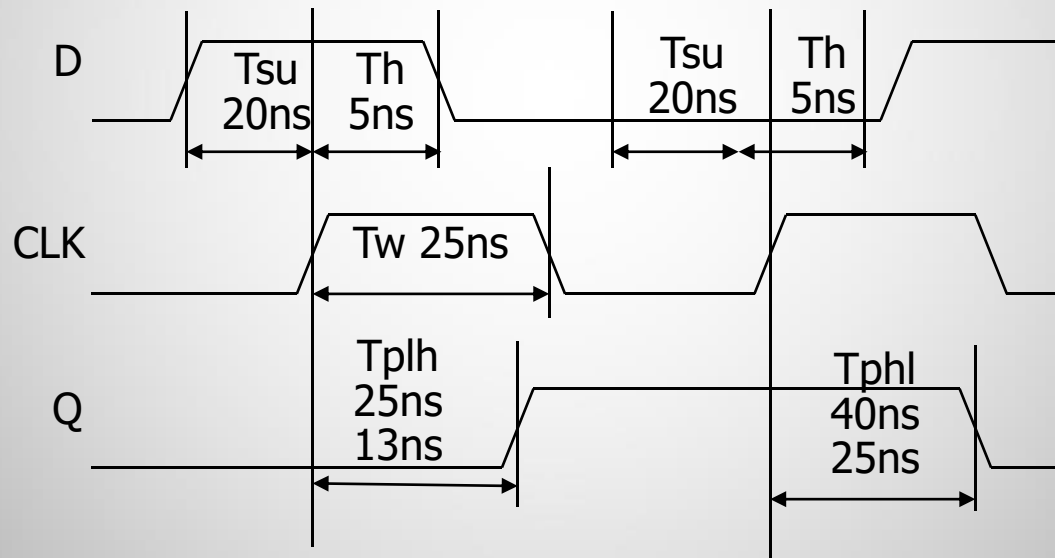
三、传输延迟时间



四、最高时钟频率

Typical Timing Specifications

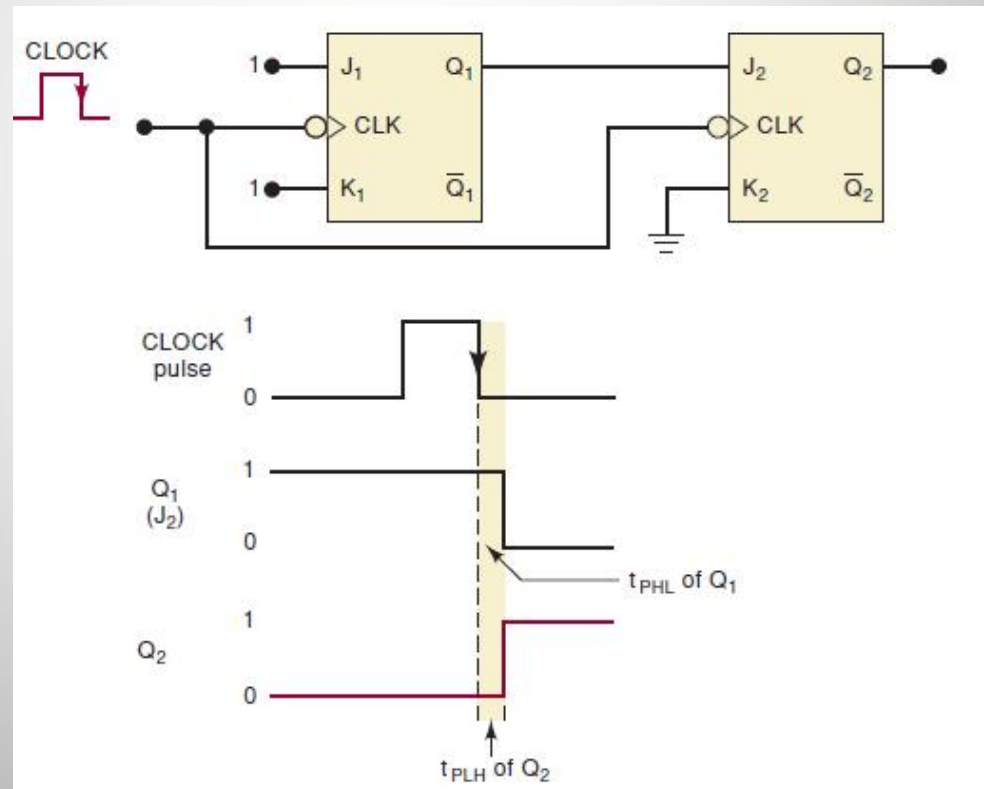
- **Positive edge-triggered D flip-flop**
 - Setup and hold times
 - Minimum clock width
 - Propagation delays (low to high, high to low, max and typical)



all measurements are made from the clocking event that is,
the rising edge of the clock

POTENTIAL TIMING PROBLEM IN FF CIRCUITS

- The potential timing problem is this:
 - because Q1 will change on the NGT of the clock pulse, the J2 input of Q2 will be changing as it receives the same NGT. This could lead to an unpredictable response at Q2.

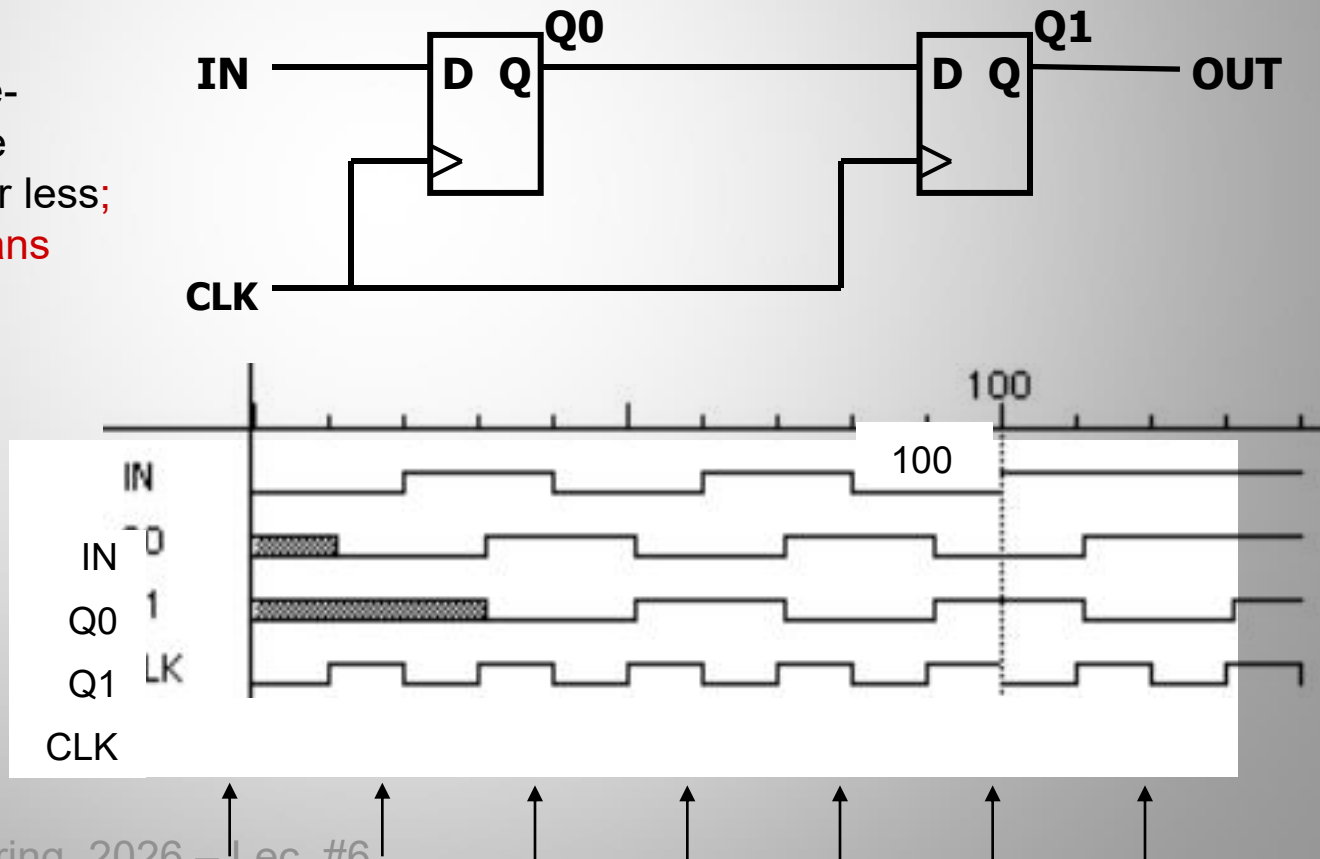


Cascading Edge-triggered Flip-Flops级联问题

- **Shift register**

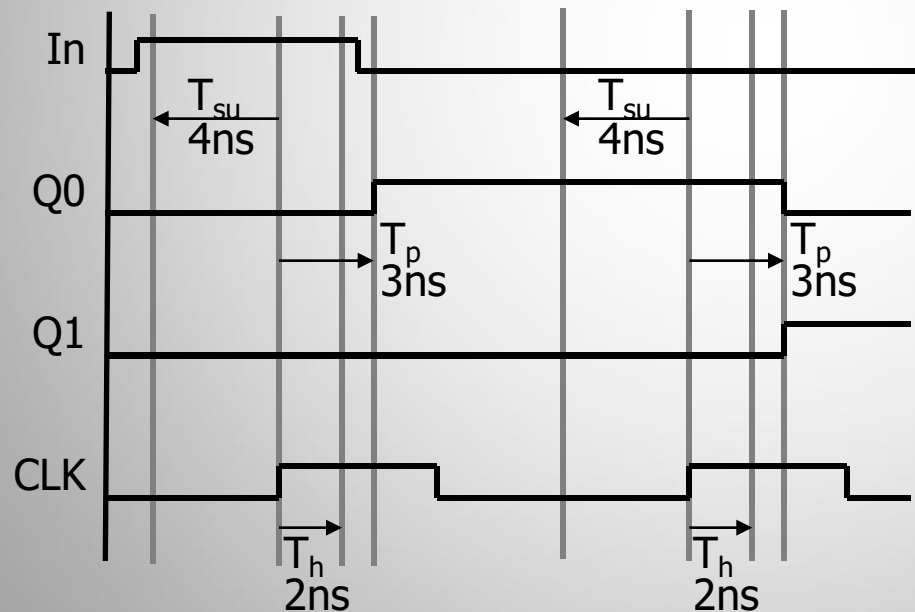
- New value goes into first stage
- While previous value of first stage goes into second stage
- Consider setup/hold/propagation delays (prop must be $>$ hold)

Fortunately, all modern edge-triggered FFs have hold time requirements that are 5 ns or less; most have $t_H = 0$, which means that they have no hold time requirement.



Cascading Edge-triggered Flip-Flops (cont'd)

- Why this works
 - Propagation delays exceed hold times
 - Clock width constraint exceeds setup time
 - This guarantees following stage will latch current value before it changes to new value



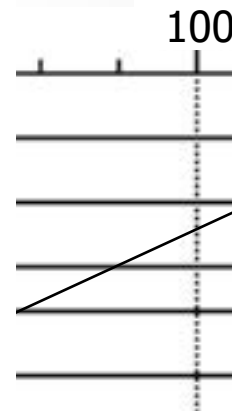
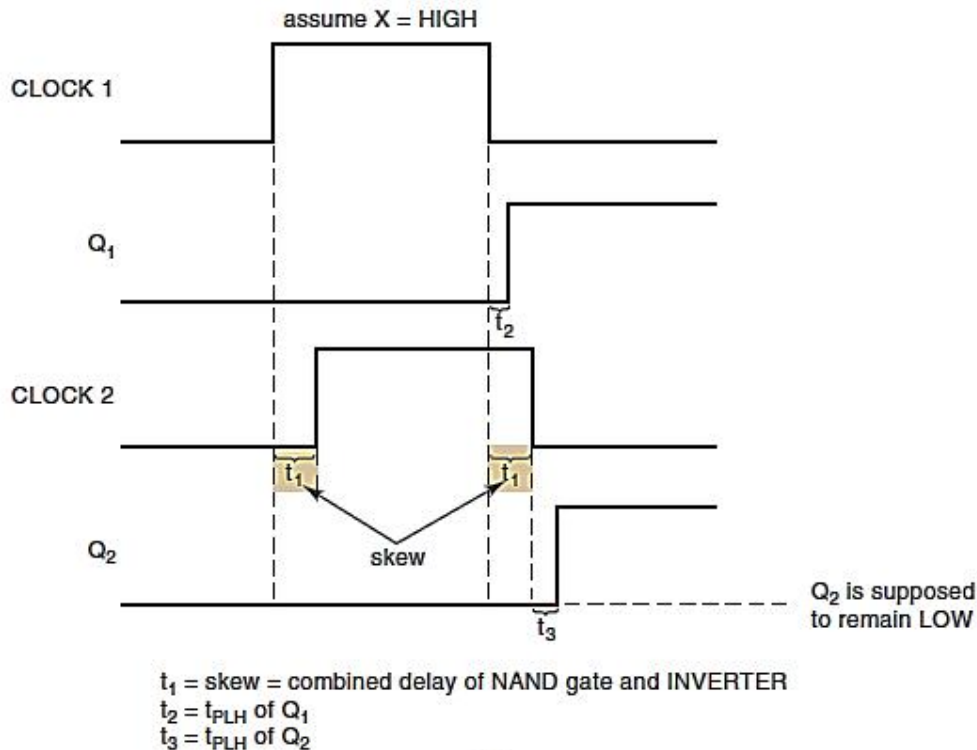
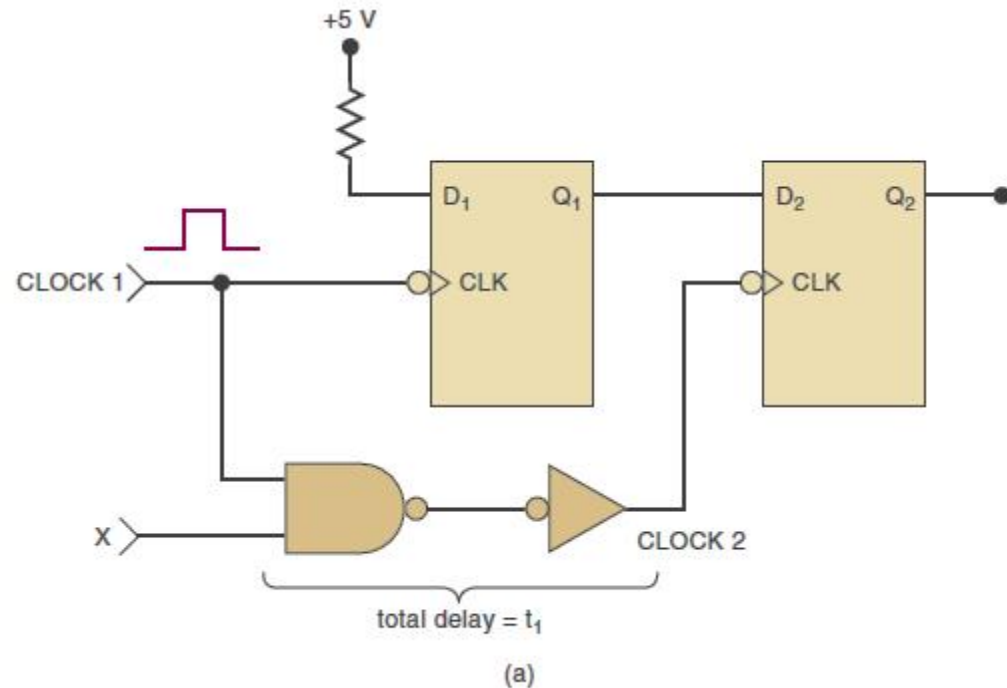
timing constraints
guarantee proper
operation of
cascaded components

assumes infinitely fast
distribution of the clock

Clock Skew时钟偏移

• The problem

- Correct behavior assumes n is determined by all storage elements
- Difficult in high-performance arrive at flip-flop is comparable
- soon become greater than logic
- Effect of skew on cascaded



CLK1 is a delayed version of CLK0

= 0, and not $Q_0 = 0, Q_1 = 1$

Summary of Latches and Flip-Flops

- **Development of D-FF**
 - Level-sensitive used in custom integrated circuits
 - » can be made with 4 switches
 - Edge-triggered used in programmable logic devices
 - Good choice for data storage register
- **Historically J-K FF was popular but now never used**
 - Similar to R-S but with 1-1 being used to toggle output (complement state)
 - Good in days of TTL/SSI (more complex input function:
 $D = JQ' + K'Q$)
 - Not a good choice for PLAs as it requires two inputs
 - Can always be implemented using D-FF
- **Preset and clear inputs are highly desirable on flip-flops**
 - Used at start-up or to reset system to a known state

Flip-Flop Features

- **Reset (set state to 0): R**
 - Synchronous: $D_{\text{new}} = R' \cdot D_{\text{old}}$ (when next clock edge arrives)
 - Asynchronous: doesn't wait for clock, quick but dangerous
- **Preset or set (set state to 10: S (or sometimes P))**
 - Synchronous: $D_{\text{new}} = D_{\text{old}} + S$ (when next clock edge arrives)
 - Asynchronous: doesn't wait for clock, quick but dangerous
- **Both reset and preset**
 - $D_{\text{new}} = R' \cdot D_{\text{old}} + S$ (set-dominant)
 - $D_{\text{new}} = R' \cdot D_{\text{old}} + R'S$ (reset-dominant)
- **Selective input capability (input enable/load): LD or EN**
 - Multiplexer at input: $D_{\text{new}} = LD' \cdot Q + LD \cdot D_{\text{old}}$
 - Load may/may not override reset/set (usually R/S have priority)
- **Complementary outputs: Q and Q'**

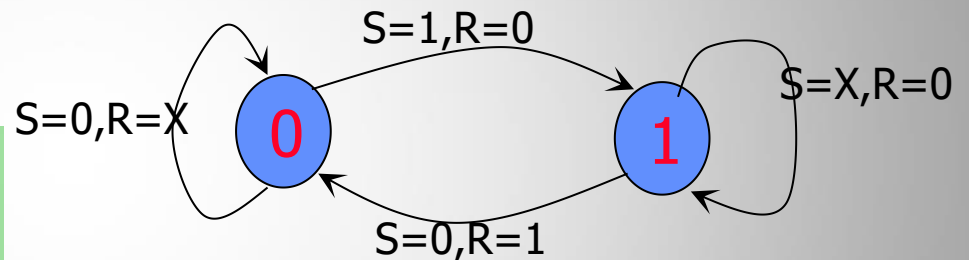
触发器Flip-Flop分类

- 逻辑功能分类
 - RS锁存器
 - JK触发器
 - T触发器
 - D触发器
- 逻辑功能指按触发器的次态和现态及输入信号之间的逻辑关系.
 - 特性表
 - 特性方程
 - 状态转换图

RS 锁存器

- 特性方程 $Q^{n+1} = S + R'Q^n$
- RS Latch的状态转换图
- 特性表/真值表

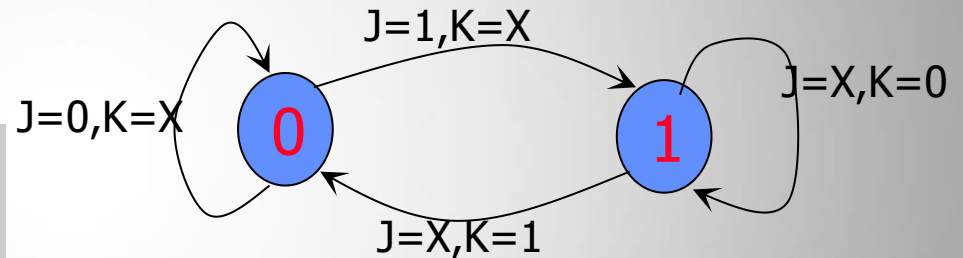
S	R	Q^n	Q^{n+1}	
0	0	0	0	} 保持
0	0	1	1	
0	1	0	0	} 复位
0	1	1	0	
1	0	0	1	} 置位
1	0	1	1	
1	1	0	0	} 不定
1	1	1	0	



JK 触发器

- 特性方程: $Q^{n+1} = JQ^n + K'Q^n$
- JK FF的状态转换图
- 特性表/真值表

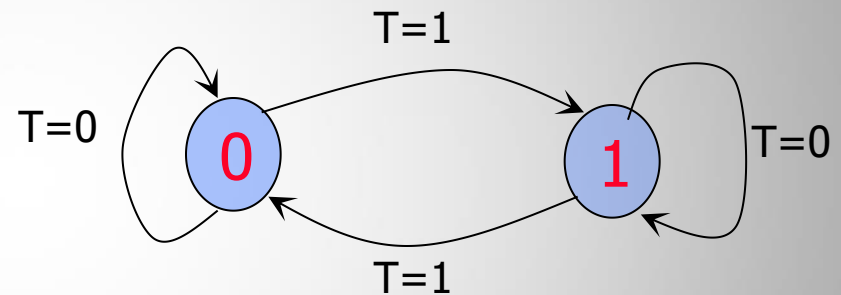
J	K	Q^n	Q^{n+1}	
0	0	0	0	} 保持
0	0	1	1	
0	1	0	0	} 复位
0	1	1	0	
1	0	0	1	} 置位
1	0	1	1	
1	1	0	1	} 翻转
1	1	1	0	



T 触发器

- 特性方程: $Q^{n+1} = TQ^n + T'Q^n$
- T FF的状态转换图
- 特性表/真值表

T	Q^n	Q^{n+1}	
0	0	0	} 保持
0	1	1	
1	0	1	} 翻转
1	1	0	



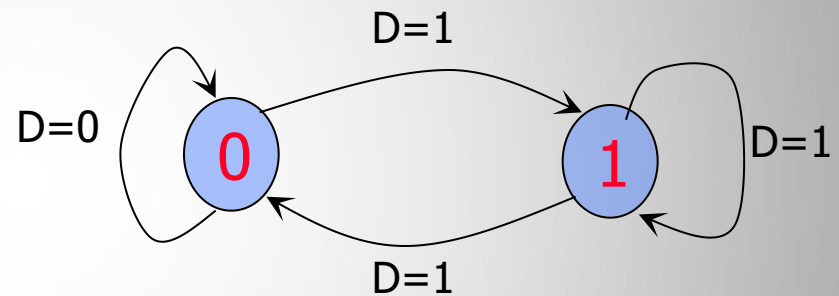
- T'触发器: $T=1, Q^{n+1}=Q^n$

JK触发器的两个输入端连在一起作为T端，可以构成T Flip-flop

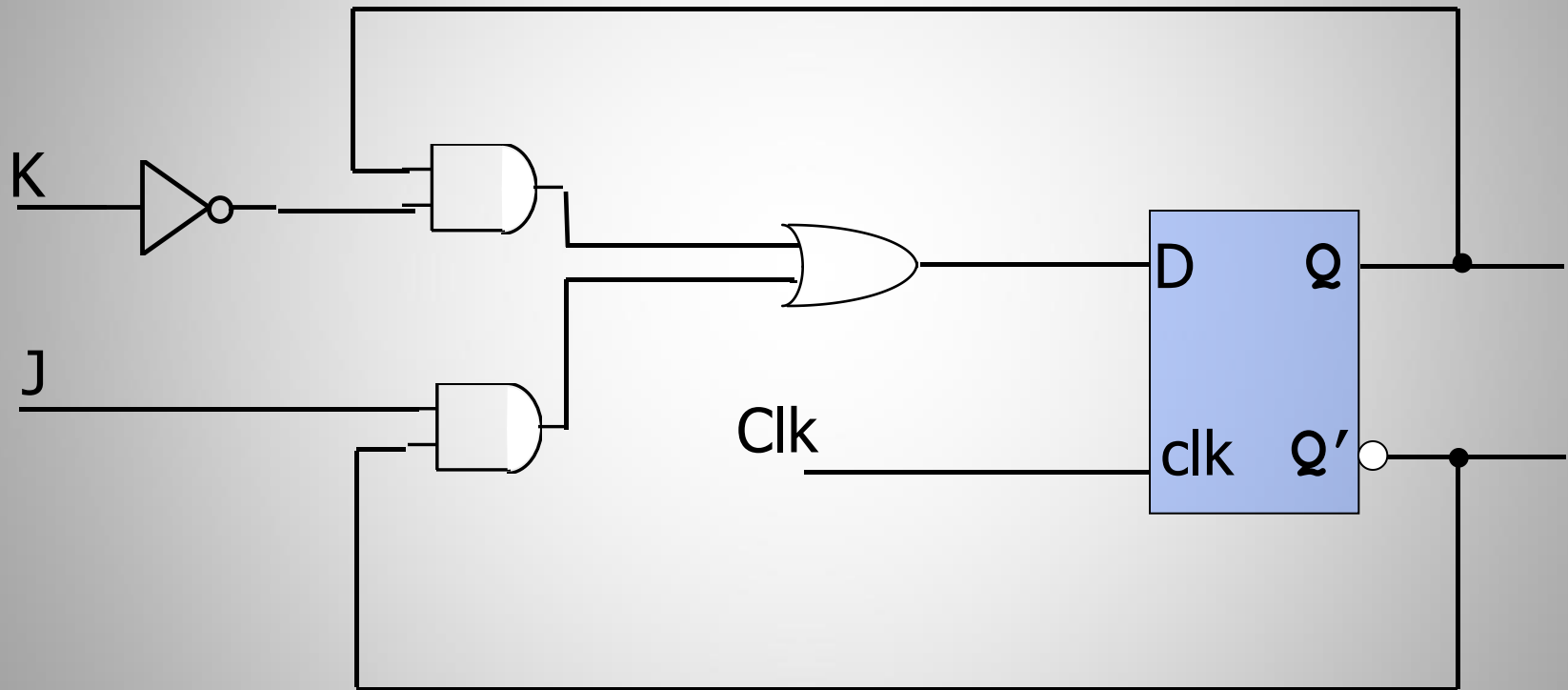
D 触发器

- 特性方程: $Q^{n+1}=D$
- D FF的状态转换图
- 特性表/真值表

D	Q^n	Q^{n+1}	
0	0	0	} reset
0	1	0	
1	0	1	} set
1	1	1	



采用D 触发器实现JK触发器



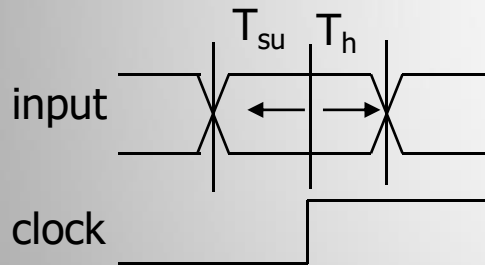
Timing Methodologies

- **Rules for interconnecting components and clocks**
 - Guarantee proper operation of system when strictly followed
- **Approach depends on building blocks used for memory elements**
 - Focus on systems with edge-triggered flip-flops
 - » Found in programmable logic devices
 - Many custom integrated circuits focus on level-sensitive latches
- **Basic rules for correct timing:**
 - (1) Correct inputs, with respect to time, are provided to the flip-flops
 - (2) No flip-flop changes state more than once per clocking event

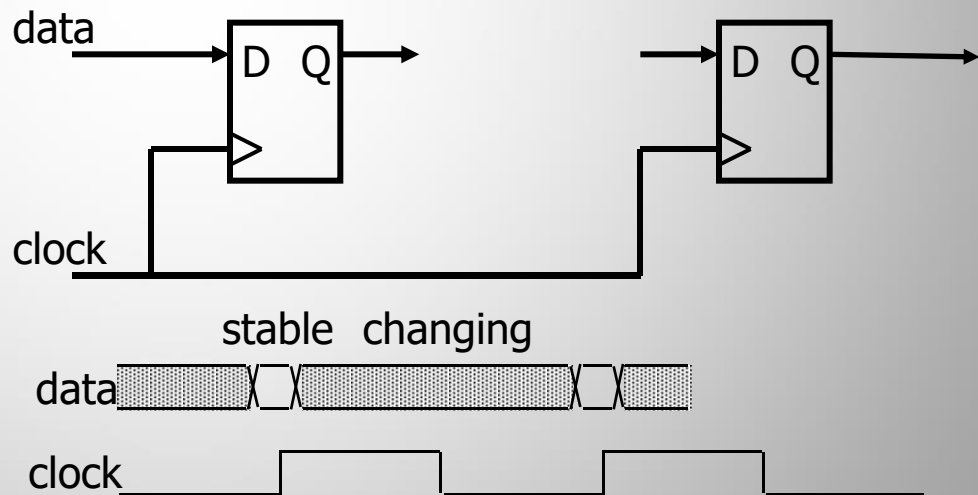
Timing Methodologies (cont'd)

- **Definition of terms**

- **clock:** periodic event, causes state of memory element to change; can be rising or falling edge, or high or low level
- **setup time:** minimum time before the clocking event by which the input must be stable (T_{su})
- **hold time:** minimum time after the clocking event until which the input must remain stable (T_h)

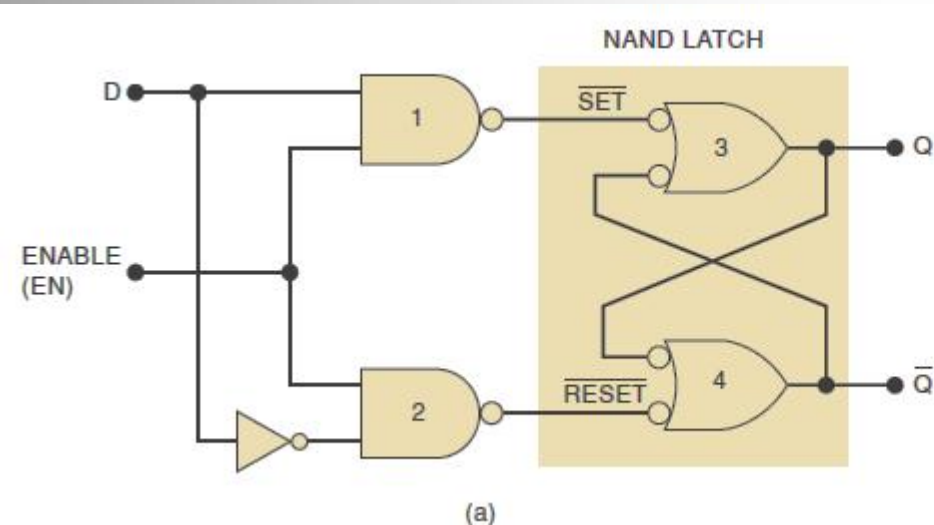
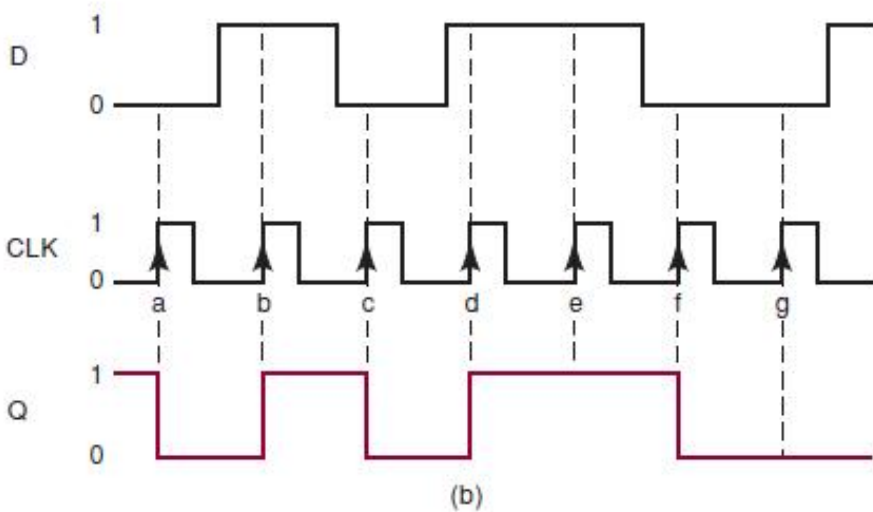
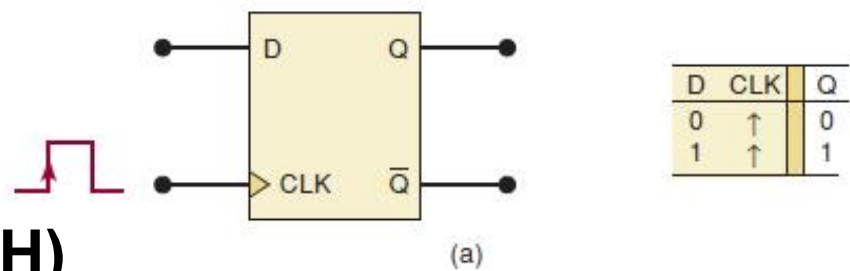
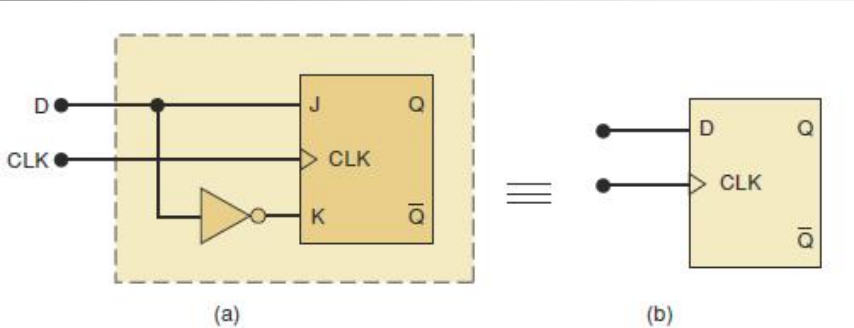


there is a timing "window" around the clocking event during which the input must remain stable and unchanged in order to be recognized



5.4 寄存器

D LATCH (TRANSPARENT LATCH)



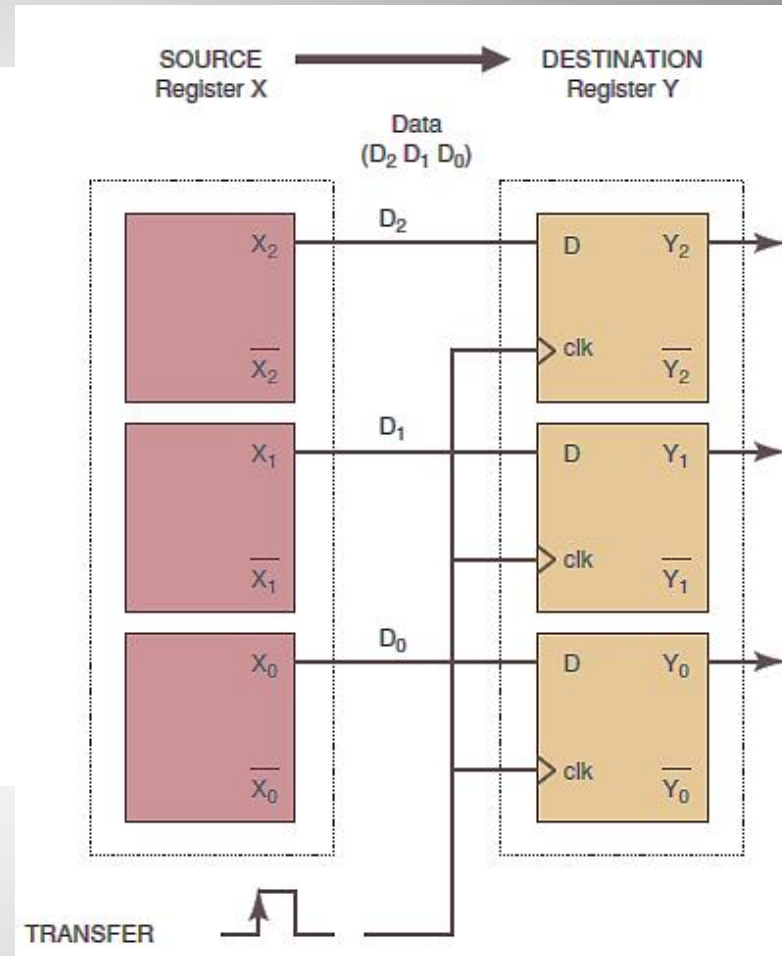
Inputs		Output
EN	D	Q
0	X	Q_0 (no change)
1	0	0
1	1	1

"X" Indicates "don't care."
 Q_0 is state Q just prior to EN going LOW.

Actual Ics

多位成组

	TTL		CMOS	
	7474	74LS112	74C74	74HC112
t_S	20	20	60	25
t_H	5	0	0	0
t_{PHL}	from CLK to Q	24	200	31
t_{PLH}	from CLK to Q	16	200	31
t_{PHL}	from $\overline{CLR}$ to Q	24	225	41
t_{PLH}	from $\overline{PRE}$ to Q	16	225	41
$t_{W(L)}$	CLK LOW time	15	100	25
$t_{W(H)}$	CLK HIGH time	20	100	25
$t_{W(L)}$	at $\overline{PRE}$ or $\overline{CLR}$	15	60	25
f_{MAX}	in MHz	30	5	20



SERIAL DATA TRANSFER: SHIFT REGISTERS

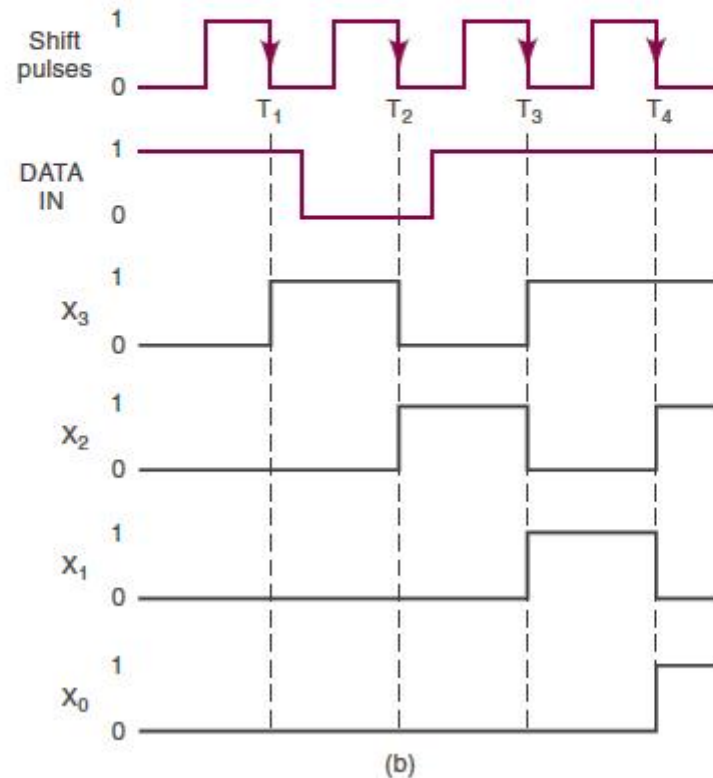
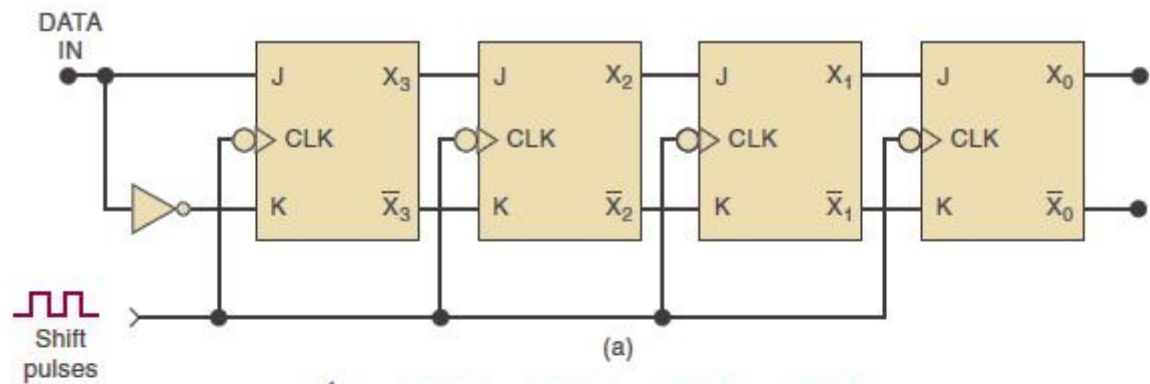


FIGURE 5-45 Four-bit shift register.

5.5 半导体存储器

A major advantage of digital over analog systems is the ability to store easily large quantities of digital information and data for short or long periods.

5.5 半导体存储器 memory devices

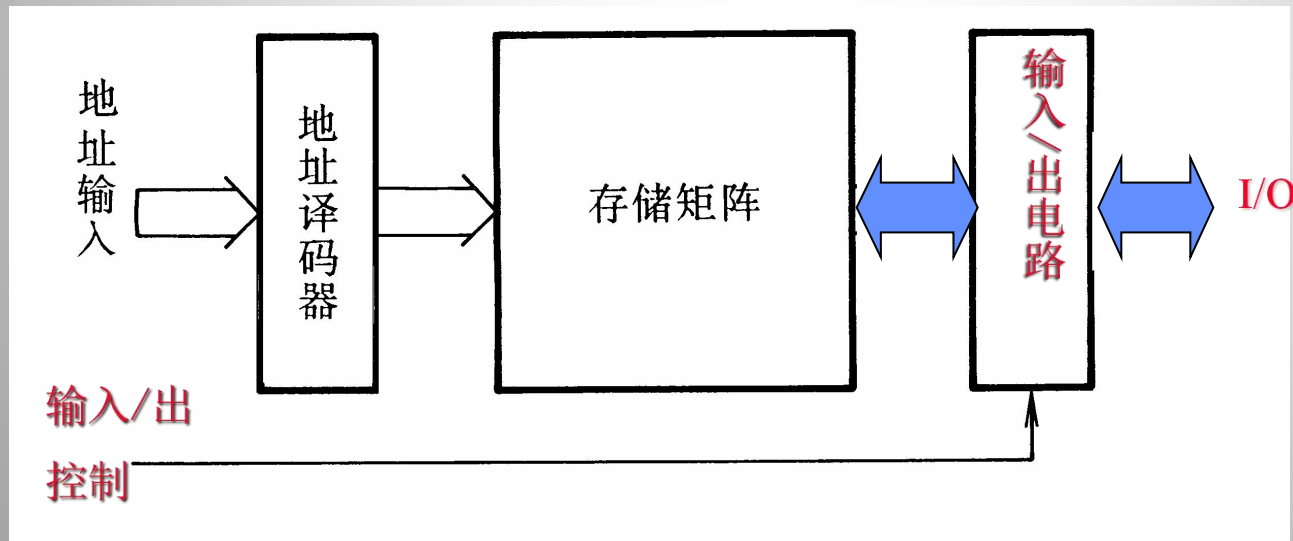
1 概述

flip-flop?

能存储大量二值信息的器件

一、一般结构形式 ! 单元数庞大

! 输入/输出引脚数目有限



二、分类

1、从存/取功能分：

①只读存储器

(Read-Only-Memory)

掩模 **ROM**

可编程 **ROM**

可擦除的可编程 **EPROM**

②随机读/写

(Random-Access-Memory)

静态 **RAM**

动态 **RAM**

2、从工艺分：

①双极型

②MOS型

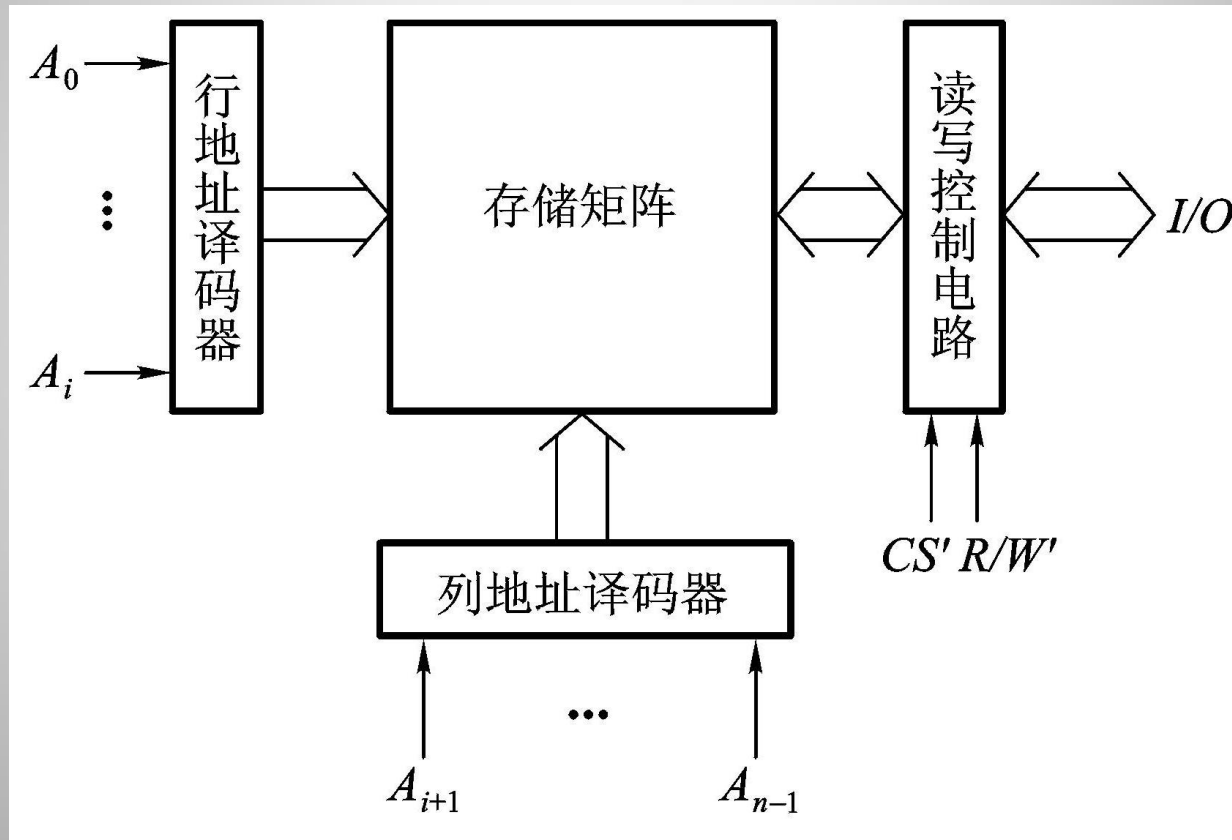
Addresses

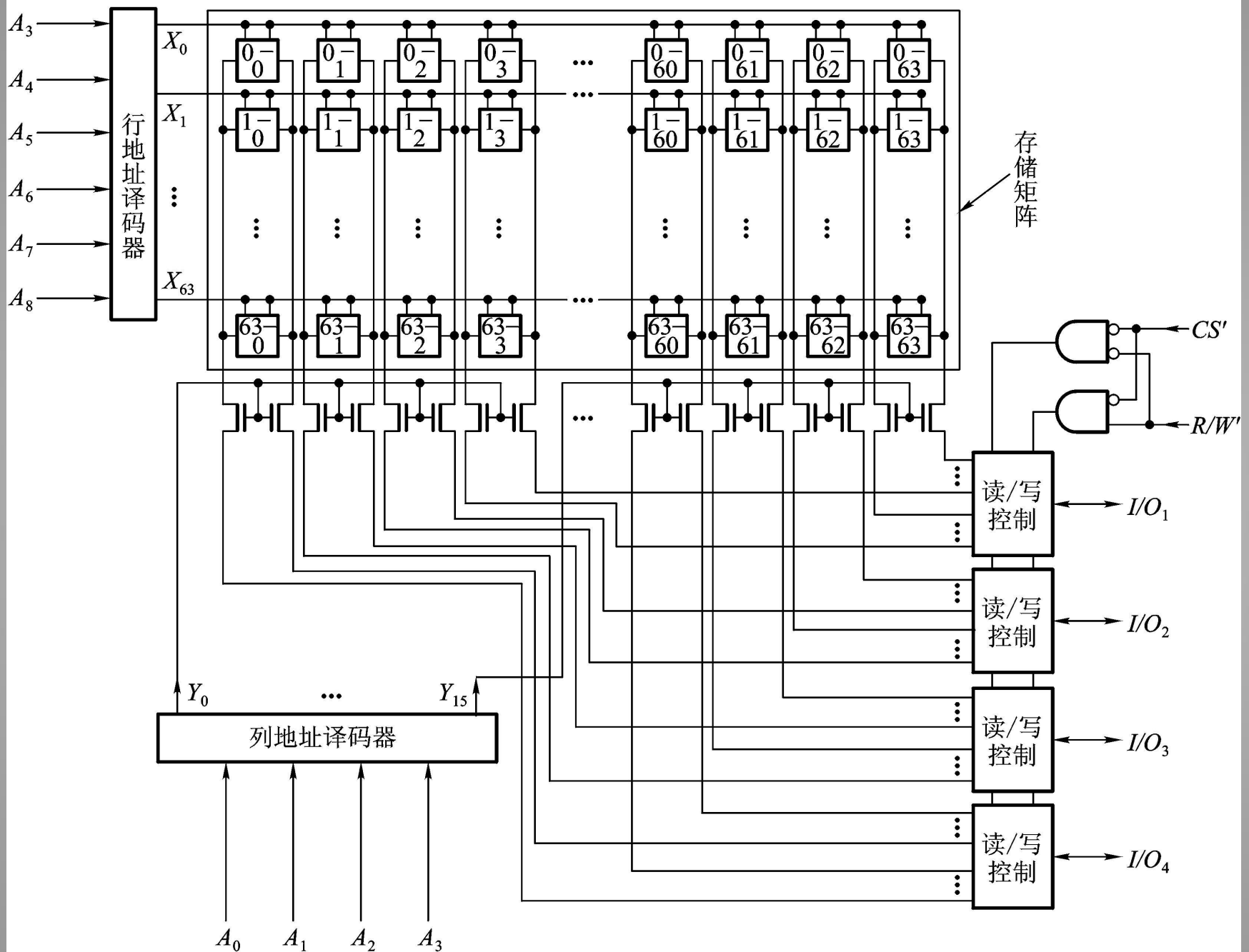
000	Word 0
001	Word 1
010	Word 2
011	Word 3
100	Word 4
101	Word 5
110	Word 6
111	Word 7

5.5 随机存储器RAM random-access memory

5.5.1 静态随机存储器（SRAM）STATIC RAM

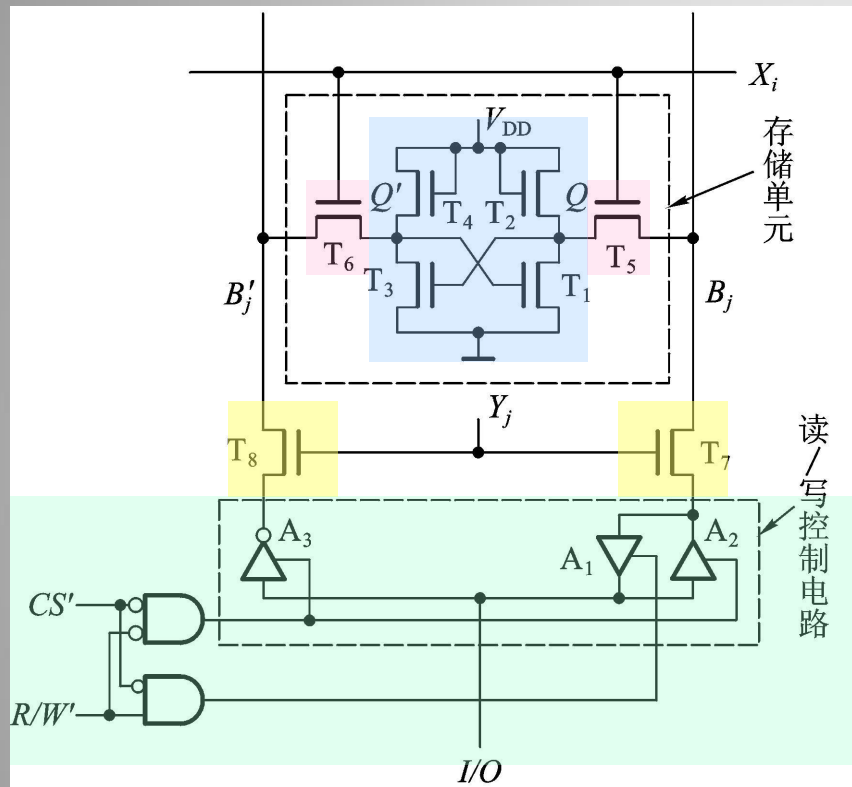
一、结构与工作原理





二、SRAM的存储单元

六管N沟道增强型MOS管



$T_1 \sim T_4$ 为基本 RS 触发器，
作存储单元

$X_i = 1$ 时，能在1行中被选中，
 T_5, T_6 导通， Q 、 Q' 与 B_j 、 B'_j 相通

当 $CS' = 0$ 时，

若 $R/W' = 1$ ，则 A_1 导通， A_2 与 A_3 截止，
 $Q \rightarrow I/O$ ，读操作

若 $R/W' = 0$ ，则 A_1 截止， A_2 与 A_3 导通，

$I/O \rightarrow Q$ ，写操作

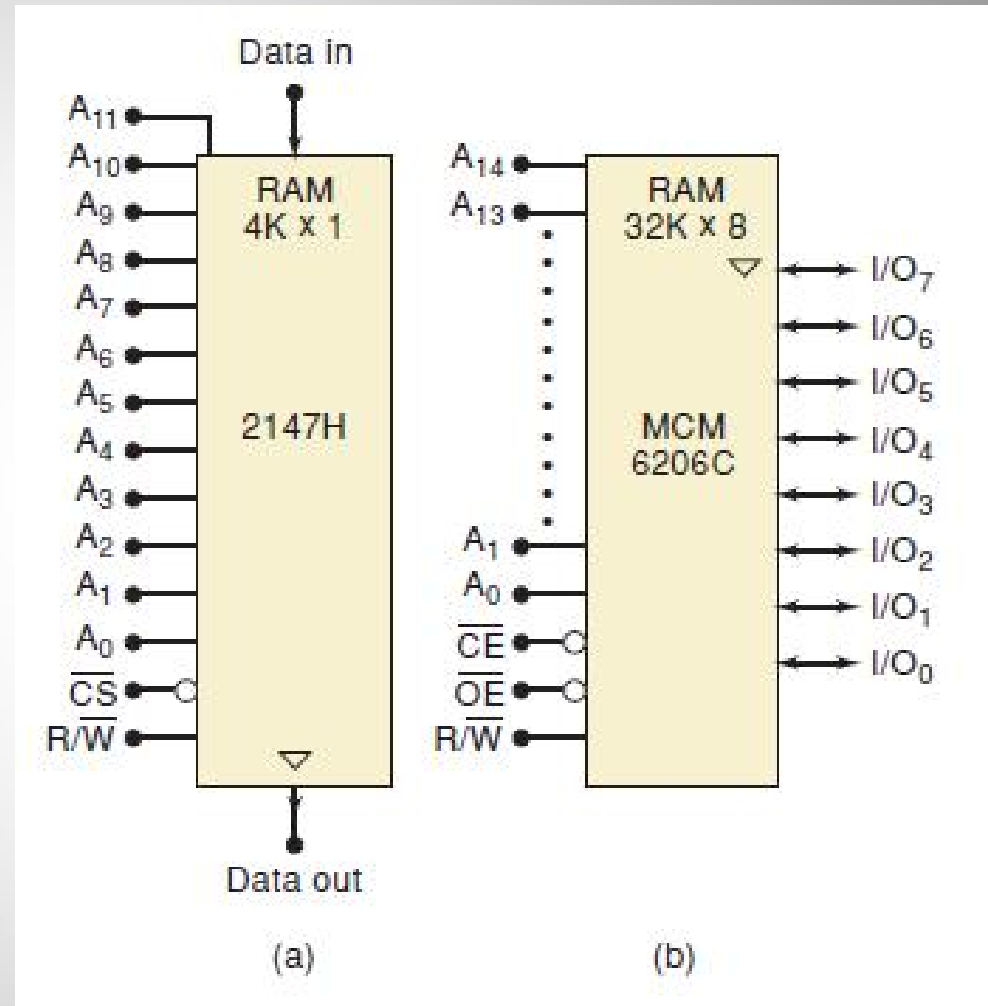
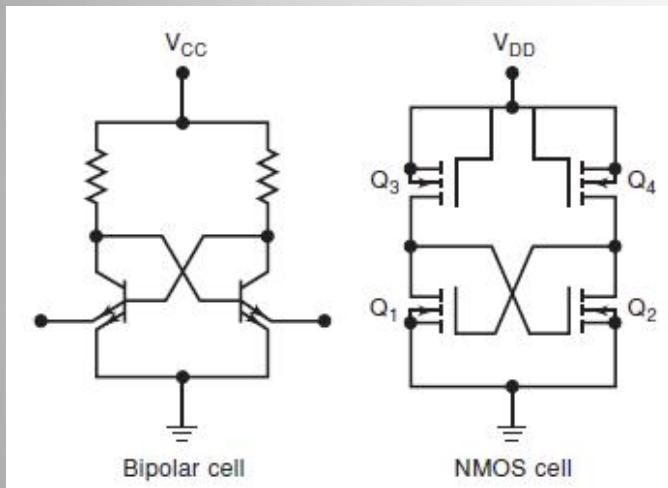
$Y_j = 1$ 时，所在列被选中，

T_7, T_8 导通，这时

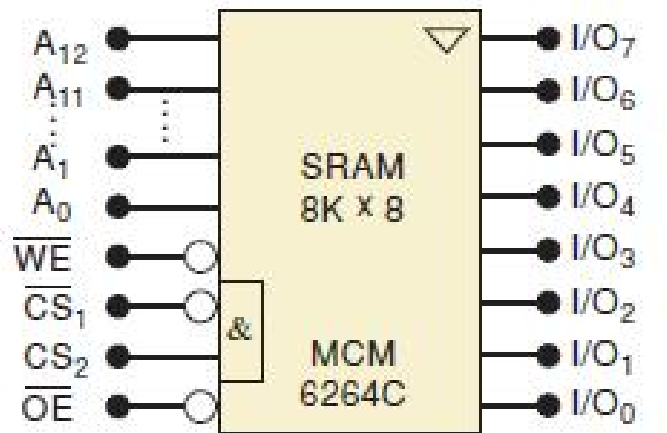
{	第 <i>i</i> 行	}	单元与缓冲器相连
	第 <i>j</i> 列		

Logic symbols

- (a) the 2147H RAM chip;
- (b) the MCM6206C RAM.

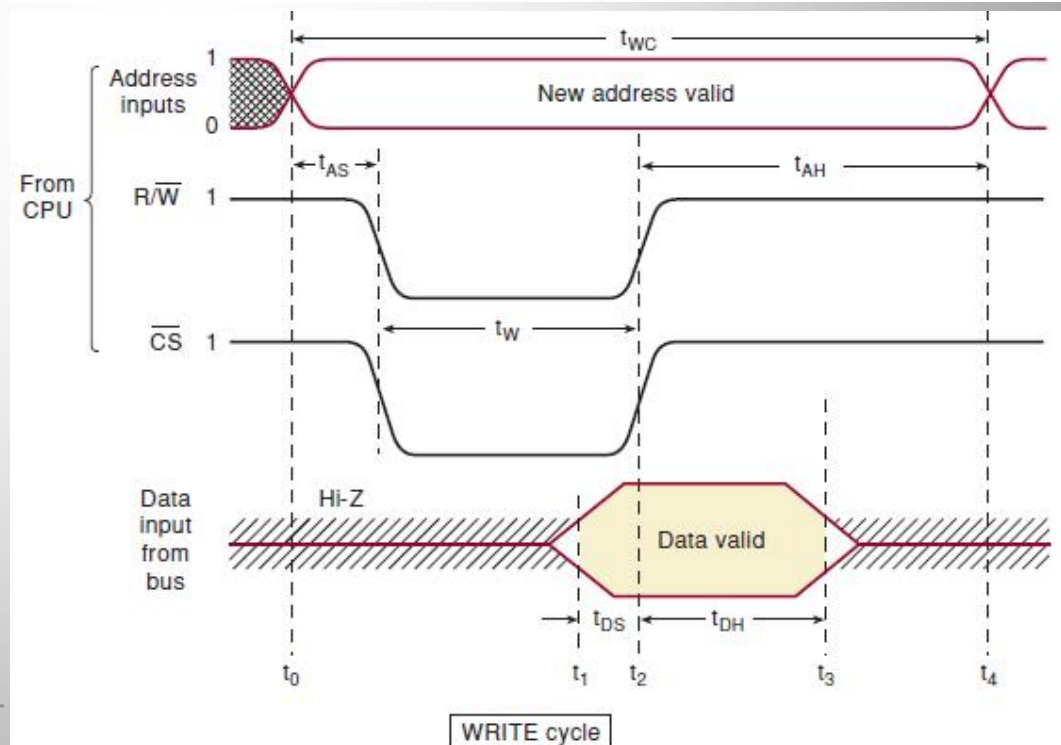
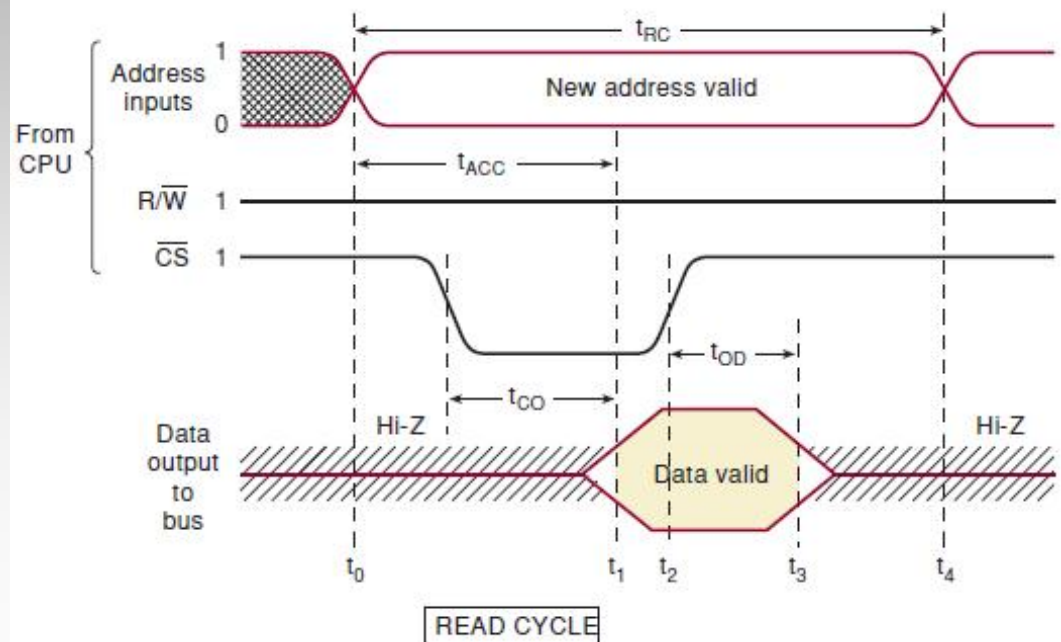


Symbol and mode table for the CMOS MCM6264C.



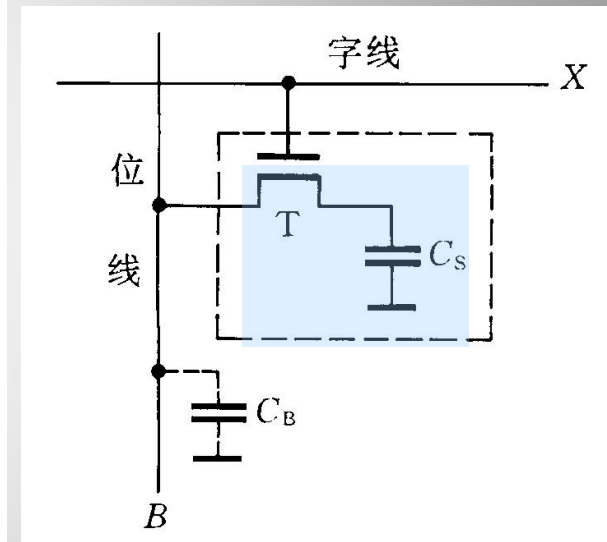
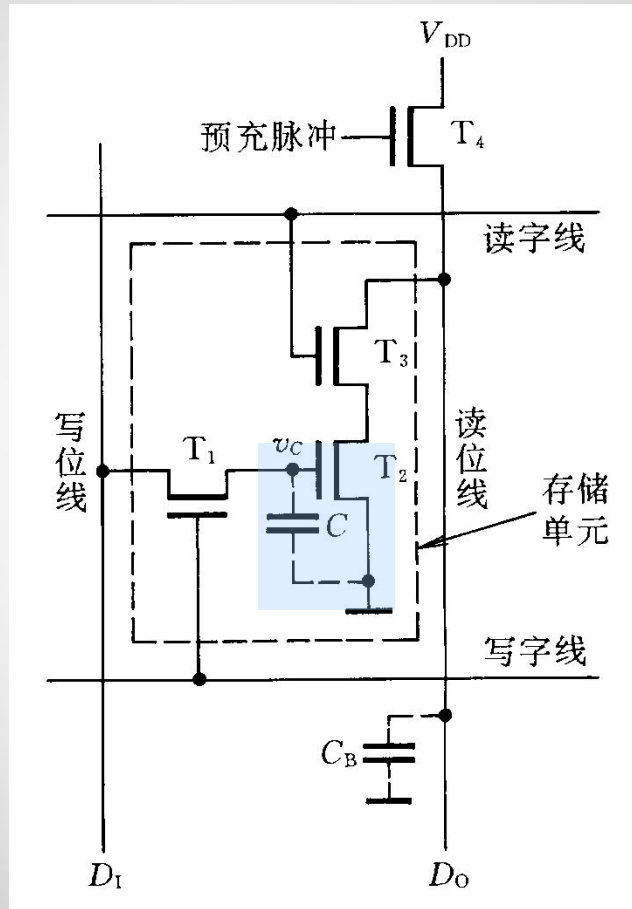
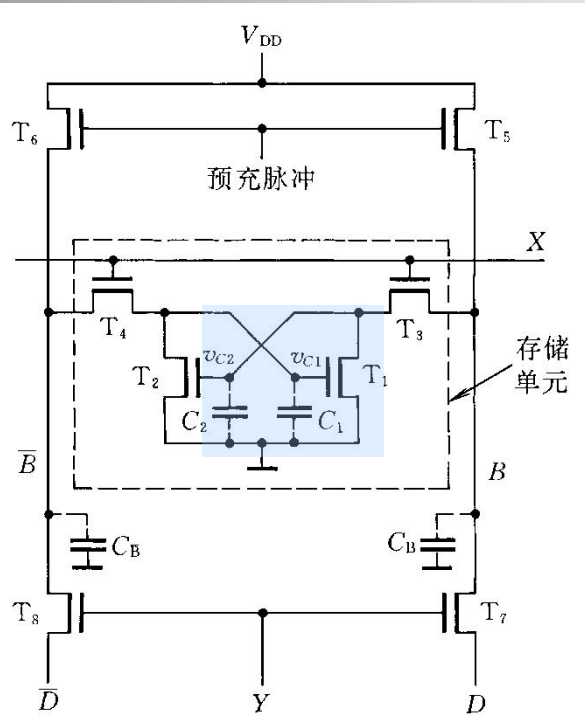
Mode	Inputs				I/O pins
	$\overline{WE}$	$\overline{CS}_1$	CS_2	$\overline{OE}$	
READ	1	0	1	0	DATA _{OUT}
WRITE	0	0	1	X	DATA _{IN}
Output disable	1	X	X	1	High Z
Not selected (power down)	X	1	X	X	High Z

X = don't care



5.5.2* 动态随机存储器 (DRAM) DYNAMIC RAM

动态存储单元是利用MOS管栅极电容可以存储电荷的原理



Dynamic RAMs store 1s and 0s as charges on a small MOS capacitor (typically a few picofarads). In modern DRAM chips, each memory cell must be refreshed typically every 2, 4, or 8 ms, or its data will be lost.

Cell arrangement in a dynamic RAM

- DRAM TECHNOLOGY

- Memory Modules

- » single-in-line memory module (SIMM)
 - » dual-in-line memory module (DIMM)
 - » SODIMM
 - » RIMM
 - » DRDRAM

- FPM DRAM

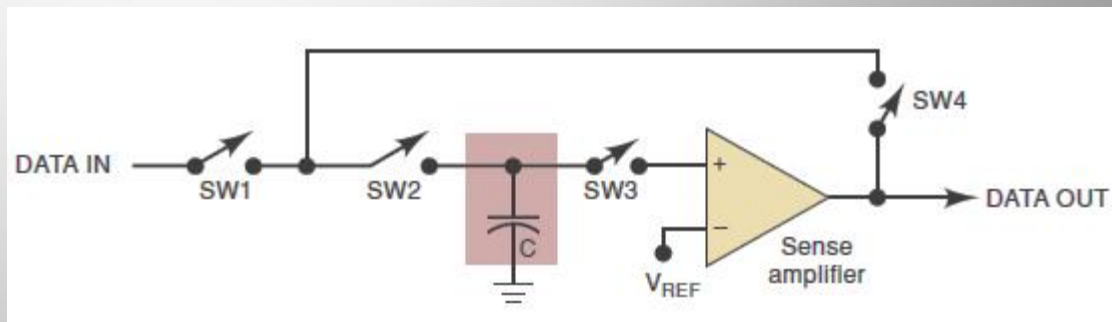
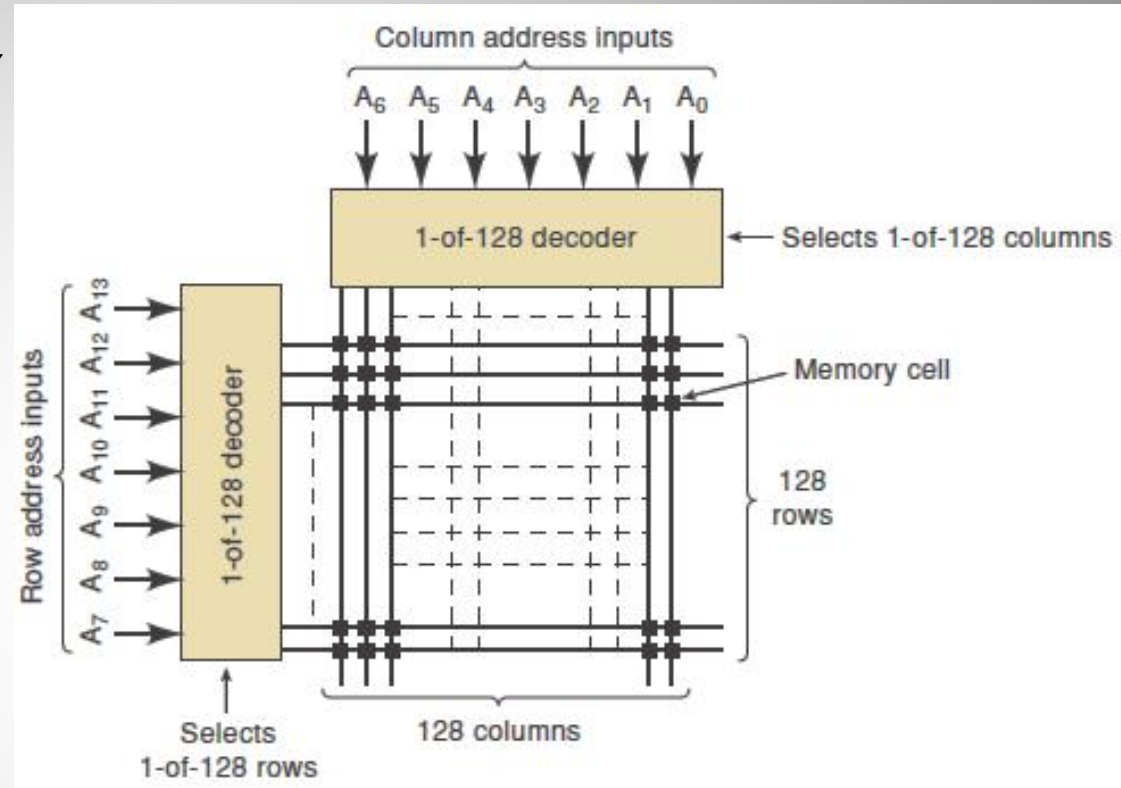
- EDO DRAM

- SDRAM

- DDRSDRAM

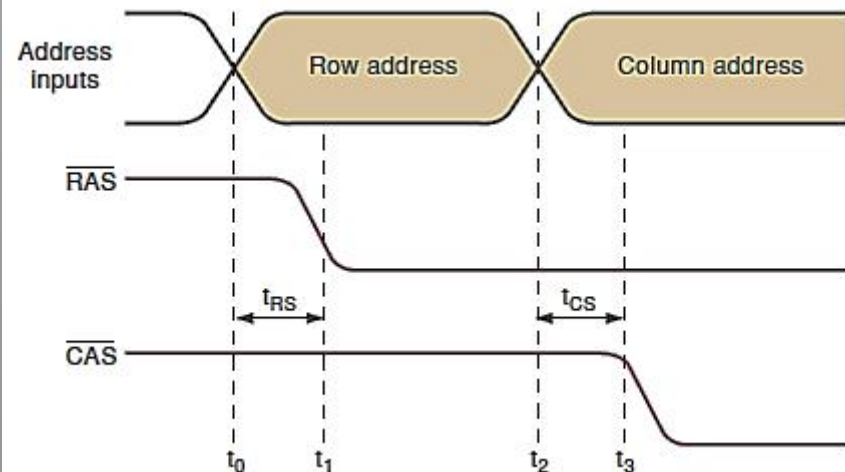
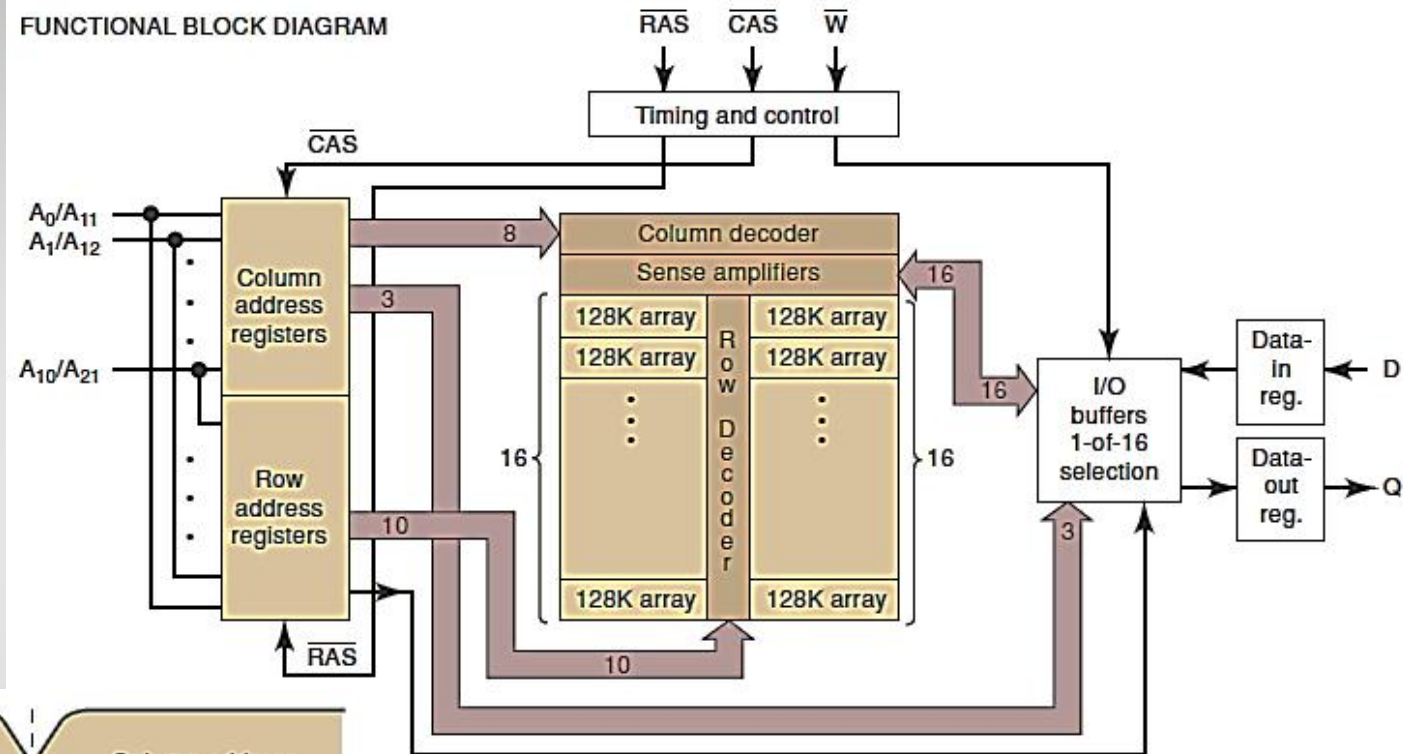
- SLDRAM

- DRDRAM



Simplified architecture of the TMS44100 ; RAS>CAS timing

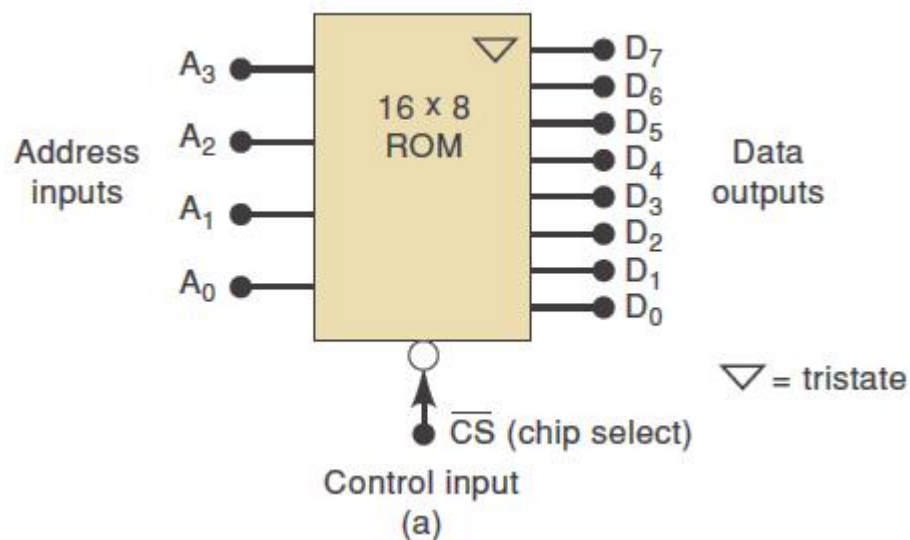
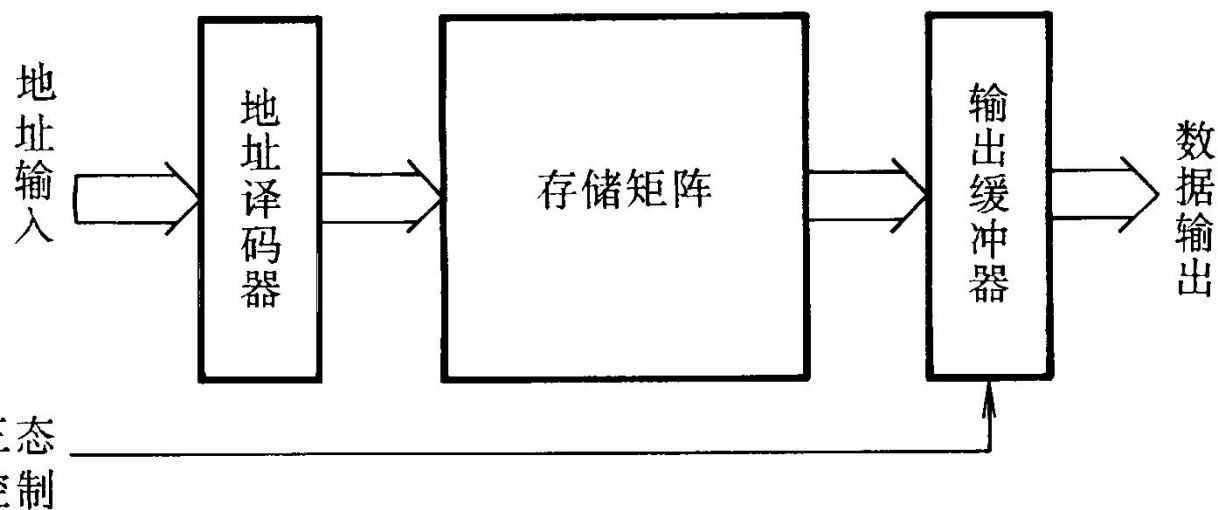
FUNCTIONAL BLOCK DIAGRAM



5.5.3 ROM-READ-ONLY MEMORIES

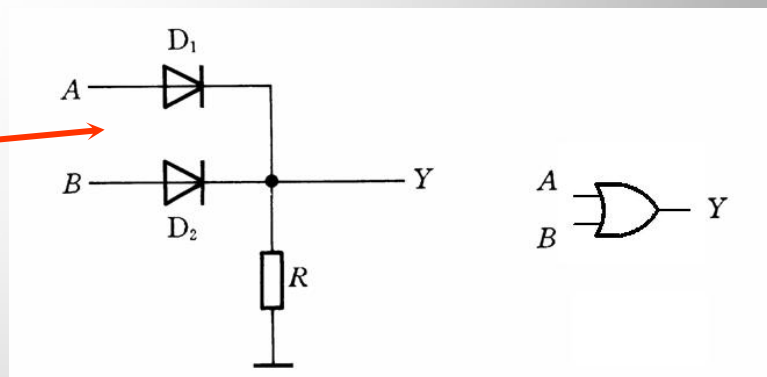
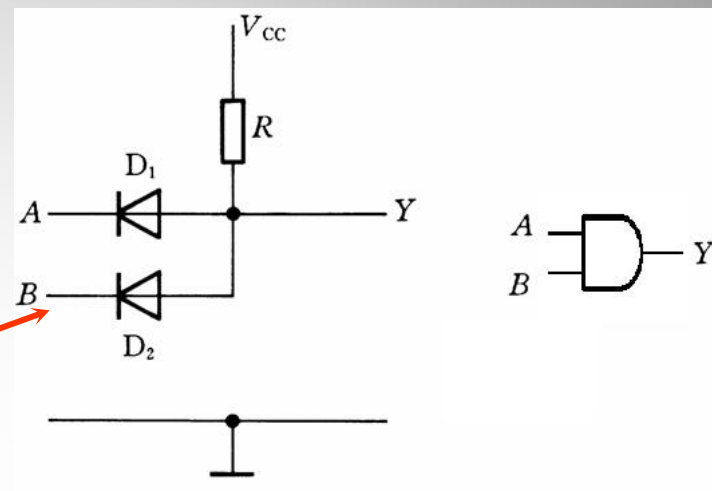
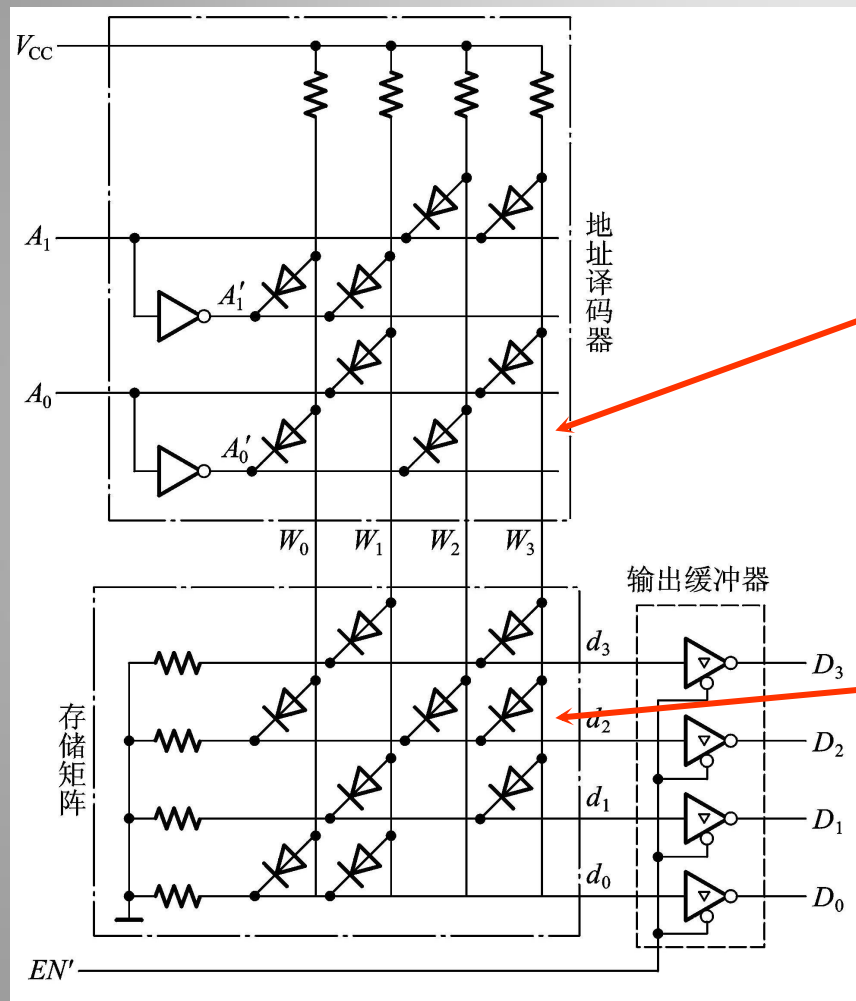
1 掩模ROM

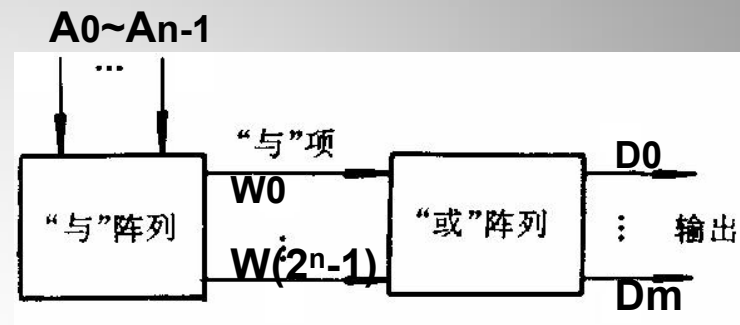
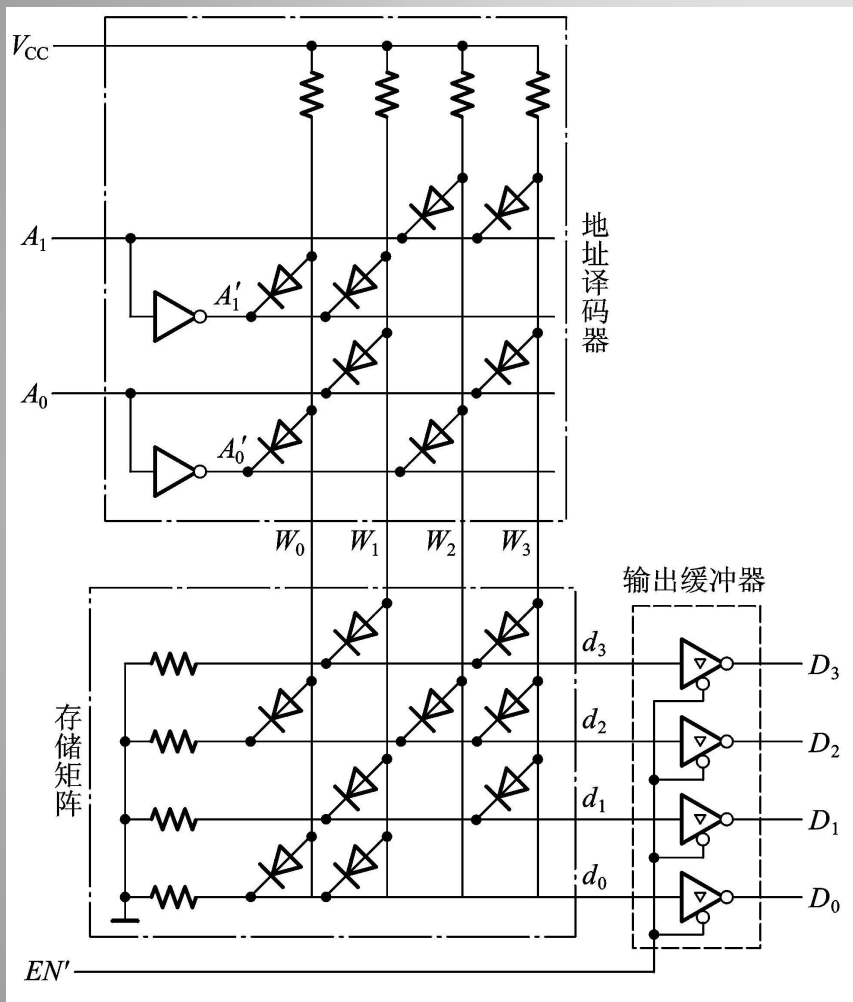
一、结构



Word	Address				Data
	A_3	A_2	A_1	A_0	D_7-D_0
0		0			DE
1		1			3A
2		2			85
3		3			AF
4		4			19
5		5			7B
6		6			00
7		7			ED
8		8			3C
9		9			FF
10		A			B8
11		B			C7
12		C			27
13		D			6A
14		E			D2
15		F			5B

二、举例

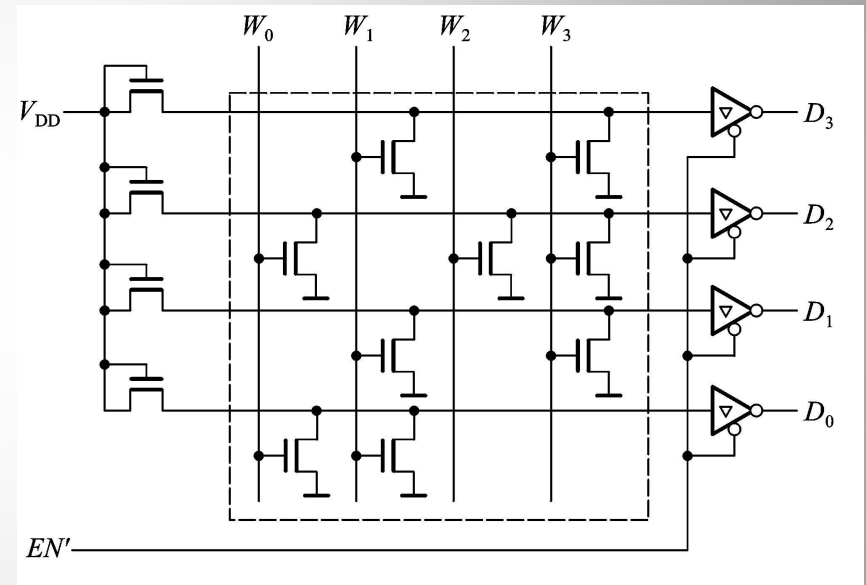
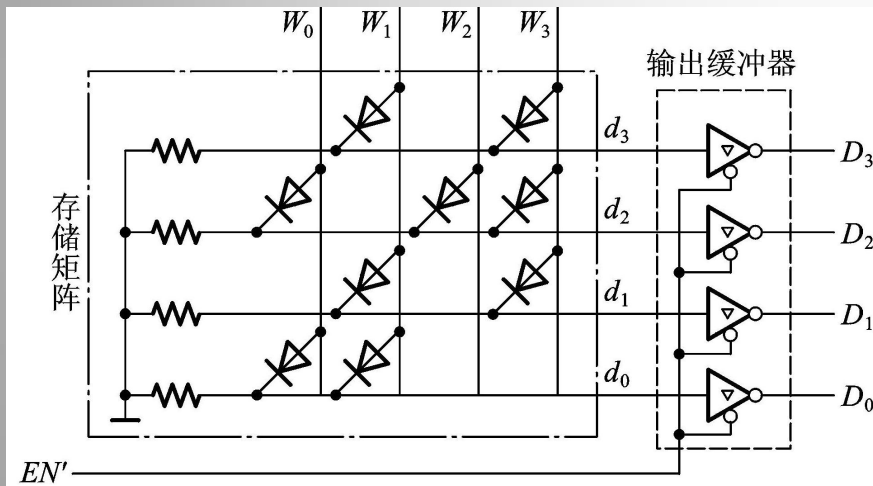




地 址		数 据			
A ₁	A ₀	D ₃	D ₂	D ₁	D ₀
0	0	0	1	0	1
0	1	1	0	1	1
1	0	0	1	0	0
1	1	1	1	1	0

两个概念：

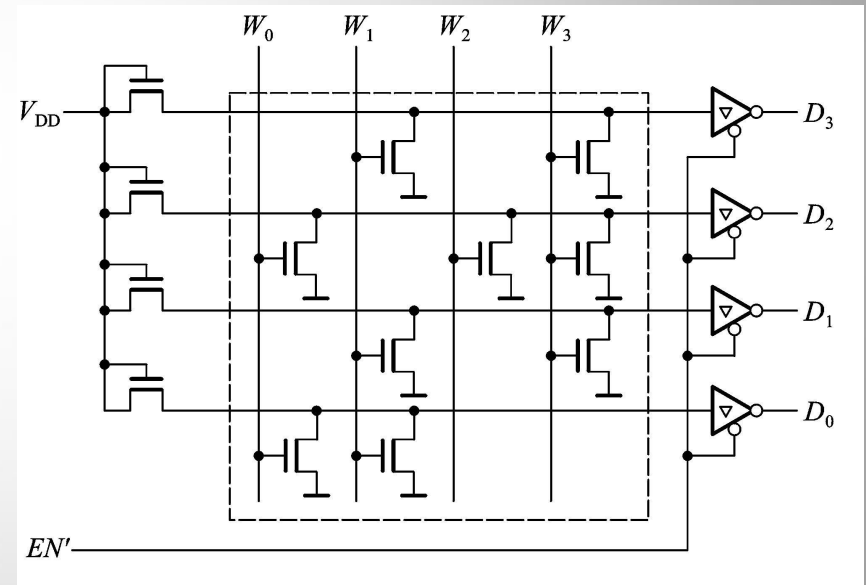
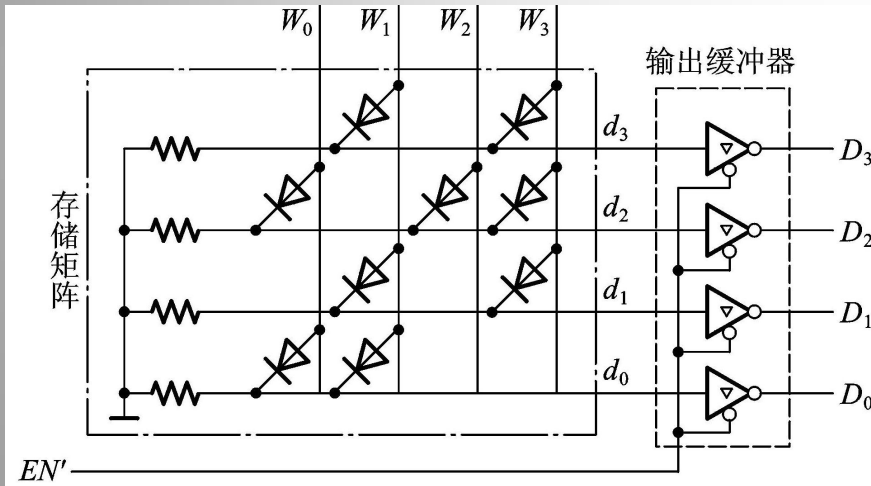
- 存储矩阵的每个交叉点是一个“存储单元”，存储单元中有器件存入“1”，无器件存入“0”



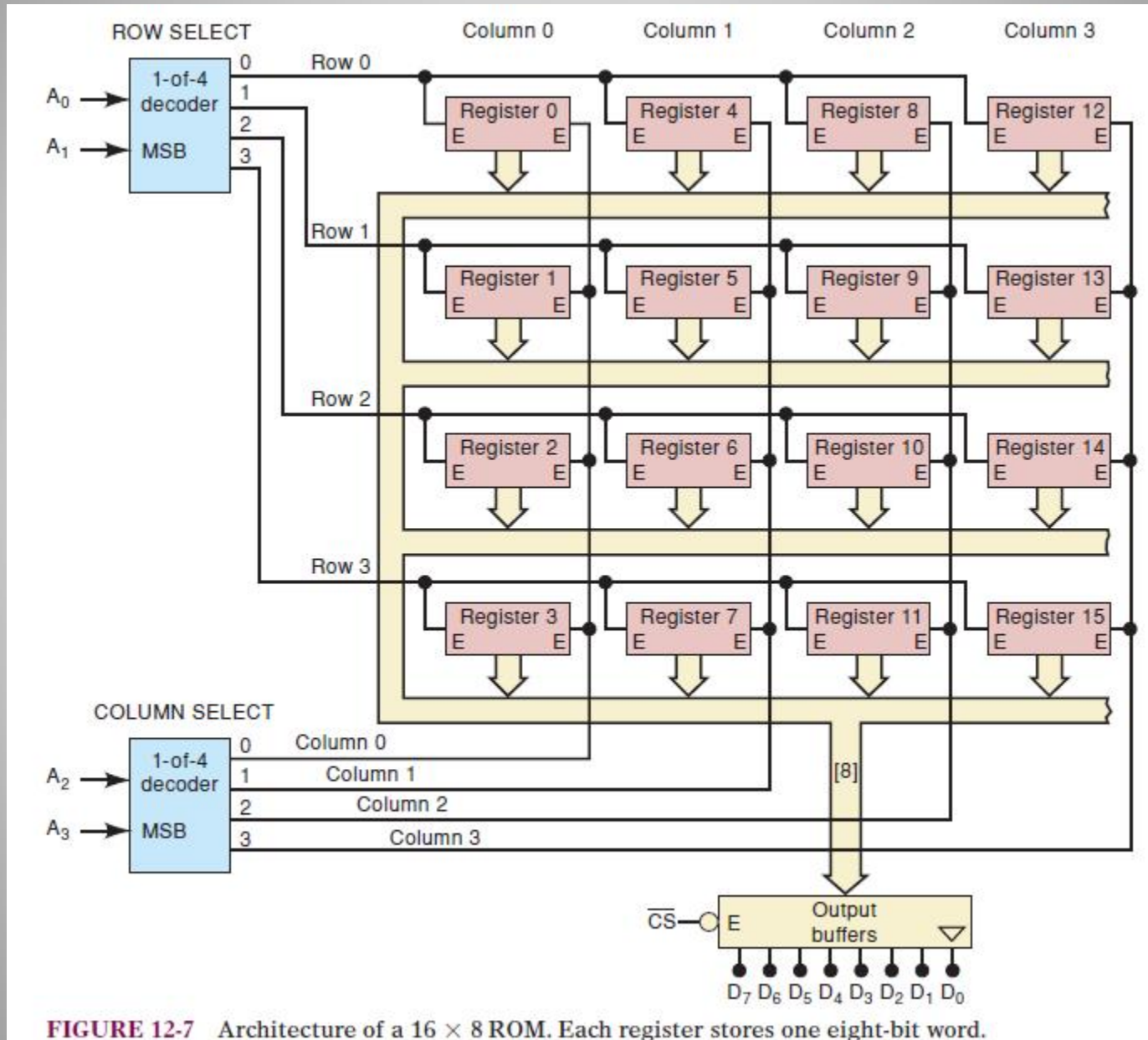
- 存储器的容量：“字数 x 位数”

掩模ROM的特点:

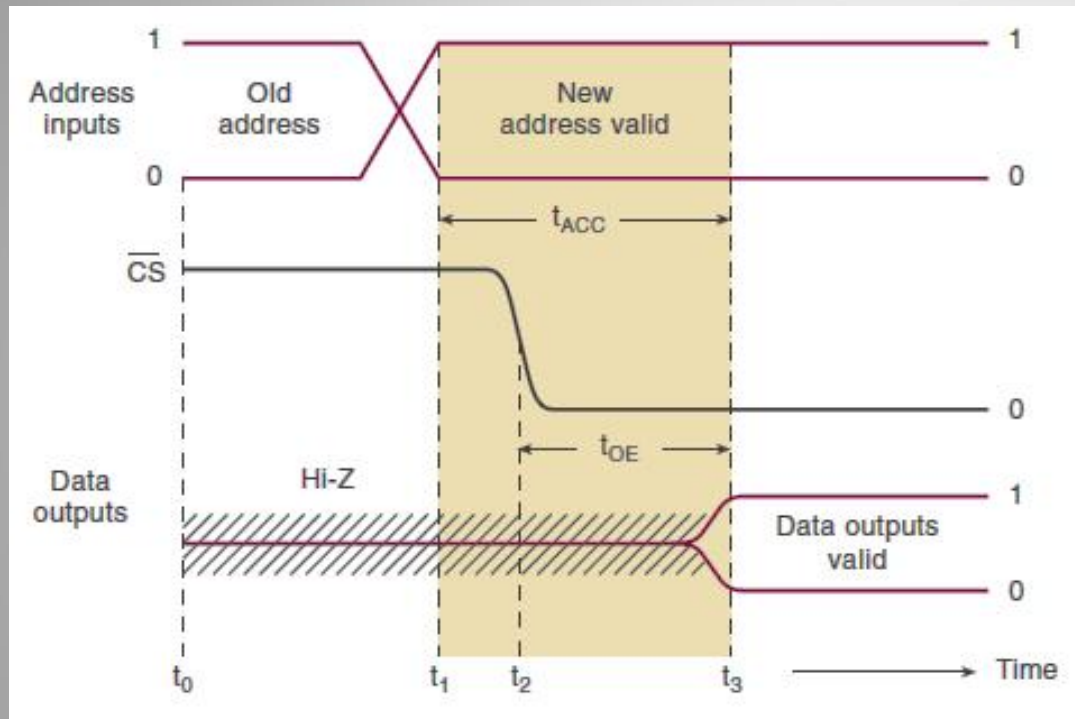
出厂时已经固定, 不能更改, 适合大量生产
简单, 便宜, 非易失性nonvolatile



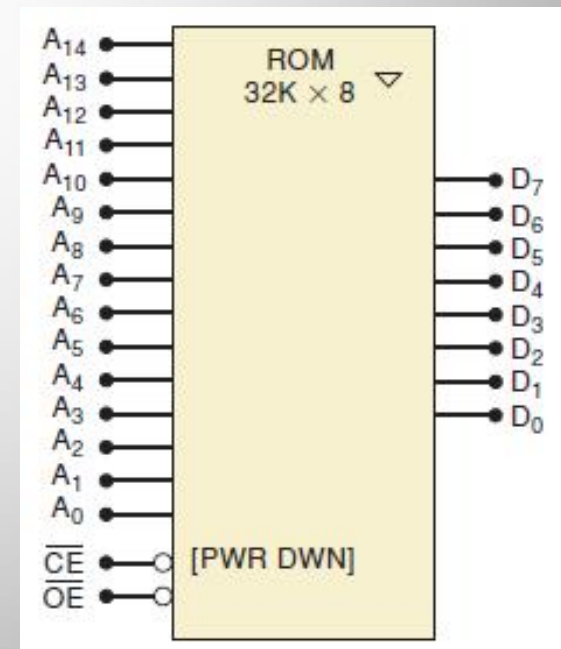
ROM ARCHITECTURE



Typical timing for a ROM read operation.

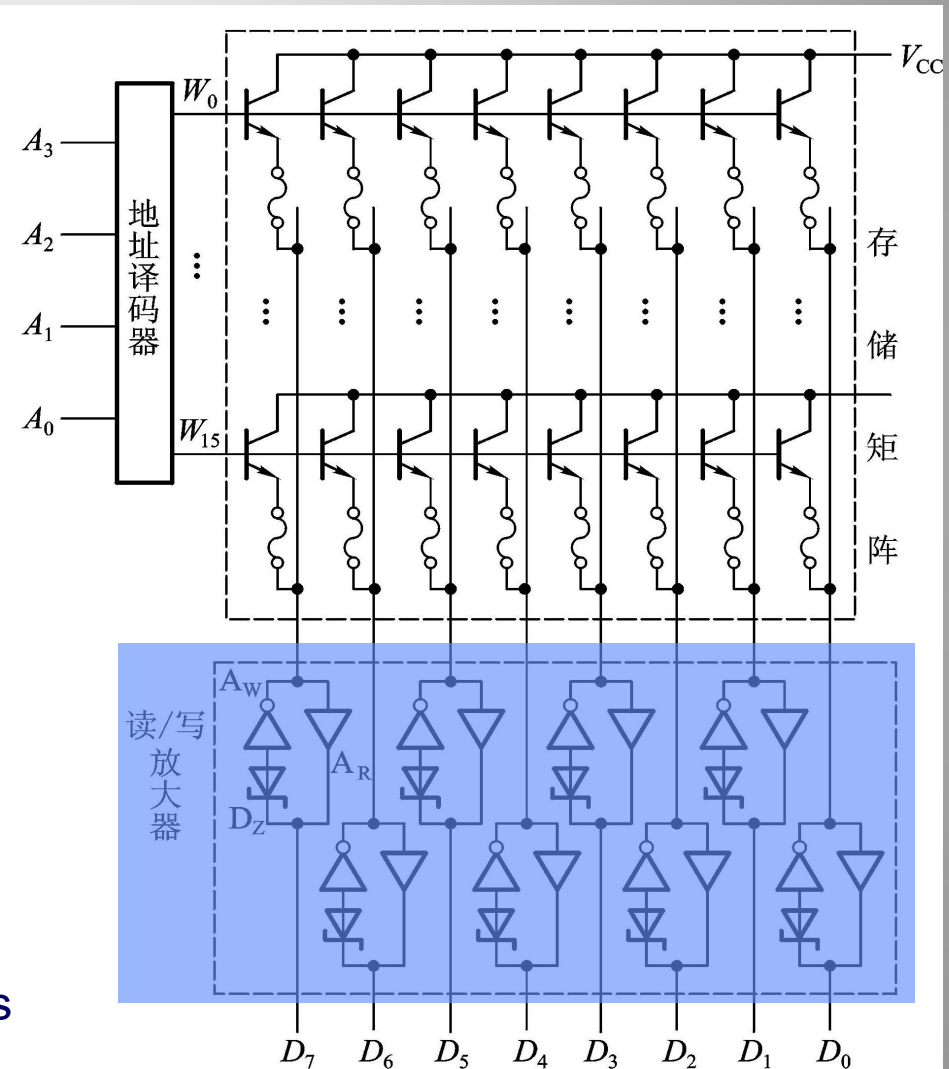
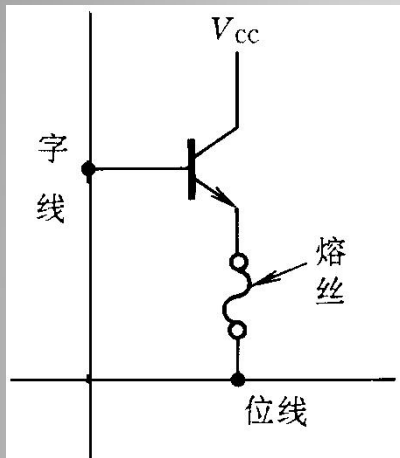


x		$y = x^2 + 3$			
A_1	A_0	D_3	D_2	D_1	D_0
0	0	0	0	1	1
0	1	0	1	0	0
1	0	0	1	1	1
1	1	1	1	0	0



2 可编程ROM (PROM) Programmable ROMs

总体结构与掩模ROM一样，但存储单元不同

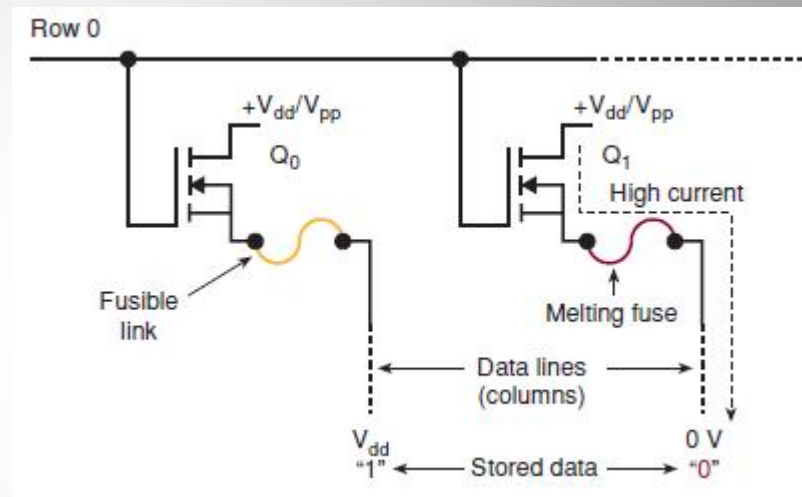
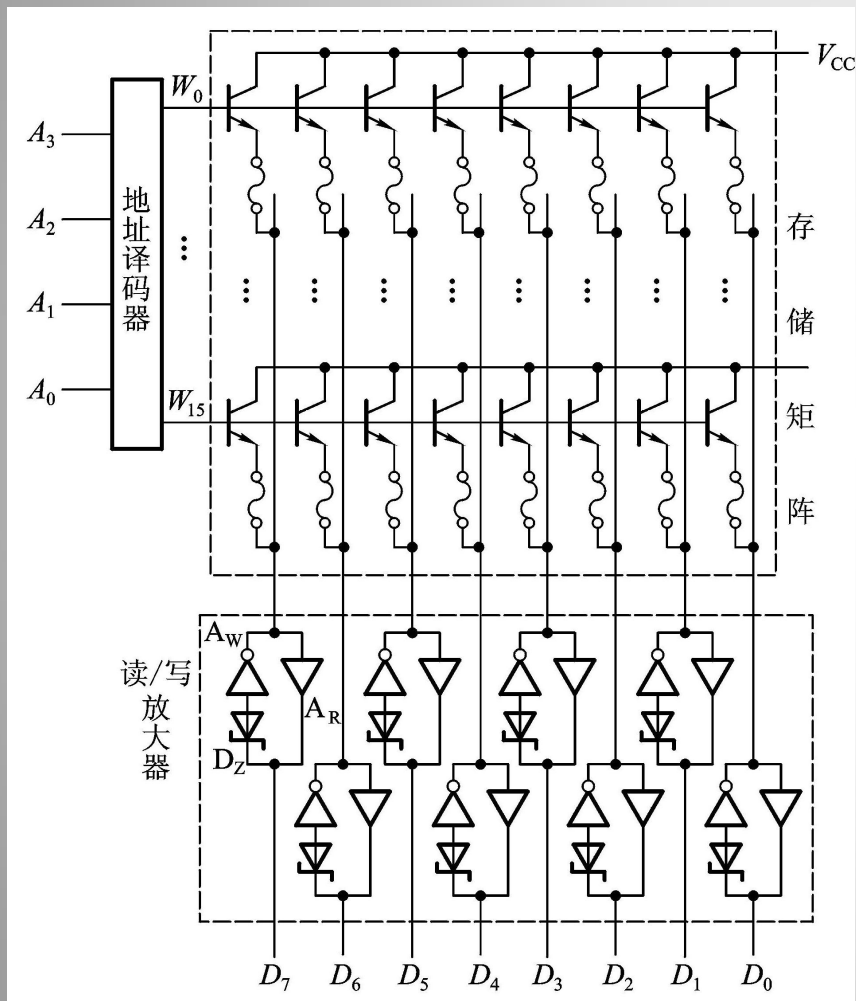


- * 熔丝由易熔合金制成
 - * 出厂时，每个结点上都 有
 - * 编程时将不用的熔断！！
- 是一次性编程，不能改 写

“one-time programmable” (OTP) ROMs

2 可编程ROM (PROM)

总体结构与掩模ROM一样，但存储单元不同



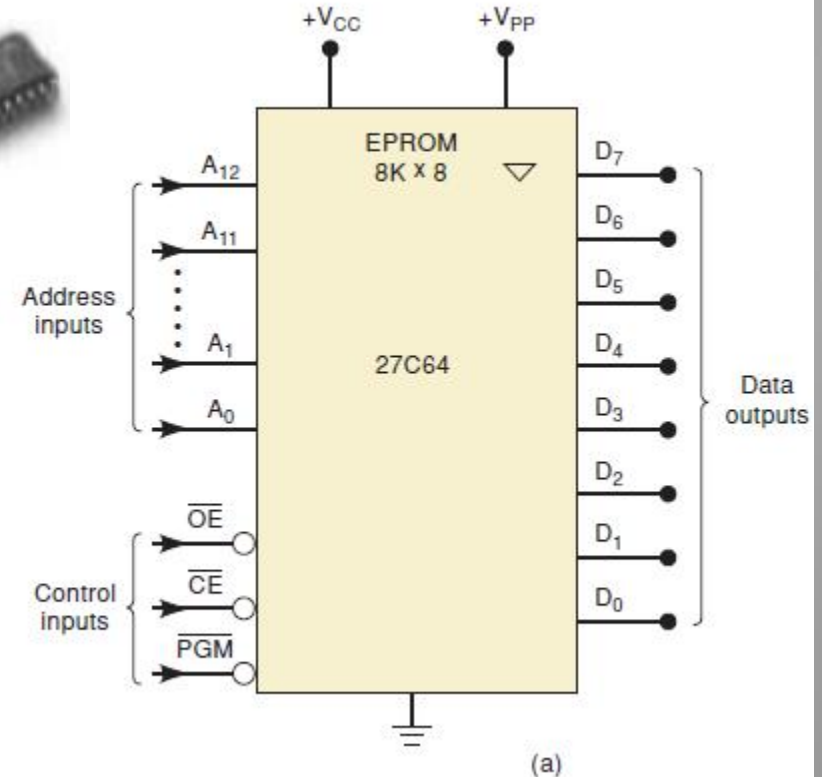
写入时，要使用编程器

3 可擦除的可编程ROM (EPROM) Erasable Programmable ROM

总体结构与掩模ROM一样，但存储单元不同

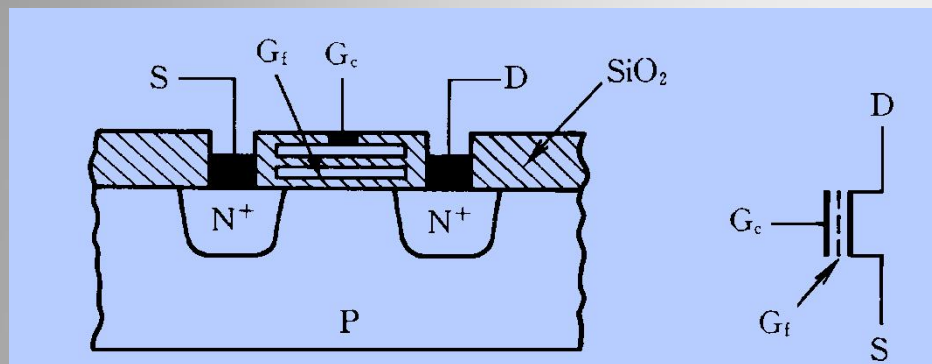
一、用紫外线擦除的PROM (UVEPROM)

nonvolatile



SIMOS(Stacked – gate Injection MOS)

叠栅注入MOS管



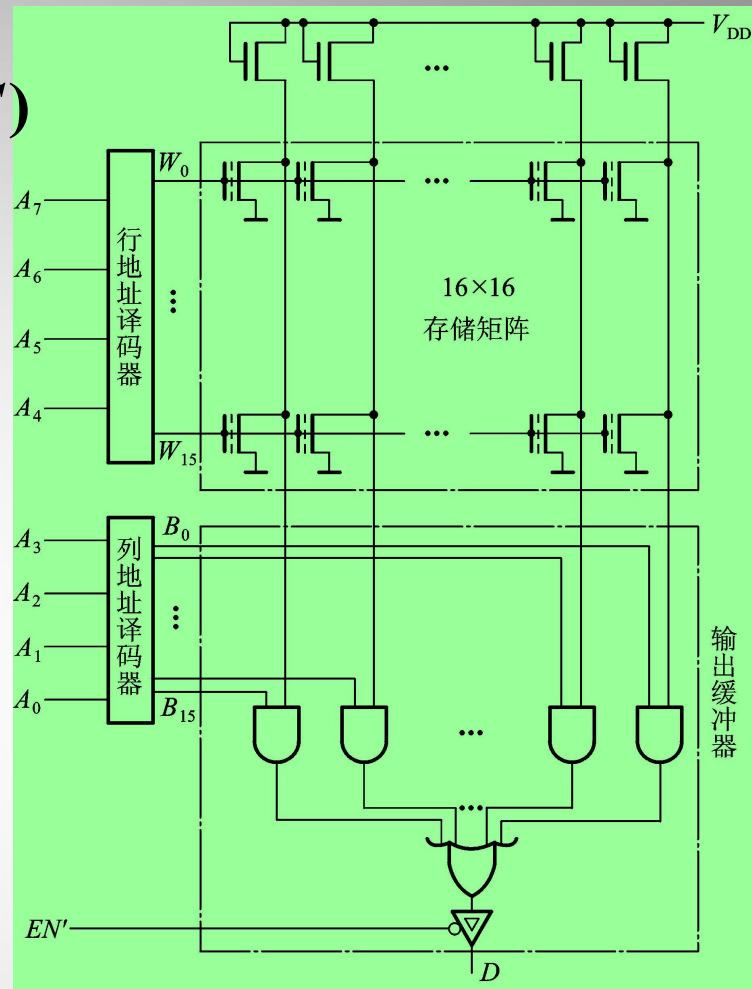
G_c : 控制栅

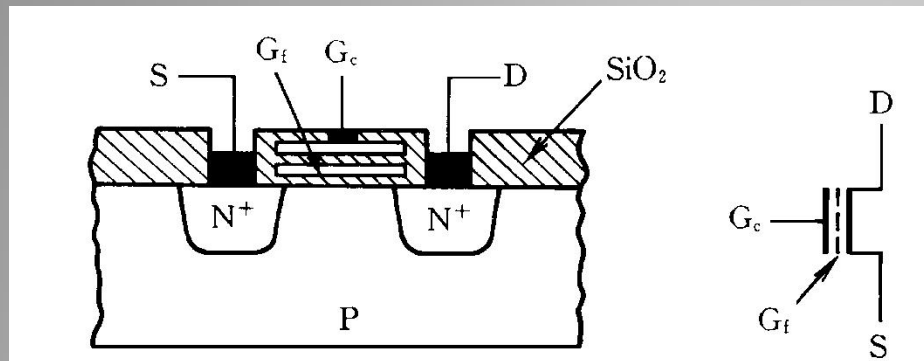
G_f : 浮置栅

工作原理:

若 G_f 上充以负电荷, 则 G_c 处正常逻辑高电平下不导通

若 G_f 上未充负电荷, 则 G_c 处正常逻辑高电平下导通





	Inputs				Outputs
Mode	$\overline{CE}$	$\overline{OE}$	$\overline{PGM}$	VPP	D ₇ – D ₀
Read	0	0	1	0–5V	DATA _{out}
Output Disable	0	1	1	0–5V	High Z
Standby	1	X	X	X	High Z
Program	0	1	0	12.75 V	DATA _{in}
PGM Verify	0	0	1	12.75 V	DATA _{out}

“写入”：雪崩注入, $D-S$ 间加高压 ($20 \sim 25V$), 发生雪崩击穿
 同时在 G_c 上加 $25V, 50ms$ 宽的正脉冲,
 吸引高速电子穿过 SiO_2 到达 G_f , 形成注入电荷

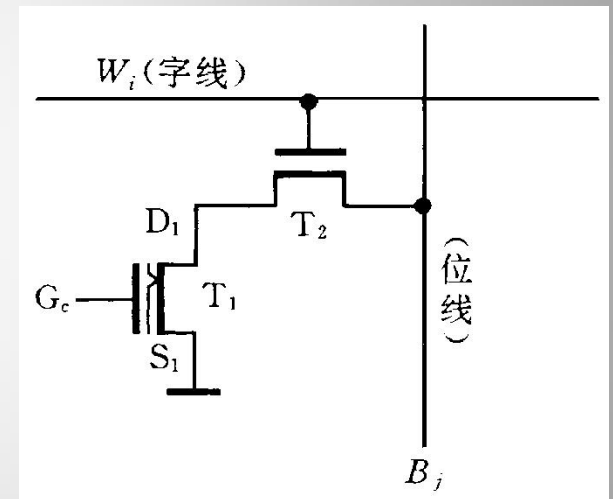
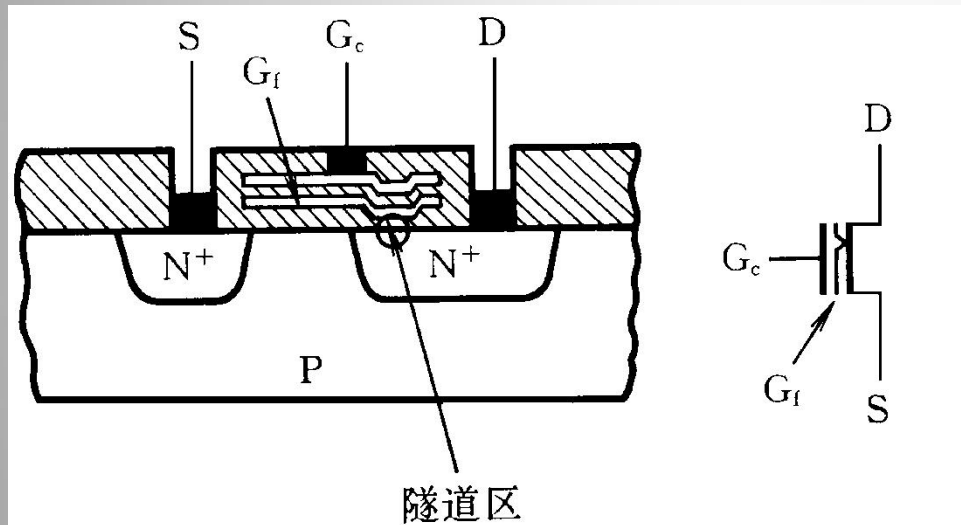
“擦除”：通过照射产生电子-空穴对, 提供泄放通道
 紫外线照射 $20 \sim 30$ 分钟 (阳光下一周, 荧光灯下3年)

二、电可擦除的可编程ROM (E²PROM)

Electrically Erasable PROM

总体结构与掩模ROM一样，但存储单元不同
为克服 *UVEPROM* 擦除慢，操作不便的缺点
采用 *FLOTOX* (浮栅隧道氧化层 *MOS* 管)

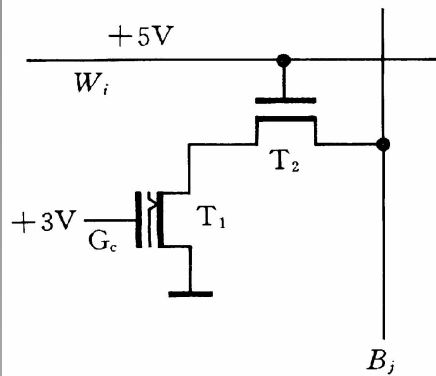
Without a UV light source
and a special PROM
programmer unit



G_f 与 D 之间有小的隧道区, SiO_2 厚度 $< 2 \times 10^{-8} \text{ m}$

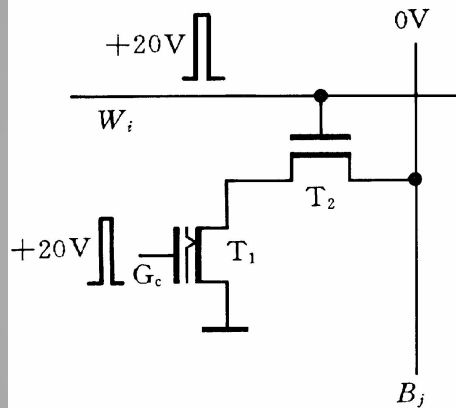
当场强达到一定大小 (10^7 V/cm), 电子会穿越隧道

——“隧道效应”

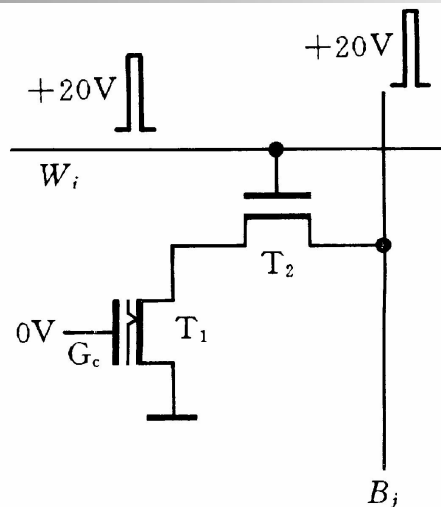


工作原理:

G_f 充电荷后, 正常读出 G_c 电压 ($3V$) 下, T 截止
未充电荷时, 正常读出 G_c 电压 ($3V$) 下, T 导通



充电: W_i, G_c 加 $20V, 10ms$ 的正脉冲, B_j 接 0
电子隧道区 $\rightarrow G_f$

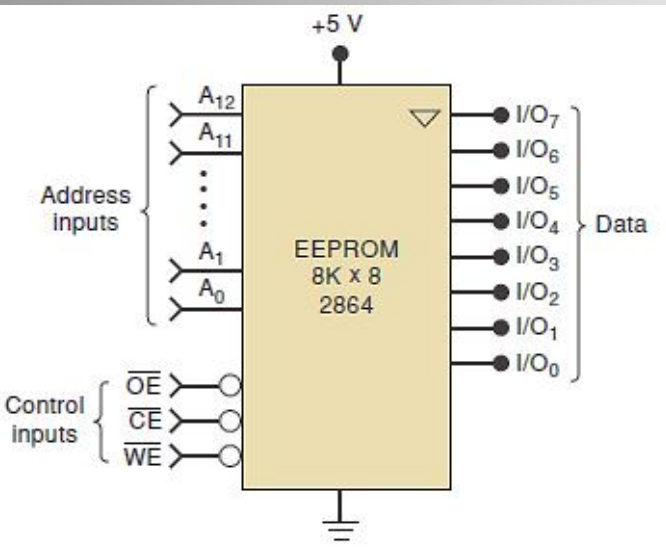


放电: G_c 接 0 , W_i, B_j 加正脉冲,
 G_f 上电荷经隧道区放电

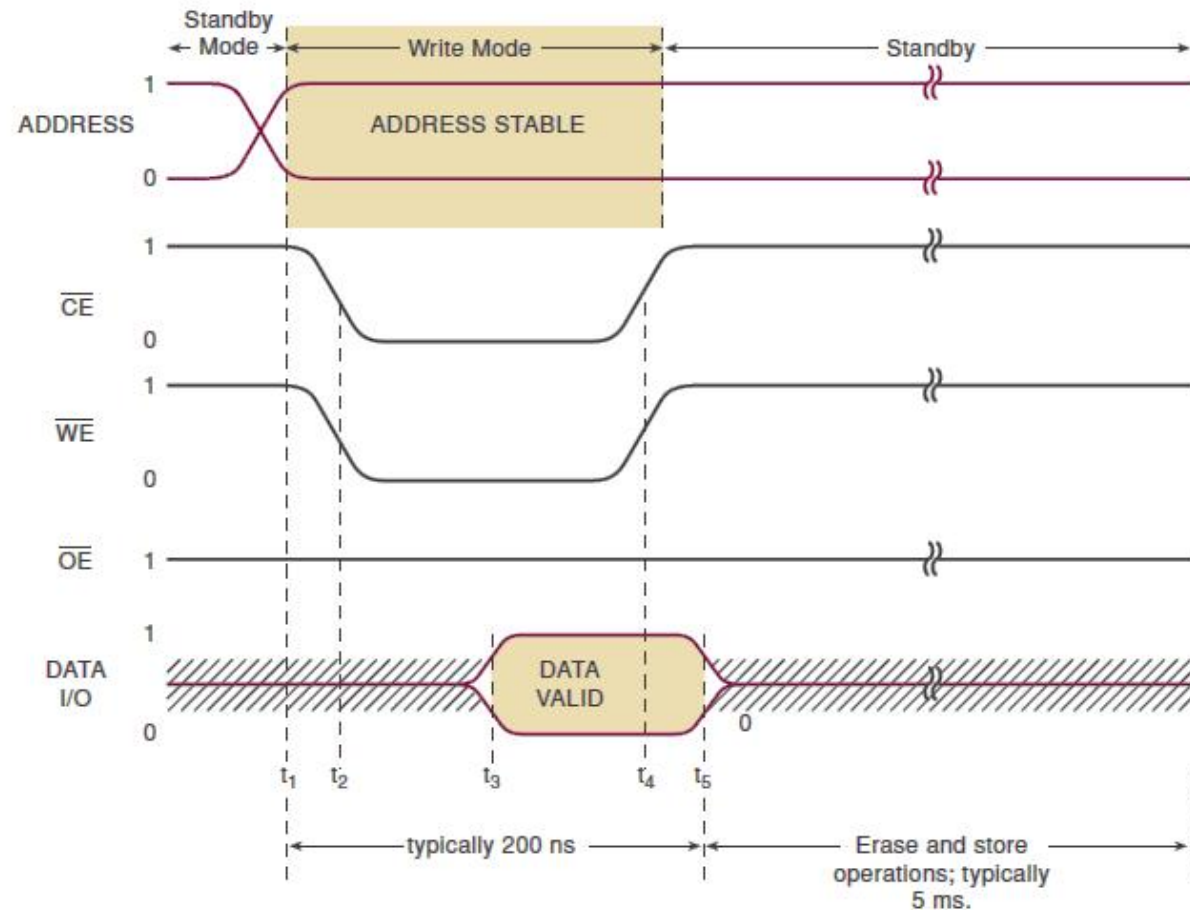
Electrically Erasable PROM

- **Another advantage** of the EEPROM over the EPROM is the ability to erase and rewrite individual bytes (eight-bit words) in the memory array electrically.
- This **byte erasability** makes it much easier to make changes in the data stored in an EEPROM.
- The **early EEPROMs**, such as Intel's 2816, required appropriate support circuitry external to the memory chips, included the 21-V programming voltage (*VPP*).
- The **newer devices**, such as the Intel 2864, have integrated this support circuitry onto the same chip with the memory array, it requires only a single 5-V power pin.

Timing for the write operation.

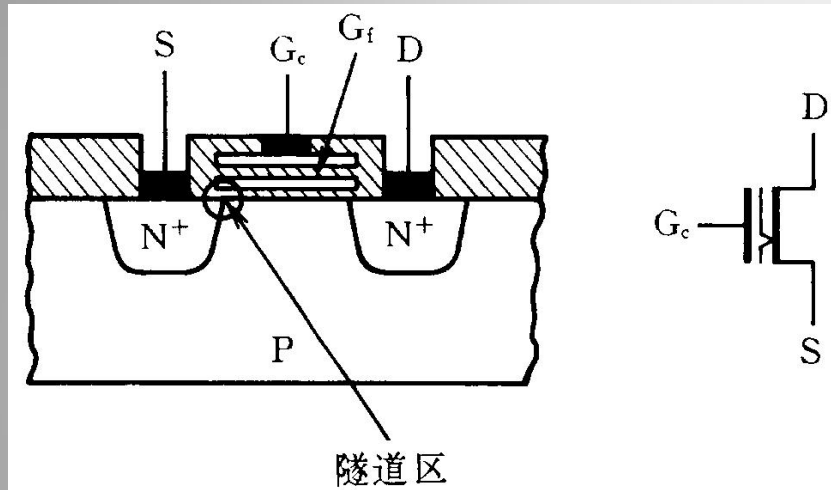


Mode	Inputs			I/O pins
	$\overline{CE}$	$\overline{OE}$	$\overline{WE}$	
READ	LOW	LOW	HIGH	DATA _{OUT}
WRITE	LOW	HIGH	LOW	DATA _{IN}
STANDBY	HIGH	X	X	High Z



三、快闪存储器 (Flash Memory)

为提高集成度，省去T2（选通管）
改用叠栅MOS管（类似SIMOS管）



G_f 与衬底间 S_iO_2 更薄 (10~15nm)

G_f 与S区有极小的重叠区—(隧道区)

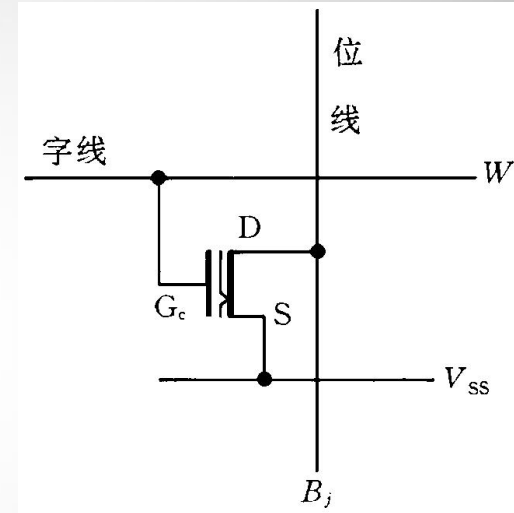
G_f 放电，利用隧道效应

$G_c = 0, V_{ss}$ 加12V, 100ns的正脉冲

G_f 上电荷经隧道区放电

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The challenge for semiconductor engineers was to fabricate a nonvolatile memory with the EEPROM's in-circuit electrical erasability, but with densities and costs much closer to those of EPROMs, while retaining the high-speed read access of both.



*工作原理:

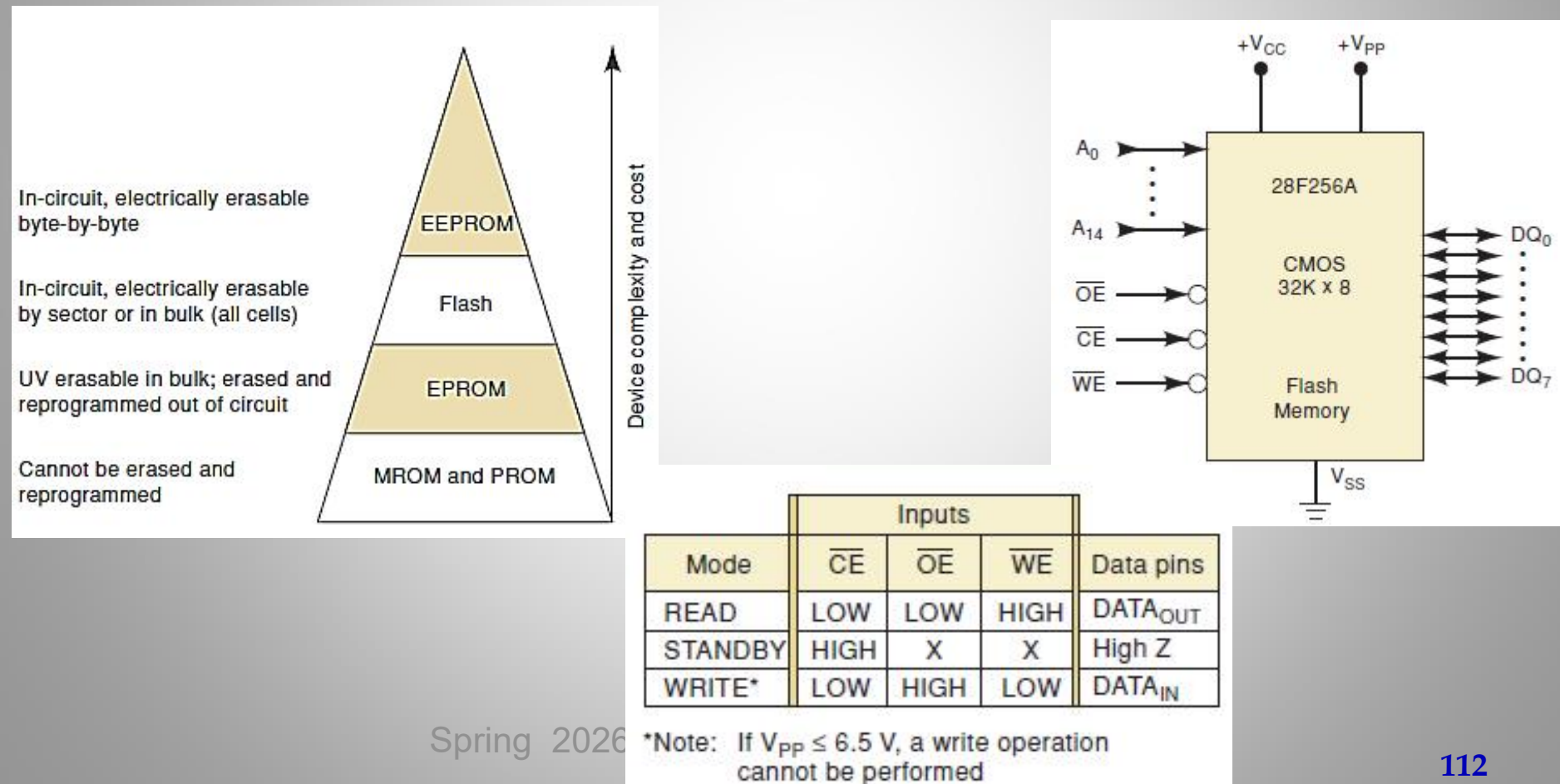
向 G_f 充电利用雪崩注入方式,

$D-S$ 加正压 (6V), V_{ss} 接0

G_c 加12V, 10us的正脉冲

Trade-offs for semiconductor nonvolatile memories show that complexity and cost increase as erase and programming flexibility increases.

- A typical flash memory has a write time of 10 ms per byte compared to 100 for the most advanced EPROM and 5 ms for EEPROM (which includes automatic byte erase time).



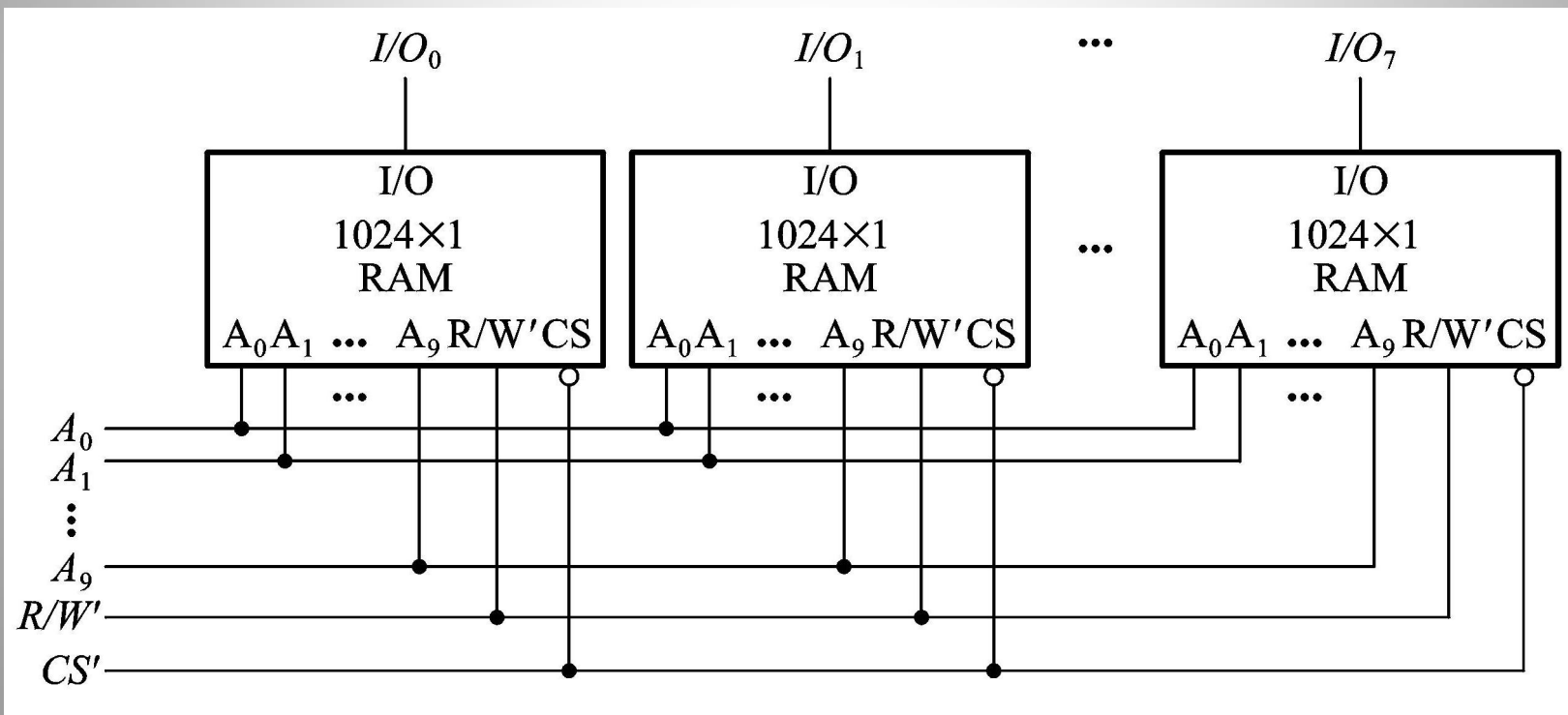
5.5.4 存储器容量的扩展

1. 位扩展方式

适用于每片RAM,ROM字数够用而位数不够时

接法：将各片的地址线、读写线、片选线并联即可

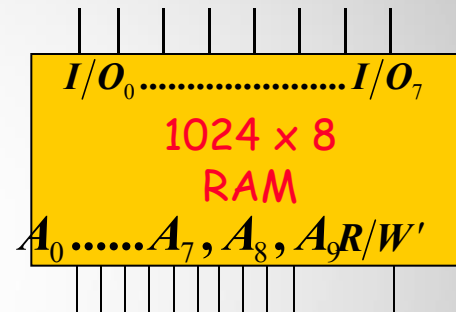
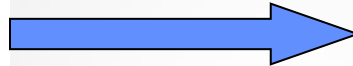
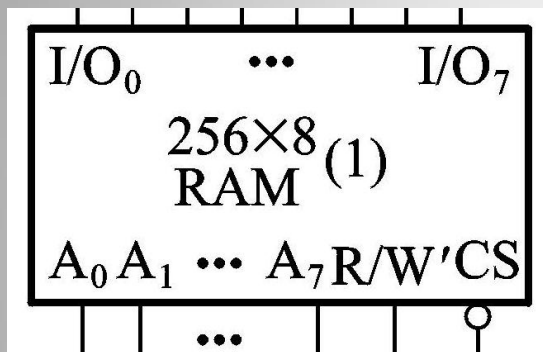
例：用八片 1024×1 位 $\rightarrow 1024 \times 8$ 位的RAM



2. 字扩展方式

适用于每片RAM,ROM位数够用而字数不够时

例：用四片 256×8 位 $\rightarrow 1024 \times 8$ 位 RAM

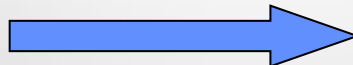


数据线: $I/O_0 \sim I/O_7$

地址线: $A_0 \sim A_7$

读/写信号: R/W'

片选信号: CS'



数据线: $I/O_0 \sim I/O_7$

地址线: $A_0 \sim A_7, A_8, A_9$

读/写信号: R/W'

每一片提供256个字，需要256个地址($A_{0\sim7} : 0 \sim 0\text{---}1 \sim 1$)

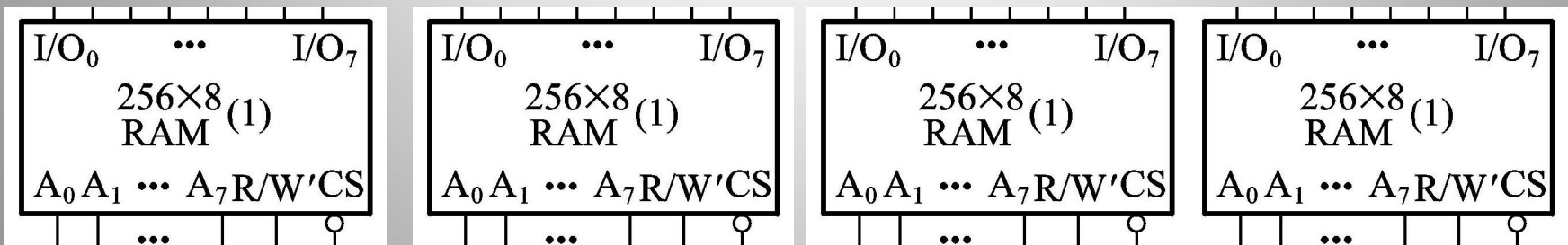
用 A_9, A_8 两位代码区分四片

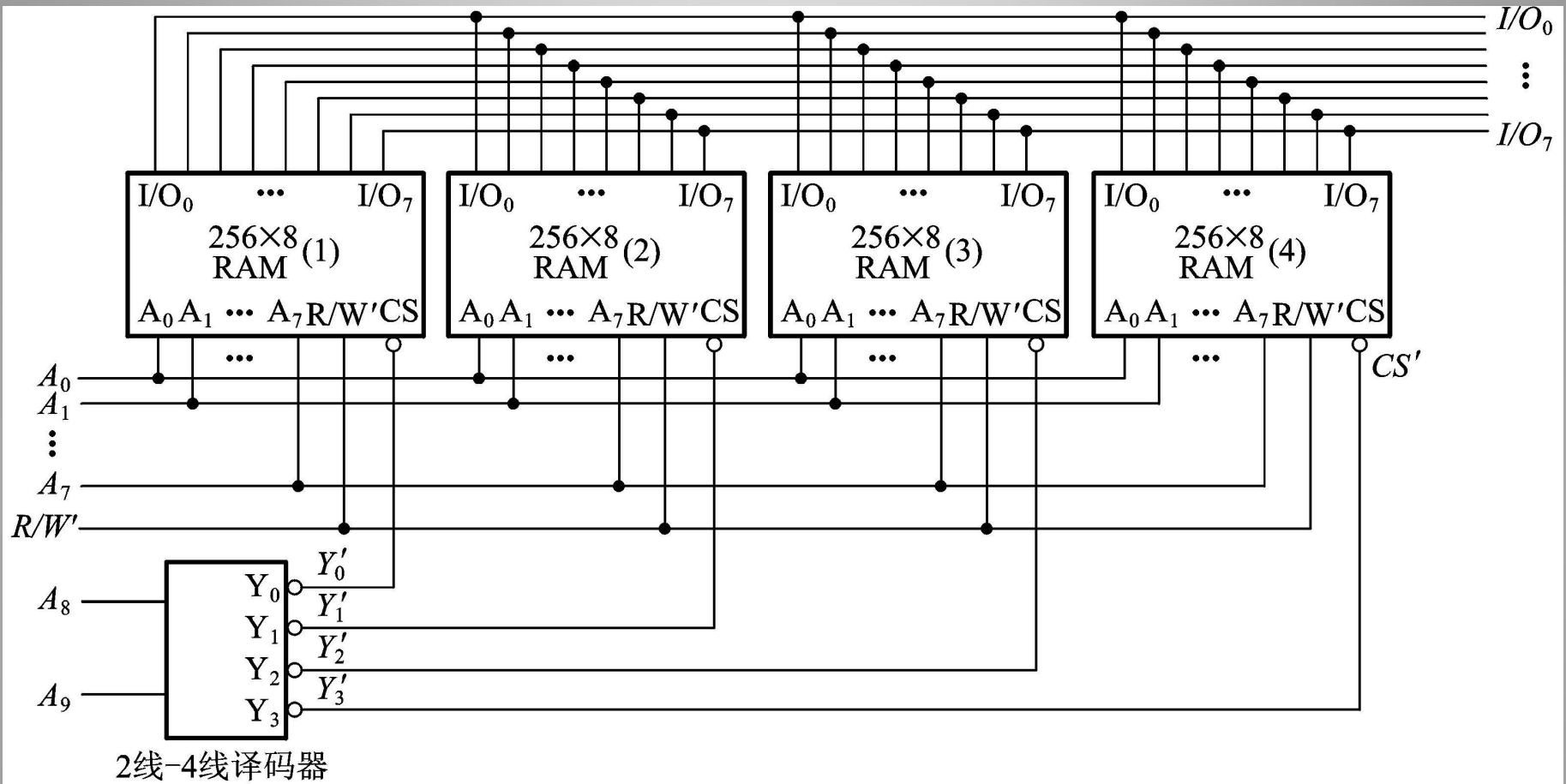
即将 $A_9 A_8$ 译成 $Y'_0 \sim Y'_3$, 分别接四片的 CS'

A_9	A_8	CS'_1	CS'_2	CS'_3	CS'_4
0	0	0	1	1	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	1	0

四片的地址分配就是：

00 $A_7 \sim A_0$,	01 $A_7 \sim A_0$,	10 $A_7 \sim A_0$,	11 $A_7 \sim A_0$
0 ~ 255	256 ~ 511	512 ~ 767	768 ~ 1023

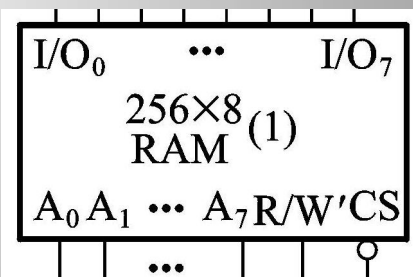
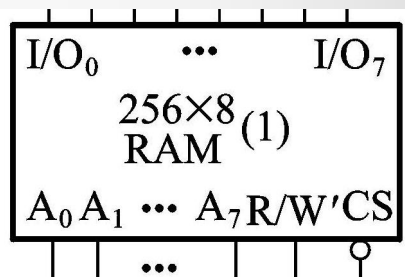
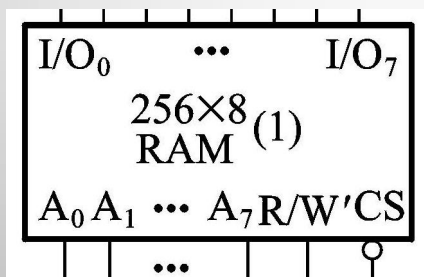
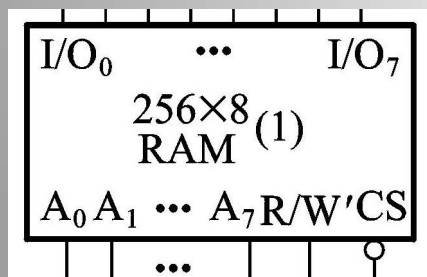


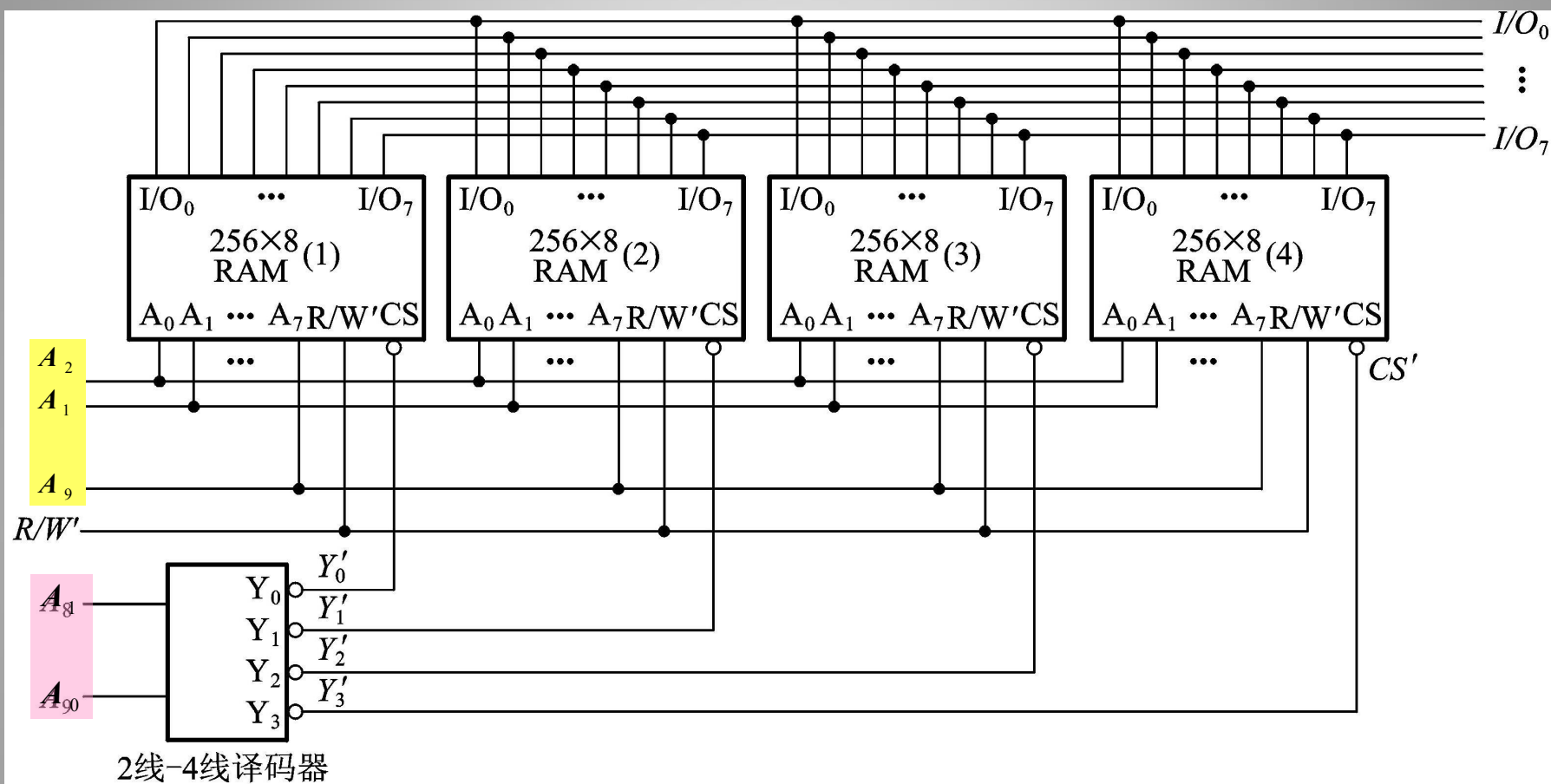


A_1	A_0	CS'_1	CS'_2	CS'_3	CS'_4
0	0	0	1	1	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	1	0

四片的地址分配就是：

$A_9 \sim A_2 00$, $A_9 \sim A_2 01$, $A_9 \sim A_2 10$, $A_9 \sim A_2 11$



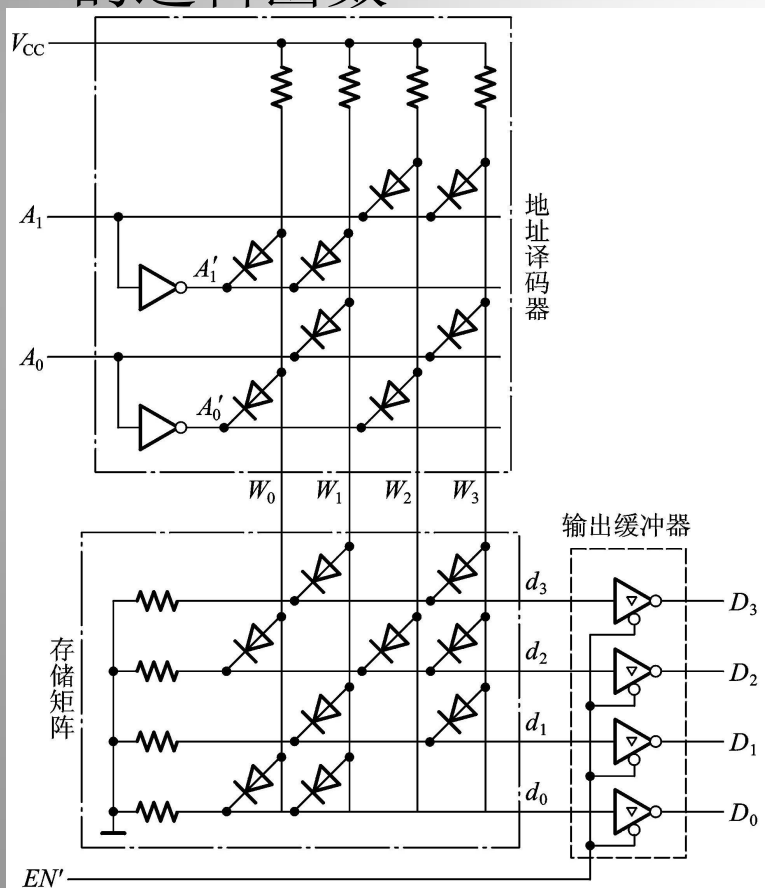


5.5.5 用存储器实现组合逻辑函数

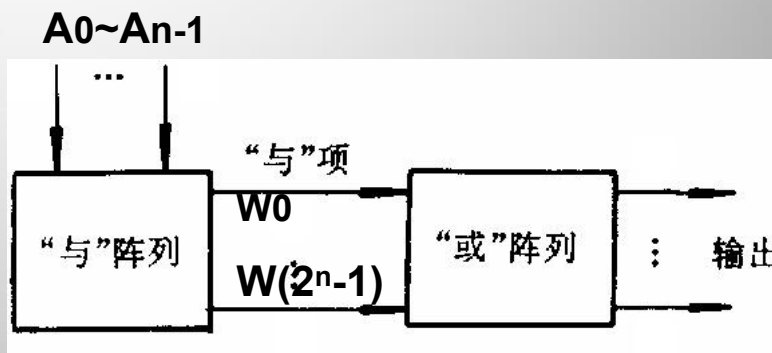
一、基本原理

从ROM的数据表可见：

若以地址线为输入变量，则数据线即为一组关于地址变量的逻辑函数



地 址		数 据			
A ₁	A ₀	D ₃	D ₂	D ₁	D ₀
0	0	0	1	0	1
0	1	1	0	1	1
1	0	0	1	0	0
1	1	1	1	1	0

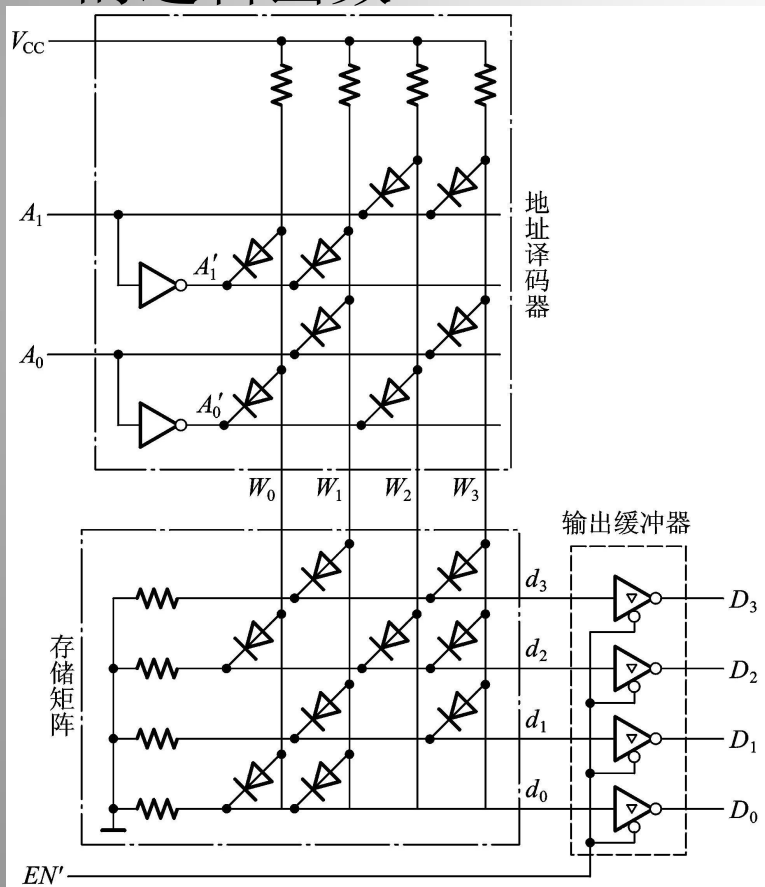


5.5.5 用存储器实现组合逻辑函数

一、基本原理

从ROM的数据表可见：

若以**地址线为输入变量**，则数据线即为一组关于地址变量的逻辑函数



地 址		数 据			
A_1	A_0	D_3	D_2	D_1	D_0
0	0	0	1	0	1
0	1	1	0	1	1
1	0	0	1	0	0
1	1	1	1	1	0

地 址		数 据			
A_1	A_0	D_3	D_2	D_1	D_0
0	0	0	1	0	1
0	1	1	0	1	1
1	0	0	1	0	0
1	1	1	1	1	0

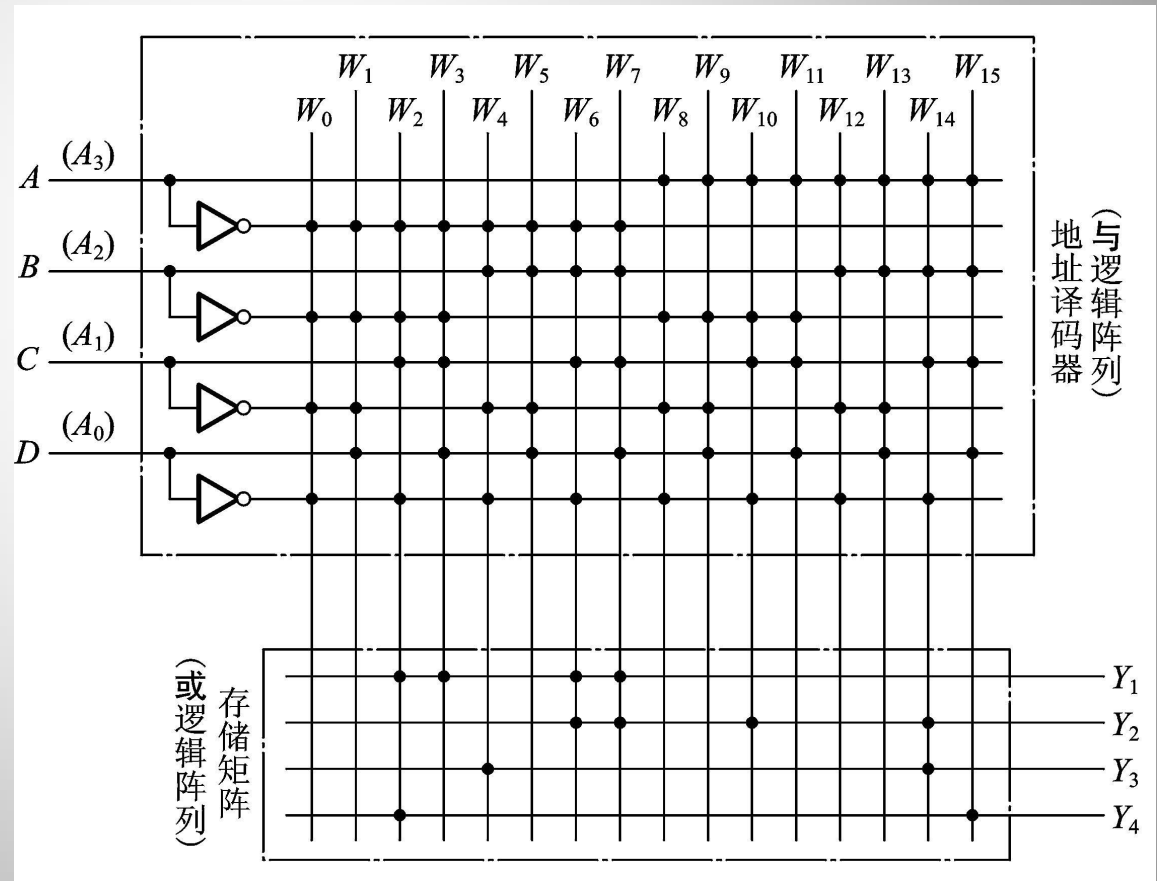
二、举例

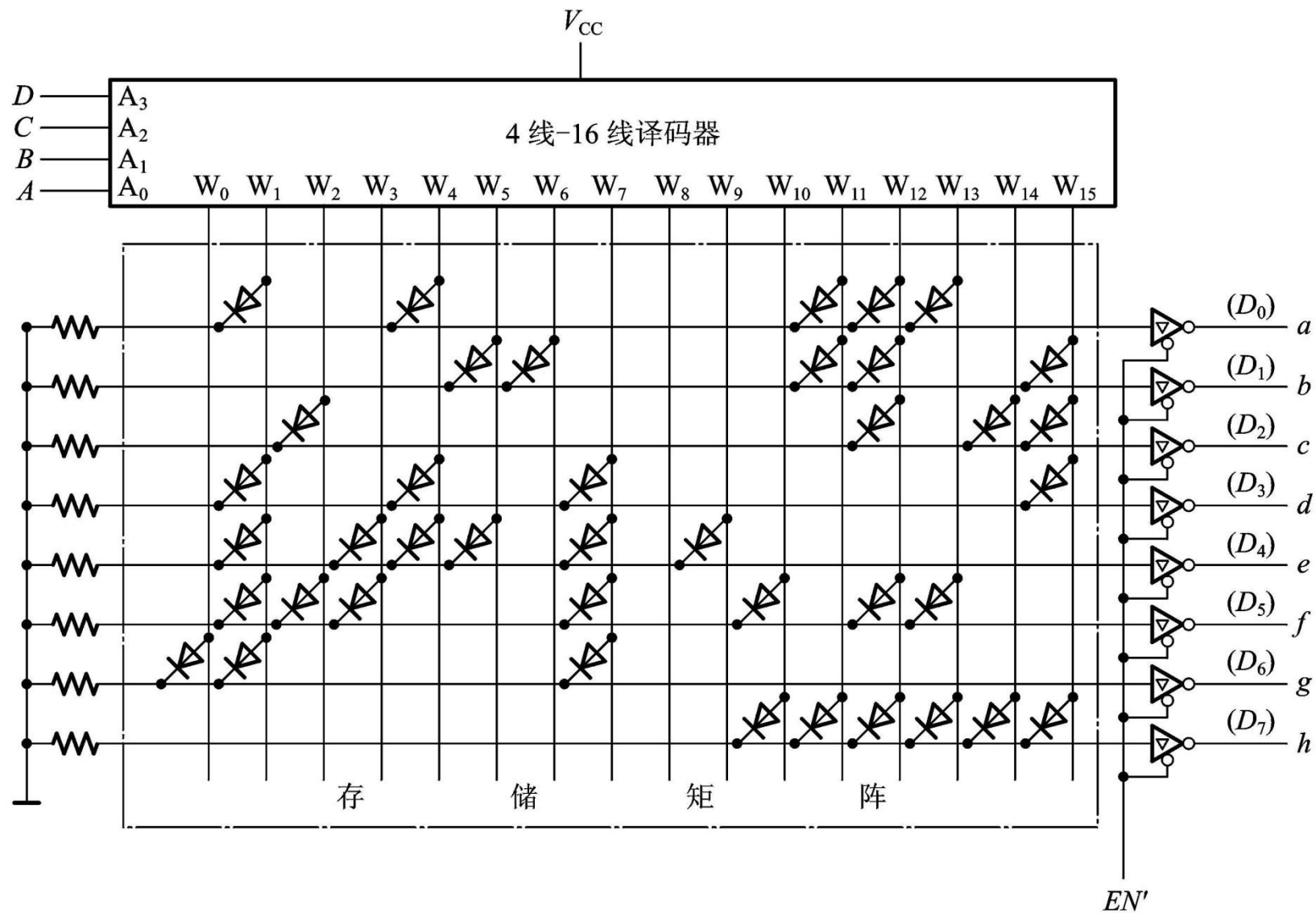
用ROM产生：

$$\begin{cases} Y_1 = A'BC + A'B'C \\ Y_2 = AB'CD' + BCD' + A'BCD \\ Y_3 = ABCD' + A'BC'D' \\ Y_4 = A'B'CD' + ABCD \end{cases}$$



$$\begin{cases} Y_1 = \sum m(2,3,6,7) \\ Y_2 = \sum m(6,7,10,14) \\ Y_3 = \sum m(4,14) \\ Y_4 = \sum m(2,15) \end{cases}$$



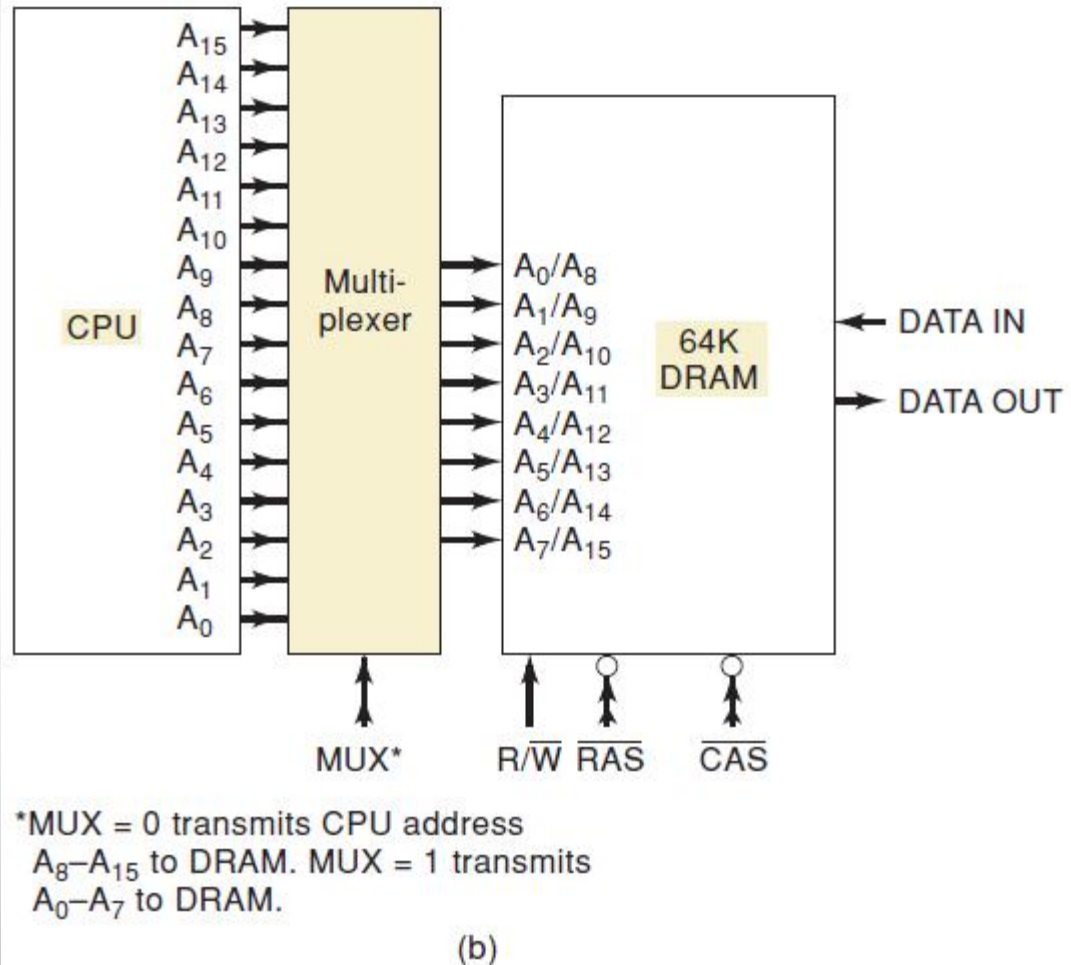
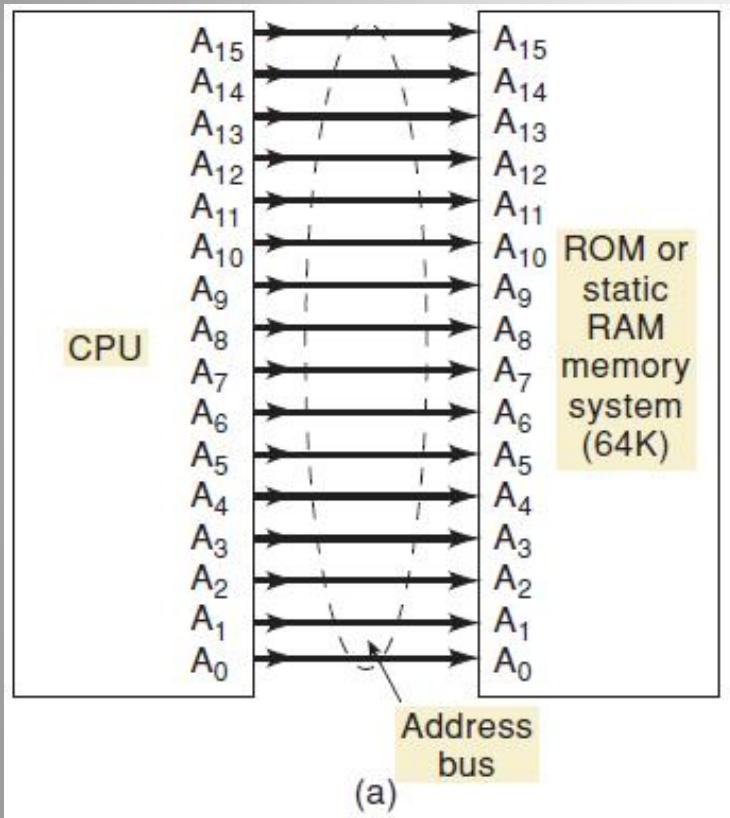


SPECIAL MEMORY FUNCTIONS

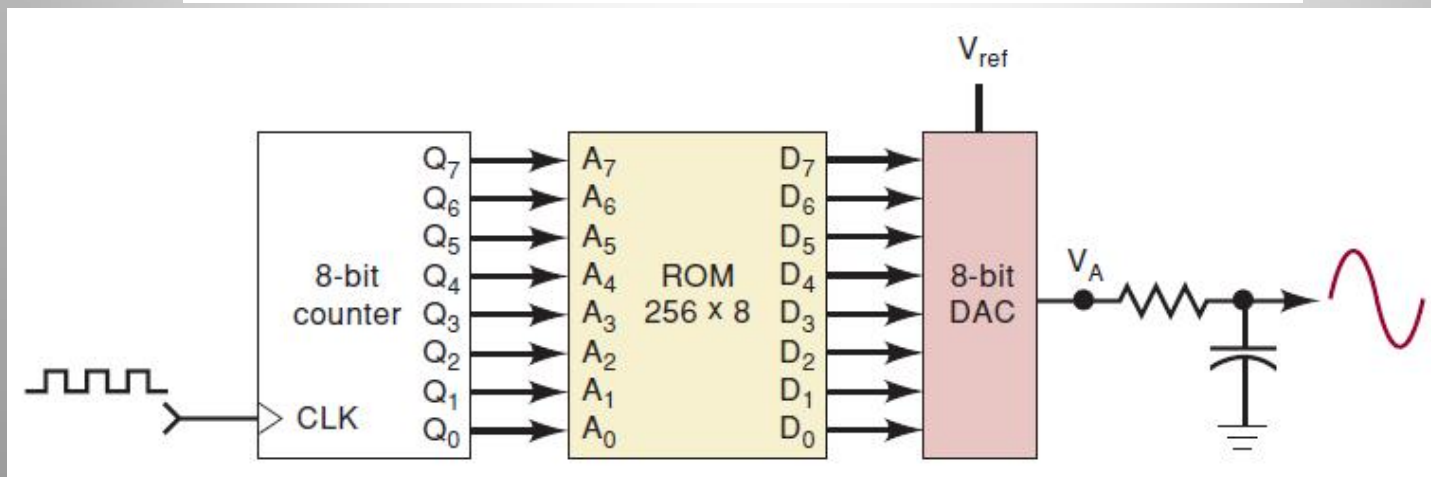
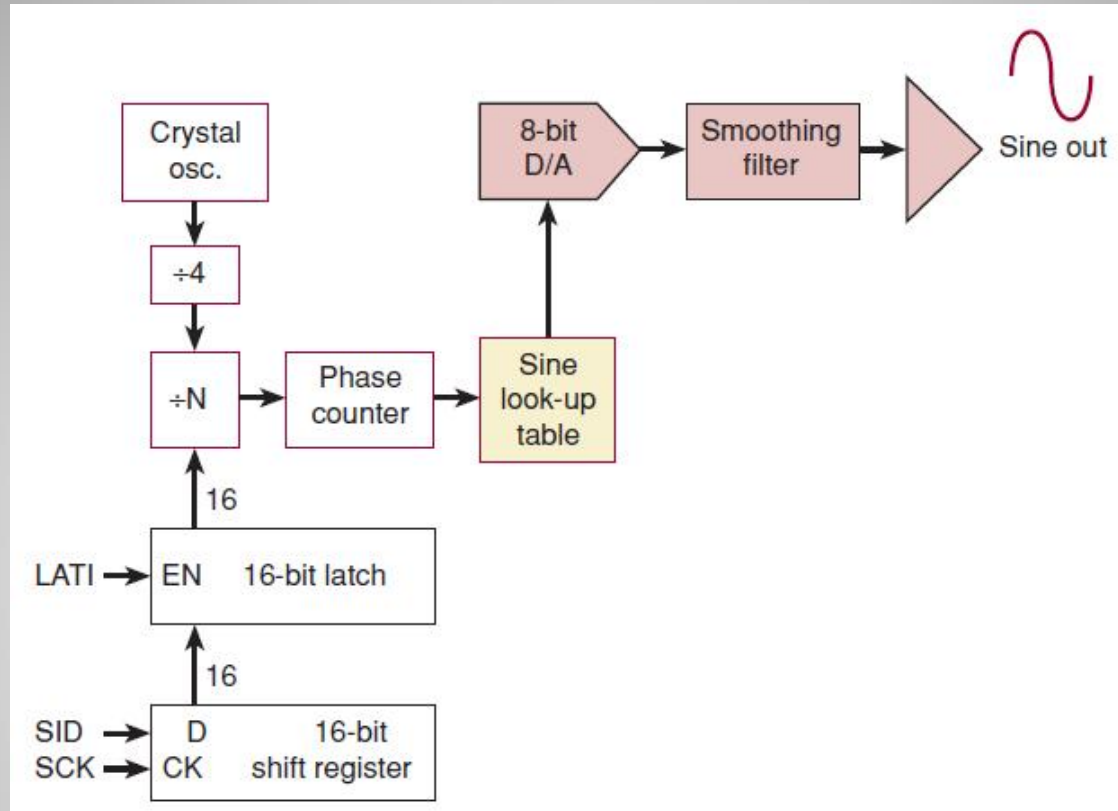
- **Cache Memory**
- **First-In, First-Out Memory (FIFO)**
- **Circular Buffers**

实例：PC to Memory

- (a) CPU address bus driving ROM or static-RAM memory;
- (b) CPU addresses driving a multiplexer that is used to multiplex the CPU address lines into the DRAM.



实例：Function Generator



课后作业

- 查阅：
 - 国际电路公司NXP, onsemi, fairchild, TI等的
 - » 触发器、同步计数器、异步计数器芯片的型号手册、电参数、速度.....
 - 图书馆资源: 电子器件天地, 软件
- 作业：
 - 《见学在浙大》
- 自学Verilog HDL语言
- 阅读：
 - 推荐英文版电子书, “Digital_Systems_Principles_and_Applications”, CHAPTER 5
 - » 详细、实用较强;
 - ch6.1-6.3;
 - Ch6.4-6.6